

The Endocrine System

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CHAPTER CONTENTS

Pituitary Gland, 971

Clinical Manifestations of Pituitary Disease, 972

Pituitary Neuroendocrine Tumors and Hyperpituitarism, 972

Lactotroph PitNET, 974

Somatotroph PitNET, 975

Corticotroph PitNET, 976

Other Anterior Pituitary Tumors, 976

Hypopituitarism, 977

Posterior Pituitary Syndromes, 977

Hypothalamic Suprasellar Tumors, 978

Thyroid Gland, 978

Hyperthyroidism, 979

Hypothyroidism, 980

Congenital Iodine Deficiency and Myxedema, 981

Thyroiditis, 981

Hashimoto Thyroiditis, 982

Chronic Lymphocytic (Painless) Thyroiditis, 983

Subacute Granulomatous Thyroiditis, 983

Graves Disease, 984

Diffuse and Multinodular Goiter, 985

Diffuse Nontoxic (Simple) Goiter, 985

Multinodular Goiter, 985

Neoplasms of the Thyroid, 987

Thyroid Adenomas, 987

Thyroid Carcinomas, 988

Papillary Carcinoma, 989

Follicular Carcinoma, 990

Noninvasive Follicular Thyroid Neoplasm With Papillary-like

Nuclear Features and Invasive Encapsulated Follicular Variant of PTC, 991

High Grade Differentiated, Poorly Differentiated, and Anaplastic (Undifferentiated) Carcinoma, 992

Medullary Carcinoma, 992

Congenital Anomalies, 993

Parathyroid Glands, 993

Hyperparathyroidism, 994

Primary Hyperparathyroidism, 994

Secondary Hyperparathyroidism, 996

Hypoparathyroidism, 997

Pseudohypoparathyroidism, 997

Endocrine Pancreas, 997

Diabetes Mellitus, 997

Diagnosis, 998

Classification, 999

Glucose Homeostasis, 999

Regulation of Insulin Release, 1000

Insulin Action and Insulin-Signaling Pathways, 1000

Pathogenesis of Type 1 Diabetes, 1001

Genetic Susceptibility, 1001

Environmental Factors, 1002

Mechanisms of β -Cell Destruction, 1002

Pathogenesis of Type 2 Diabetes, 1003

Genetic Factors, 1003

Environmental Factors, 1003

Metabolic Defects in Type 2 Diabetes, 1003

Monogenic Forms of Diabetes, 1004

Genetic Defects in β -Cell Function, 1004

Genetic Defects That Impair Tissue Response to Insulin, 1004

Diabetes and Pregnancy, 1005

Clinical Features of Diabetes, 1005

The Classic Triad of Type 1 Diabetes, 1005

Acute Metabolic Complications of Diabetes, 1005

Chronic Complications of Diabetes, 1006

Morphology of Chronic Complications of Diabetes, 1007

Clinical Manifestations of Chronic Diabetes, 1010

Pancreatic Neuroendocrine Neoplasms, 1011

Hyperinsulinism (Insulinoma), 1011

Zollinger-Ellison Syndrome (Gastrinoma), 1012

Other Rare Pancreatic Endocrine Neoplasms, 1012

Adrenal Glands, 1012

Adrenal Cortex, 1012

Adrenocortical Hyperfunction (Hyperadrenalism), 1012

Hypercortisolism (Cushing Syndrome), 1012

Primary Hyperaldosteronism, 1015

Adrenogenital Syndromes, 1017

Adrenocortical Insufficiency, 1018

Primary Acute Adrenocortical Insufficiency, 1018

Waterhouse-Friderichsen Syndrome, 1019

Primary Chronic Adrenocortical Insufficiency (Addison Disease), 1019

Secondary Adrenocortical Insufficiency, 1021

Adrenocortical Neoplasms, 1021

Other Adrenal Lesions, 1022

Adrenal Medulla, 1023

Pheochromocytoma, 1023

Multiple Endocrine Neoplasia Syndromes, 1025

Multiple Endocrine Neoplasia, Type 1, 1025

Multiple Endocrine Neoplasia, Type 2, 1026

Pineal Gland, 1026

The endocrine system consists of a highly integrated and widely distributed group of organs, called *glands*, that orchestrate a state of metabolic equilibrium among the various organs of the body. Signaling by secreted molecules can be classified into three types—autocrine, paracrine, or endocrine—on the basis of the distance over which the signal acts. **In endocrine signaling, the secreted molecules, also known as hormones, act on target cells that are distant from their sites of synthesis.** An endocrine hormone is carried by the blood from its site of release to its target organ. Interestingly, the production of several (but not all) hormones from endocrine glands is stimulated by trophic factors released from the pituitary. The endocrine hormone generally inhibits production of the trophic factors, a process known as *feedback inhibition*, thus maintaining physiologic levels of the hormone.

Endocrine diseases can be generally classified as (1) diseases of underproduction or overproduction of hormones and their resulting biochemical and clinical consequences, and (2) diseases associated with the development of mass lesions. Such lesions might be nonfunctional, or they might be associated with overproduction or underproduction of hormones. The study of endocrine diseases requires integration of morphologic findings with biochemical measurements of the levels of hormones, their regulators, and other metabolites.

PITUITARY GLAND

The pituitary gland is a small, bean-shaped structure that lies at the base of the brain within the sella turcica. Its function is controlled by the hypothalamus, to which it is connected by a stalk containing axons extending from the hypothalamus and a rich venous plexus. Along with the hypothalamus, the pituitary has a central role in regulating the function of many of the other endocrine glands.

The pituitary gland is composed of two morphologically and functionally distinct components: the anterior lobe (adenohypophysis) and the posterior lobe (neurohypophysis). **The anterior pituitary, which constitutes about 80% of the gland, produces trophic hormones that stimulate the production of hormones from the thyroid, adrenal glands, ovaries, and testes.** The anterior pituitary is composed of epithelial cells derived embryologically from the developing oral cavity. In routine histologic sections, it contains a colorful array of cells that variously have eosinophilic cytoplasm (*acidophils*), basophilic cytoplasm (*basophils*), or poorly staining cytoplasm (*chromophobe cells*) (Fig. 24.1). Detailed studies have demonstrated that the distinct staining properties of these cells are related to the presence of different polypeptide hormones within their cytoplasm that control the activity of other endocrine glands. There are six terminally differentiated cell types in the anterior pituitary, each of which is defined by the hormones that it synthesizes:

- *Somatotrophs* produce growth hormone (GH).
- *Mammotrophs* produce GH and prolactin (PRL).
- *Lactotrophs* produce PRL.
- *Corticotrophs* produce adrenocorticotrophic hormone (ACTH or corticotropin).
- *Thyrotrophs* produce thyroid-stimulating hormone (TSH).
- *Gonadotrophs* produce follicle-stimulating hormone (FSH) and luteinizing hormone (LH). In females, FSH stimulates the formation of graafian follicles in the ovary, and LH induces ovulation and the formation of corpora lutea in the ovary. The same two hormones also regulate spermatogenesis and testosterone production in males.

The production of most pituitary hormones is controlled by positively and negatively acting factors made in the hypothalamus (Fig. 24.2) that are carried to the anterior pituitary by the hypophyseal portal venous plexus. While most hypothalamic factors promote

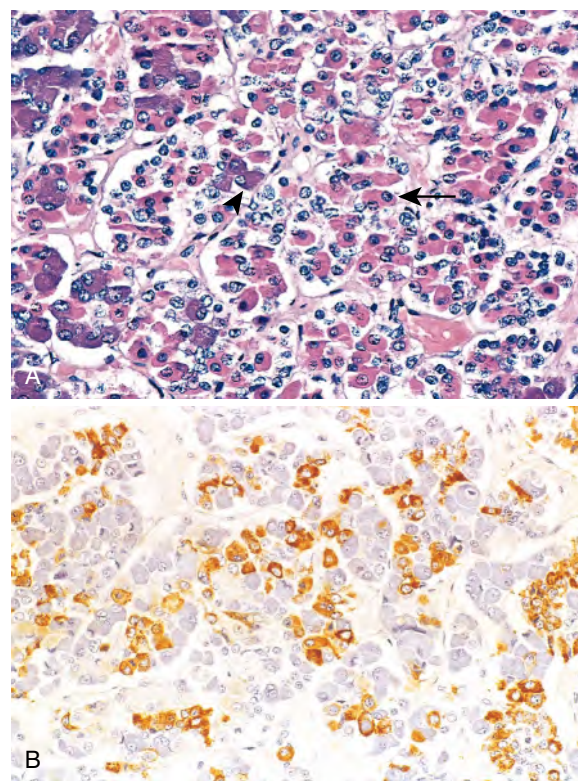


Fig. 24.1 (A) Photomicrograph of a normal pituitary. The gland is populated by several distinct cell populations containing a variety of stimulating (trophic) hormones. Each hormone has different staining characteristics, resulting in a mixture of cell types in routine histologic preparations (*arrow* denotes acidophil, *arrowhead* denotes basophil). (B) Immunostain for human growth hormone.

pituitary hormone release, others (e.g., somatostatin and dopamine) are inhibitory. Rarely, signs and symptoms of pituitary disease may be caused by over- or underproduction of hypothalamic factors, rather than a primary pituitary abnormality. During embryogenesis, specific transcription factors regulate the differentiation of multipotent stem cells within the Rathke pouch into the various cell types of the anterior pituitary. For example, somatotrophs, mammotrophs, lactotrophs, and thyrotrophs are all derived from a common precursor that expresses the transcription factor POU1F1, also known as PIT-1. By contrast, corticotrophs are derived from progenitor cells expressing the transcription factor TPIT (also known as T-box protein 19 or Tbx19), and gonadotrophs are derived from precursor cells expressing steroidogenic factor-1 (SF-1) and GATA-2. The expression of these lineage-specific transcription factors is retained in pituitary neuroendocrine tumors and is used to classify these tumors (see later).

The posterior pituitary is essentially an extension of the hypothalamus and consists of modified glial cells (termed pituicytes) and axonal processes extending from the hypothalamus through the pituitary stalk to the posterior lobe (axon terminals). Two peptide hormones are secreted from the posterior pituitary, *oxytocin* and *antidiuretic hormone* (ADH, also called *arginine vasopressin* or AVP). These hormones are actually synthesized in the hypothalamus and are transported through axons to the posterior pituitary. In response to appropriate stimuli, the preformed hormones are then released directly into the systemic circulation. For example, dilation of the cervix in pregnancy results in oxytocin release, leading to contraction of the uterine smooth muscle, facilitating parturition (uterine labor). Similarly, oxytocin released on nipple stimulation in the

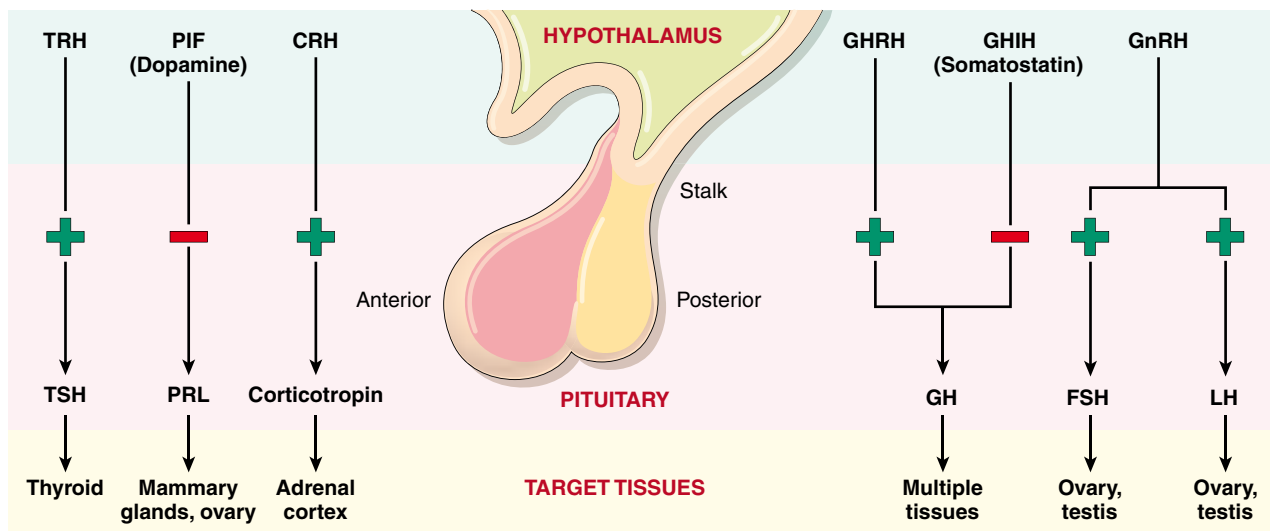


Fig. 24.2 The hypothalamic-pituitary axis. The hypothalamus regulates the secretion of hormones from the adenohypophysis (anterior pituitary gland) by releasing stimulatory (*plus signs*) and inhibitory factors (*minus signs*). These in turn modulate the release of six hormones from the anterior pituitary: adrenocorticotropic hormone (*corticotropin*); follicle-stimulating hormone (*FSH*); growth hormone (*GH*, or somatotropin); luteinizing hormone (*LH*); prolactin (*PRL*); and thyroid-stimulating hormone (*TSH*, or thyrotropin). *CRH*, Corticotropin-releasing hormone; *GHIH*, growth hormone-inhibiting hormone; *GHRH*, growth hormone-releasing hormone; *GnRH*, gonadotropin-releasing hormone; *PIF*, prolactin-inhibiting factor; *TRH*, thyrotropin-releasing hormone.

postnatal period acts on the smooth muscles surrounding the lactiferous ducts of the mammary glands and facilitates lactation. Synthetic oxytocin can be given to pregnant females to induce labor. The most important function of ADH is to conserve water by restricting diuresis during periods of dehydration and hypovolemia. Decreased blood pressure, sensed by baroreceptors (pressure-sensing receptors) in the cardiac atria and carotid arteries, stimulates ADH release. An increase in plasma osmotic pressure detected by osmoreceptors also triggers ADH secretion. In contrast, states of hypervolemia and increased atrial distention result in inhibition of ADH secretion.

CLINICAL MANIFESTATIONS OF PITUITARY DISEASE

The manifestations of pituitary disorders are related to either excess or deficiency of pituitary hormones, or to mass effects.

- **Hyperpituitarism** arises from excess secretion of trophic hormones. Hyperpituitarism is usually caused by hormone-secreting pituitary neuroendocrine tumors. Other causes include hypersecretion of hormones by nonpituitary tumors and due to certain hypothalamic disorders. The symptoms of hyperpituitarism are discussed later in the context of individual tumors.
- **Hypopituitarism** results from deficiency of trophic hormones. It may be caused by ischemic injury, surgery or radiation, inflammatory disorders, and the mass effects of nonfunctional pituitary tumors.
- **Symptoms related to local mass effects.** Because of the close proximity of the optic nerves and chiasm to the sella, expanding pituitary lesions often compress crossing fibers in the optic chiasm. This gives rise to visual field abnormalities, classically in the form of defects in the lateral (temporal) visual fields, so-called bitemporal hemianopia. Like any expanding intracranial mass, pituitary tumors can produce signs and symptoms of elevated intracranial pressure, including headache, nausea, and vomiting. On occasion, hemorrhage into a tumor can lead to rapid worsening of hypopituitarism because of acute enlargement of the lesion, a situation termed *pituitary apoplexy*. Pituitary apoplexy is often a neurosurgical emergency because it can cause sudden death (see later).

Diseases of the posterior pituitary often come to clinical attention because of increased or decreased secretion of ADH and associated changes in fluid and electrolyte balances.

PITUITARY NEUROENDOCRINE TUMORS AND HYPERPITUITARISM

The most common cause of hyperpituitarism is a pituitary neuroendocrine tumor (PitNET) arising in the anterior lobe. Pituitary tumors are classified on the basis of the hormones and cell type-specific transcription factors that are expressed by the tumor cells (Table 24.1). Some PitNETs secrete two hormones (GH and prolactin being the most common combination), and rarely pituitary tumors are plurihormonal. Less common causes of hyperpituitarism include some hypothalamic disorders. By contrast, large PitNETs, particularly nonfunctioning ones, may cause hypopituitarism by destroying the adjacent normal anterior pituitary parenchyma.

PitNETs are usually found in adults; the peak incidence is from 35 to 60 years of age. Nonfunctional tumors are likely to come to clinical attention at a later stage than those associated with endocrine abnormalities and are therefore more likely to be larger in size. Based on autopsy studies, the prevalence of pituitary tumors in the population has been estimated to be about 14%, but most of these lesions are clinically silent. As behavior is notoriously difficult to predict, PitNETs are no longer classified as adenomas or carcinomas, but as localized or metastatic PitNETs.

Pathogenesis

As with other neoplasms, PitNETs are caused by mutations in cancer genes, which are most commonly acquired somatic mutations but which may also be germline mutations associated with an inherited predisposition to pituitary neoplasms (Table 24.2):

- **Activating G-protein mutations** are one of the most common alterations in PitNETs. As described in Chapter 1, G-proteins normally play a critical role in signal transduction, transmitting signals from particular cell surface receptors (e.g., growth hormone releasing hormone [GHRH] receptor) to intracellular effectors (e.g., adenylyl cyclase), which then generate second messengers (e.g., cyclic adenosine monophosphate,

TABLE 24.1 Classification of Pituitary Neuroendocrine Tumors (PitNETs)

PitNET Type	Hormone	Transcription Factor	Morphologic Variant	Associated Syndrome ^a
Somatotroph PitNET	GH	POUF1	Densely granulated	Gigantism (children)
	GH	POUF1	Sparsely granulated	Acromegaly (adults)
	GH, PRL (in same cells)	POUF1, ER α	Mammotroph	
	GH, PRL (in different cells)	POUF1, ER α	Mixed somatotroph-lactotroph	
Lactotroph PitNET	PRL	POUF1, ER α	Sparsely granulated	Galactorrhea and amenorrhea (in females)
	PRL	POUF1, ER α	Densely granulated	
	PRL, GH (focal and variable)	POUF1, ER α	Acidophilic stem cell	Sexual dysfunction, infertility
Thyrotroph PitNET	TSH	POUF1	Thyrotroph	Hyperthyroidism
Corticotroph PitNET	Corticotropin	TBX19	Densely granulated	Cushing syndrome
	Corticotropin	TBX19	Sparsely granulated	Nelson syndrome
	Corticotropin	TBX19	Crooke cell (prominent intracytoplasmic cytokeratin filaments)	Mass effects (20% of corticotroph tumors are hormonally silent)
Gonadotroph PitNET	FSH, LH	SF-1, GATA-2, ER α	Gonadotroph	Mass effects and hypopituitarism (most gonadotroph tumors are hormonally silent)
Null cell PitNET	None	None		Mass effects
Plurihormonal PitNET	GH, PRL, TSH	POUF1		

^aNote that nonfunctional ("silent") tumors in each category express the corresponding hormone(s) and transcription factor within the neoplastic cells, as determined by special immunohistochemical staining on tissues. However, these tumors do not produce the associated clinical syndrome and typically present with *mass effects* accompanied by *hypopituitarism* due to destruction of normal pituitary parenchyma. These features are particularly common with gonadotroph tumors and up to one-fifth of corticotroph tumors, while rarely observed with somatotroph or lactotroph tumors. Null cell tumors, by definition, only present with mass effects. FSH, Follicle-stimulating hormone; GH, growth hormone; LH, luteinizing hormone; PRL, prolactin; TSH, thyroid-stimulating hormone.

Modified from Lopes MBS. The 2017 World Health Organization classification of tumors of the pituitary gland: a summary. *Acta Neuropathol.* 2017;134:521-535.

TABLE 24.2 Genetic Alterations in Pituitary Tumors

Gene	Protein Function	Oncogenic Mutations	Most Commonly Associated Pituitary Tumor
<i>GNAS</i>	α subunit of stimulatory G-protein, G α	Somatic activating mutation	Somatotroph tumor
<i>USP8</i>	Deubiquitinase	Somatic activating mutation	Corticotroph tumor
Protein kinase A, regulatory subunit 1 α (<i>PRKAR1A</i>) ^a	Negative regulator of protein kinase A (PKA), leading to increased cAMP production	Germline inactivating mutations (Carney complex)	Somatotroph or lactotroph tumor
<i>MEN1</i> ^a	Transcription regulator	Germline inactivating mutations (multiple endocrine neoplasia, type 1)	Somatotroph, lactotroph, or corticotroph tumor
<i>CDKN1B</i> (p27/KIP1) ^a	Negative cell cycle regulator	Germline inactivating mutations ("MEN-1-like" syndrome)	Corticotroph tumor
Aryl hydrocarbon receptor interacting protein (<i>AIP</i>) ^a	Receptor for aryl hydrocarbons and a ligand-activated transcription factor	Germline inactivating mutations (familial isolated pituitary NET syndrome)	Somatotroph or lactotroph tumor (especially in patients younger than 35 years)
<i>HRAS</i>	Mitogenic signaling, cell growth and survival	Somatic activating mutation	Metastatic pituitary tumor
<i>DICER1</i> ^a	MicroRNA processing	Germline inactivating mutation	Pituitary blastoma

^aGenetic alterations associated with familial or germline predisposition to pituitary neuroendocrine tumors.

cAMP). They are heterotrimeric proteins, composed of α -subunits that bind guanine nucleotide and interact with cell surface receptors and intracellular effectors (Fig. 24.3), and β - and γ -subunits that noncovalently bind α -subunits. G α is a stimulatory G-protein with a pivotal role in signal transduction in several endocrine organs, including the pituitary. The α -subunit of G α (G α) is encoded by the guanine nucleotide binding protein, alpha stimulating activity (GNAS) gene. In the basal state, G α exists in an inactive state, with guanosine diphosphate (GDP) bound to the guanine nucleotide-binding site of G α . On interaction with the ligand-bound cell surface receptor, GDP dissociates, and guanosine triphosphate (GTP) binds to G α , activating the G-protein. The activation of G α generates cAMP, a potent mitogen for several types of endocrine cells (e.g., pituitary somatotrophs and corticotrophs, thyroid follicular cells, parathyroid cells). Normally, G α activation is transient

because of an intrinsic GTPase activity in the α -subunit, which hydrolyzes GTP into GDP. Approximately 40% of somatotroph cell tumors bear somatic *GNAS* mutations that abrogate the GTPase activity of G α , leading to constitutive activation of G α , persistent generation of cAMP, and unchecked cellular proliferation (Table 24.2). *GNAS* mutations have also been described in a minority of corticotroph tumors; in contrast, *GNAS* mutations are absent in thyrotroph, lactotroph, and gonadotroph tumors, because their trophic hypothalamic release hormones act via other signaling pathways.

- Activating mutations of *ubiquitin-specific protease 8* (*USP8*) occur in 30% to 60% of corticotroph tumors. The encoded protein is an enzyme that removes ubiquitin residues from proteins like epidermal growth factor receptor (EGFR), protecting them from proteasome-dependent degradation. Aberrant activation of *USP8* thus

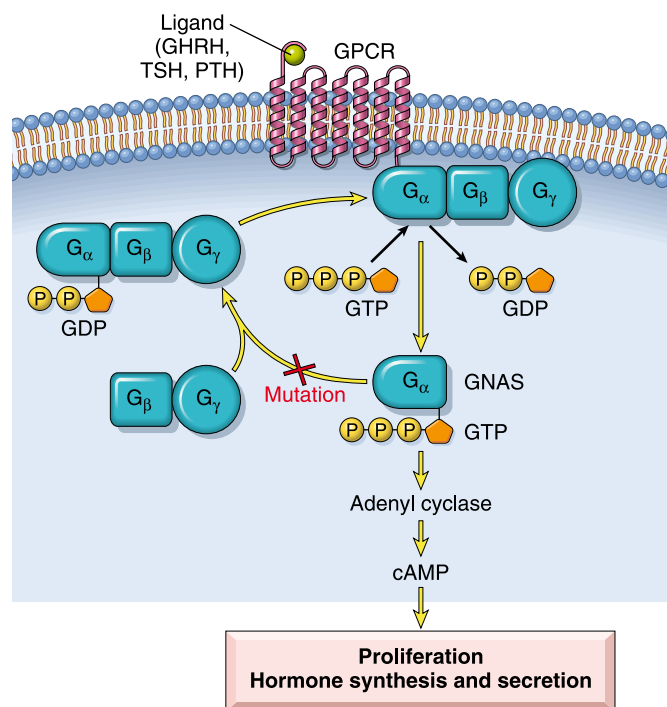


Fig. 24.3 G-protein signaling in endocrine neoplasia. Mutations that lead to G-protein hyperactivity are seen in a variety of endocrine neoplasms, including pituitary, thyroid, and parathyroid adenomas. G-proteins (composed of α , β , and γ subunits) play a critical role in signal transduction, transmitting signals from cell surface receptors (GHRH, TSH, or PTH receptor) to intracellular effectors (e.g., adenyl cyclase), which then generate second messengers (cAMP) that stimulate cellular responses. *cAMP*, Cyclic adenosine monophosphate; *GDP*, guanosine diphosphate; *GHRH*, growth hormone releasing hormone; *GPCR*, G-protein coupled receptor; *GTP*, guanosine triphosphate; *PTH*, parathyroid hormone; *Pi*, inorganic phosphate; *TSH*, thyroid stimulating hormone.

enhances the activity of EGFR and other progrowth signaling pathways in pituitary tumors.

- Approximately 5% of pituitary tumors are caused by germline loss-of-function mutations in genes such as *MEN1*, *CDKN1B*, *PRKAR1A*, or *AIP* (see Table 24.2). Of note, somatic mutations of these four genes are rarely encountered in sporadic pituitary tumors.

MORPHOLOGY

The typical **PitNET** is soft and well-circumscribed. Small neoplasms may be confined to the sella turcica, but with expansion they frequently erode the sella turcica and anterior clinoid processes. Larger lesions may extend superiorly through the diaphragm sella into the suprasellar region, compressing the optic chiasm and adjacent structures, such as cranial nerves (Fig. 24.4). In as many as 30% of cases, the tumors are not encapsulated and infiltrate neighboring tissues such as the cavernous and sphenoid sinuses, dura, and on occasion, the brain itself.

Histologically, PitNETs are usually composed of uniform (monomorphic), polygonal cells arrayed in sheets or cords. The supporting connective tissue, or reticulin, is sparse, accounting for the soft, gelatinous consistency of many of these tumors. This cellular monomorphism and the absence of a significant reticulin network distinguish pituitary tumors from normal anterior pituitary parenchyma (Fig. 24.5). Immunohistochemical stains for hormones and lineage-specific transcription factors are used to identify specific PitNET subtypes (see Table 24.1). Mitotic activity and expression of Ki-67 (MIB-1, a marker of cycling cells) correlate with tumor aggressiveness.



Fig. 24.4 Pituitary neuroendocrine tumor (PitNET). This massive, non-functional PitNET has grown beyond the confines of the sella turcica, distorting the overlying brain. On average, nonfunctional PitNETs are larger at the time of diagnosis.

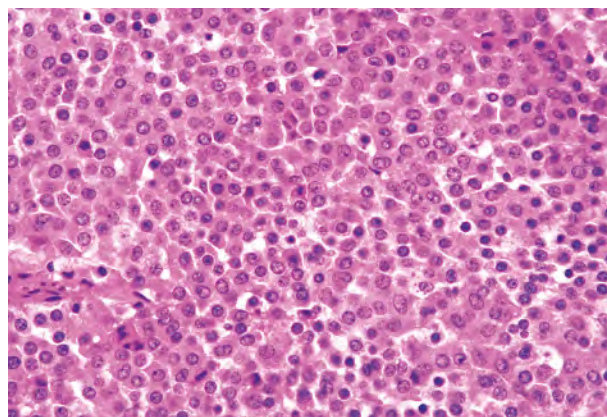


Fig. 24.5 Pituitary neuroendocrine tumor (PitNET). The monomorphism of these cells contrasts markedly with the mixture of cells seen in the normal anterior pituitary. Note also the absence of a reticulin network.

Clinical Features

The signs and symptoms of PitNETs are related to endocrine abnormalities and mass effects. The effects of excessive secretion of anterior pituitary hormones are mentioned later, when the specific types of PitNET are described. Local mass effects may be produced by any type of pituitary tumor. As mentioned earlier, these effects include radiographic abnormalities of the sella turcica, visual field abnormalities, signs and symptoms of elevated intracranial pressure, and occasionally hypopituitarism. Acute hemorrhage into a PitNET is sometimes associated with *pituitary apoplexy*, as noted earlier.

The following is a description of the individual types of tumors.

Lactotroph PitNET

Prolactin-secreting lactotroph PitNETs are the most common type of hyperfunctioning pituitary tumor, accounting for about 30% of clinically recognized cases.

MORPHOLOGY

The large majority of lactotroph tumors are composed of chromophobe cells with juxtanuclear localization of the transcription factor POU1F1; these are known as **sparsely granulated lactotroph tumors** (Fig. 24.6A). Much rarer

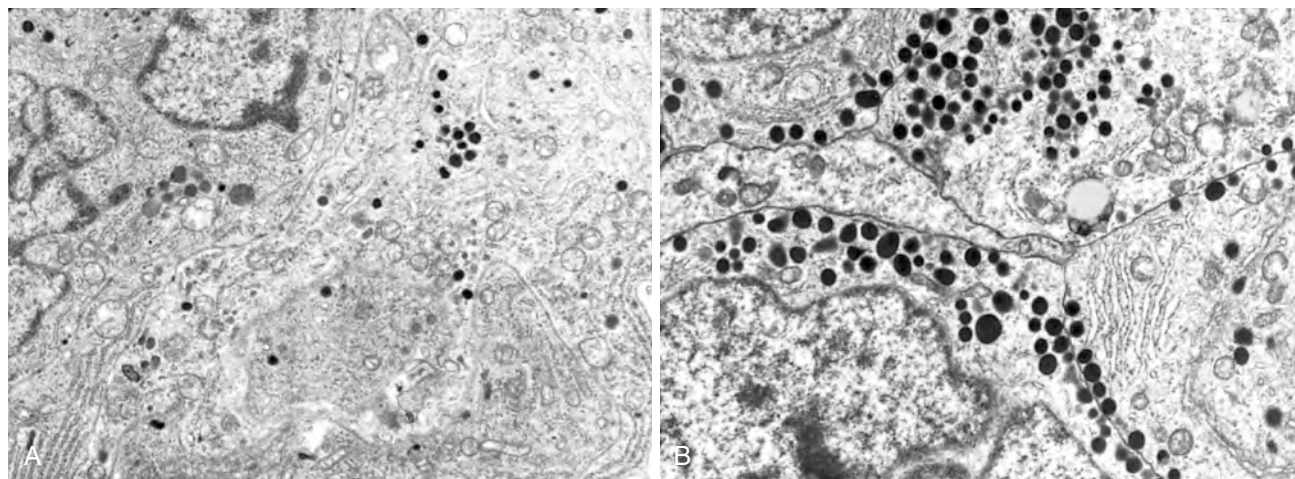


Fig. 24.6 Ultrastructural features of lactotroph pituitary neuroendocrine tumor. (A) Electron micrograph of a sparsely granulated lactotroph tumor. The tumor cells contain abundant granular endoplasmic reticulum (indicative of active protein synthesis) and small numbers of electron-dense secretory granules. (B) Electron micrograph of a densely granulated prolactin-secreting neuroendocrine tumor. The tumor cells are filled with numerous large, electron-dense secretory granules. (Courtesy Dr. Eva Horvath, St. Michael's Hospital, Toronto.)

are eosinophilic **densely granulated lactotroph tumors**, characterized by diffuse cytoplasmic POU1 localization (Fig. 24.6B). Prolactin can be demonstrated within cytoplasmic secretory granules with immunohistochemical stains, and estrogen receptor alpha (ER α) is coexpressed along with POU1, consistent with lactotroph differentiation. Lactotroph tumors often undergo dystrophic calcification, ranging from isolated psammoma bodies to extensive calcification ("pituitary stone"). Prolactin secretion by functioning tumors is usually efficient and proportional, in that serum prolactin concentrations tend to correlate with the size of the tumor.

Clinical Features

Increased serum levels of prolactin, or *hyperprolactinemia*, cause amenorrhea, galactorrhea, loss of libido, and infertility. The diagnosis of a PitNET is made more readily in females than in males, especially between 20 and 40 years of age, because hyperprolactinemia disrupts the menstrual cycle. Lactotroph PitNET is the cause of almost one-quarter of cases of amenorrhea. In contrast, in males and older females, the hormonal manifestations may be subtle, allowing the tumors to reach considerable size before being detected clinically.

Hyperprolactinemia may result from causes other than prolactin-secreting PitNETs. Physiologic hyperprolactinemia occurs in pregnancy. Prolactin levels are also elevated by nipple stimulation, as occurs during suckling in lactating females, and as a response to many types of stress. Pathologic hyperprolactinemia can also result from lactotroph hyperplasia caused by loss of dopamine-mediated inhibition of prolactin secretion. This may occur due to damage of the dopaminergic neurons of the hypothalamus or the pituitary stalk (e.g., due to head trauma), or exposure to drugs that block dopamine receptors on lactotroph cells. Any suprasellar mass (e.g., a PitNET) may disturb the inhibitory influence of the hypothalamus on prolactin secretion, resulting in hyperprolactinemia. Therefore, mild hyperprolactinemia in a person with a PitNET does not necessarily indicate a prolactin-secreting tumor. Other causes of hyperprolactinemia include renal failure and hypothyroidism. Lactotroph PitNETs are treated by surgery, or more commonly with bromocriptine or cabergoline, dopamine receptor agonists that cause the lesions to diminish in size.

Somatotroph PitNET

Growth hormone (GH)-secreting somatotroph PitNETs are the second most common type of functioning PitNET, and cause gigantism in children and acromegaly in adults. Somatotroph PitNET may be quite large by the time they come to clinical attention because the manifestations of excessive GH may be subtle.

MORPHOLOGY

Histologically, pure somatotroph PitNETs are also classified into densely granulated and sparsely granulated subtypes. **Densely granulated somatotroph PitNETs** are composed of monomorphic, eosinophilic cells with granular cytoplasm and a large central nucleus with prominent nucleoli. The cells have diffuse, strong immunoreactivity for GH. In contrast, the **sparsely granulated variants** are composed of chromophobe cells with patchy, weak staining for GH and a characteristic paranuclear glossy inclusion known as a fibrous body, composed of intermediate filaments that stain for cytokeratin. Bihormonal **mammotroph** PitNETs that synthesize GH and prolactin in the same cell are being increasingly recognized; morphologically, most resemble densely granulated somatotroph PitNETs, but have immunohistochemical reactivity for prolactin and GH. These are to be contrasted with mixed **somatotroph-lactotroph PitNETs**, which have GH and prolactin expression in different cells.

Clinical Features

In contrast to corticotroph or gonadotroph PitNETs, silent somatotroph PitNETs are rare. Persistently elevated levels of GH stimulate the hepatic secretion of insulin-like growth factor-1 (IGF-1), which causes many of the clinical manifestations.

- If a somatotroph PitNET appears in children before the epiphyses have closed, the elevated levels of GH (and IGF-1) result in *gigantism*. This is characterized by a generalized increase in body size with disproportionately long arms and legs.
- If the levels of GH are increased after closure of the epiphyses, *acromegaly* develops. In this condition, growth is most conspicuous in skin and soft tissues, viscera (thyroid, heart, liver, and adrenals), and the bones of the face, hands, and feet. Bone density may increase

(*hyperostosis*) in both the spine and the hips. Enlargement of the jaw results in its protrusion (*prognathism*) and broadening of the lower face. The feet and hands are enlarged, and the fingers become thickened and sausage-like. In most instances, gigantism is also accompanied by evidence of acromegaly. These changes may develop slowly over decades before being recognized, and hence the underlying PitNET may reach substantial size.

- GH excess can also be associated with a variety of other disturbances, including gonadal dysfunction, diabetes, generalized muscle weakness, hypertension, arthritis, congestive heart failure, and an increased risk of gastrointestinal cancers.

The diagnosis relies on documenting elevated serum IGF-1 levels. In addition, failure to suppress GH production in response to an oral load of glucose is one of the most sensitive tests for acromegaly. The causative PitNET can be removed surgically or treated by pharmacologic means. The latter includes somatostatin analogs (recall that somatostatin inhibits pituitary GH secretion) and GH receptor antagonists, which prevent hormone binding to target organs such as the liver. When effective control of high GH levels is achieved, the characteristic tissue overgrowth and related symptoms gradually recede, and the metabolic abnormalities improve. Overall, sparsely granulated somatotroph PitNETs tend to have a more aggressive course than densely granulated PitNETs and may be less responsive to somatostatin analogs. Thus, accurate subtyping of somatotroph PitNETs is of prognostic importance.

Corticotroph PitNET

Excess production of corticotropin by functioning corticotroph PitNETs leads to adrenal hypersecretion of cortisol and the development of hypercortisolism (also known as *Cushing syndrome*).

MORPHOLOGY

Corticotroph PitNETs are usually small at the time of diagnosis. These tumors are most often basophilic (**densely granulated**) and occasionally chromophobic (**sparsely granulated**). Both variants stain positively with periodic acid–Schiff (PAS) because of the presence of carbohydrate in pro-opiomelanocortin (POMC), the precursor of corticotropin. Nuclear TPIT is also positive in the neoplastic cells, consistent with corticotroph lineage. A third uncommon variant, called **Crooke cell PitNET**, is characterized by ringlike deposition of cytokeratin called Crooke change. This variant has an aggressive natural history compared with other subtypes of corticotroph PitNET.

Clinical Features

The manifestations of Cushing syndrome are discussed in more detail later with the diseases of the adrenal gland. The syndrome can be caused by a wide variety of conditions in addition to corticotropin-producing pituitary tumors. When the hypercortisolism is due to excessive production of corticotropin by the pituitary, it is designated *Cushing disease*. Large destructive pituitary growth can develop in patients after surgical removal of the adrenal glands for treatment of Cushing syndrome. This condition, known as *Nelson syndrome*, occurs because of loss of the inhibitory effect of adrenal corticosteroids on a corticotroph NET. Because the adrenals are absent in persons with Nelson syndrome, hypercortisolism does not develop. Patients present with mass effects due to the pituitary tumor, and there can be hyperpigmentation. The cause of the hyperpigmentation is potentially due to increased levels of melanocyte-stimulating hormone (MSH), which is also derived from POMC, or potentially from the high levels of corticotropin itself being able to bind the MSH receptor in melanocytes.

Other Anterior Pituitary Tumors

Several less frequently occurring types of PitNETs also merit brief comment (see Table 24.1).

- **Gonadotroph (LH-producing and FSH-producing) PitNETs** can be difficult to recognize because they secrete small amounts of hormones that usually do not cause a recognizable clinical syndrome (i.e., most are nonfunctioning). Gonadotroph PitNETs most frequently present in middle-aged males and females with neurologic symptoms, such as impaired vision, headaches, diplopia, or pituitary apoplexy. Pituitary hormone deficiencies can also be found, most commonly impaired secretion of LH, which causes decreased energy and libido in males (due to reduced testosterone) and amenorrhea in premenopausal females. The neoplastic cells express the common gonadotropin α -subunit and the specific β -FSH and β -LH subunits; FSH is usually the predominant secreted hormone. Gonadotroph PitNETs also usually express steroidogenic factor-1 (SF-1), GATA-3, and ER α , transcription factors associated with normal gonadotroph differentiation.
- **Thyrotroph (TSH-producing) PitNETs** are uncommon, accounting for approximately 1% of PitNETs. They are a rare cause of hyperthyroidism. Due to their shared lineage with lactotroph and somatotroph NETs, these tumors also express POU1F1.
- PitNETs may also secrete multiple hormones; these tumors are usually aggressive.
- **Null cell PitNETs** do not express any markers of hormonal or lineage differentiation. Not surprisingly, null cell NETs typically present with symptoms stemming from mass effects. These lesions may also compromise the residual anterior pituitary and cause hypopituitarism, which may appear slowly due to gradual enlargement of the NET or abruptly because of acute intratumor hemorrhage (pituitary apoplexy).

Metastatic PitNETs (formerly termed *pituitary carcinomas*) are rare, accounting for less than 1% of pituitary tumors. The presence of craniospinal or systemic metastases is a *sine qua non*. Most metastatic PitNETs are functional, with prolactin and corticotropin being the most commonly secreted products. Metastases usually appear late in the course, following multiple local recurrences.

Pituitary blastoma is an entity that occurs in children (typically younger than 2 years) who carry germline mutations of *DICER1*, the gene encoding a microRNA-processing protein. Morphologically, these tumors are composed of immature “blastema-like” cells (so-called “small round blue cells”) and rosette-like formations resembling the primitive Rathke epithelium from which the pituitary develops. Pituitary blastoma presents with signs and symptoms of Cushing disease. These children also develop primitive “blastema-like” neoplasms in other organs, most commonly pleuropulmonary blastoma.

KEY CONCEPTS

Hyperpituitarism

- The most common cause of hyperpituitarism is an anterior lobe pituitary PitNET.
- PitNETs can be large (formerly macroadenomas, greater than 1 cm in diameter) or small (formerly microadenomas).
- Functioning PitNETs are associated with distinct endocrine signs and symptoms, while nonfunctioning (silent) PitNETs typically present with mass effects, including visual disturbances.
- Lactotroph PitNETs secrete prolactin and can present with amenorrhea, galactorrhea, loss of libido, and infertility.
- Somatotroph PitNETs secrete GH and present with gigantism in children and acromegaly in adults, impaired glucose tolerance, and diabetes mellitus.

- Corticotroph PitNETs secrete corticotropin and present with Cushing syndrome and hyperpigmentation.
- The two distinctive morphologic features of most PitNETs are their cellular monomorphism and absence of a reticulin network.

HYPOPITUITARISM

Hypopituitarism refers to decreased secretion of pituitary hormones, which can result from diseases of the hypothalamus or the pituitary. Hypofunction of the anterior pituitary occurs when approximately 75% of the parenchyma is lost or absent.

Most cases of hypopituitarism arise from destructive processes involving the anterior pituitary:

- **Tumors and other mass lesions:** PitNETs, other tumors arising within the sella, primary and metastatic malignancies, and cysts can cause hypopituitarism. Any mass lesion in the sella can cause damage by exerting pressure on adjacent normal pituitary cells.
- **Traumatic brain injury and subarachnoid hemorrhage** are among the most common causes of pituitary hypofunction.
- **Pituitary surgery or radiation:** Surgical excision of a PitNET may inadvertently include the nonneoplastic pituitary. Radiation of the pituitary, used to prevent regrowth of residual tumor after surgery, can damage the nonneoplastic pituitary.
- **Pituitary apoplexy:** As mentioned earlier, this is caused by a sudden hemorrhage into the pituitary gland, often occurring into a PitNET. Pituitary apoplexy can lead to the acute onset of severe headache, diplopia due to effects on the oculomotor nerves, and acute hypopituitarism. Corticotropin loss can lead to acute adrenal insufficiency and hypotension. In severe cases, it can cause cardiovascular collapse, loss of consciousness, and even sudden death. The combination of mass effect from the hemorrhage and acute hypopituitarism often makes pituitary apoplexy a neurosurgical emergency.
- **Sheehan syndrome**, also known as *postpartum necrosis*, is the most common cause of ischemic necrosis of the anterior pituitary. During pregnancy, the anterior pituitary enlarges to almost twice its normal size. This physiologic expansion of the gland is not accompanied by an increase in blood supply from the low-pressure venous system; hence, there is relative hypoxia. Any further reduction in blood supply caused by obstetric hemorrhage or shock may precipitate infarction of the anterior lobe. Because the posterior pituitary is supplied directly from arterial branches, it is much less susceptible to ischemic injury and is therefore usually unaffected. Pituitary necrosis may also be encountered in disseminated intravascular coagulation and, less commonly, in sickle cell anemia, elevated intracranial pressure, traumatic injury, and shock of any origin.
- **Rathke cleft cyst:** These cysts, lined by ciliated cuboidal epithelium with occasional goblet cells and anterior pituitary cells, can accumulate proteinaceous fluid and expand, compromising the normal gland.
- **Empty sella syndrome:** Any condition or treatment that destroys part or all of the pituitary gland, such as ablation of the pituitary by surgery or radiation, can result in an empty sella and the empty sella syndrome. There are two types: (1) In a primary empty sella, a defect in the diaphragma sellae allows the arachnoid mater and cerebrospinal fluid to herniate into the sella, expanding the sella and compressing the pituitary. Sometimes the loss of functioning parenchyma is sufficient to produce hypopituitarism. (2) In secondary empty sella, a mass, such as a PitNET, enlarges the sella and is then either surgically removed or undergoes infarction, leading to loss of pituitary function.
- **Hypothalamic lesions:** As mentioned earlier, hypothalamic lesions can affect the pituitary by causing a deficiency of pituitary

hormone-releasing factors and by diminishing the secretion of ADH, resulting in diabetes insipidus (discussed later). Hypothalamic lesions that cause hypopituitarism include tumors, which may be benign or malignant; most of the latter are metastases from tumors such as breast and lung carcinoma. Hypothalamic insufficiency may also appear following irradiation of the brain.

- **Inflammatory disorders and infections**, such as sarcoidosis or tuberculous meningitis, can involve the hypothalamus and cause deficiencies of anterior pituitary hormones and diabetes insipidus.
- **Cancer immunotherapy:** Immune checkpoint blockade therapy for tumors can induce an autoimmune attack on pituitary (as well as other organs) resulting in hypopituitarism.
- **Genetic defects:** Congenital deficiency of transcription factors required for normal pituitary function is a rare cause of hypopituitarism. For example, mutation of the pituitary-specific gene *POU1F1* (*PIT1*) results in combined pituitary hormone deficiency, characterized by deficiencies of GH, prolactin, and TSH.

The clinical manifestations of anterior pituitary hypofunction vary depending on the specific hormones that are lacking.

- Children with growth hormone deficiency develop growth failure and short stature. Growth hormone deficiency in adults may lead to a decrease in lean mass and an increase in fat mass.
- Gonadotropin (LH and FSH) deficiency leads to amenorrhea and infertility in females and decreased libido, impotence, and loss of pubic and axillary hair in males.
- TSH and corticotropin deficiencies result in symptoms of hypothyroidism and hypoadrenalism, respectively, and are discussed later in this chapter.
- Prolactin deficiency results in failure of postpartum lactation.

POSTERIOR PITUITARY SYNDROMES

The clinically relevant posterior pituitary syndromes involve ADH and include diabetes insipidus and syndrome of inappropriate secretion of ADH.

- **Diabetes insipidus.** ADH deficiency is being increasingly referred to as arginine vasopressin deficiency (AVP-D). This condition is characterized by excessive urination (polyuria) due to an inability of the kidney to resorb water properly from the urine. Diabetes insipidus can occur in a variety of conditions, including head trauma, tumors, inflammatory disorders of the hypothalamus and pituitary, and surgical complications. Diabetes insipidus from ADH deficiency is designated as *central* to differentiate it from *nephrogenic* diabetes insipidus (also called arginine vasopressin resistance), which is a result of renal tubular unresponsiveness to circulating ADH. The clinical manifestations of these two disorders are similar and include the excretion of large volumes of dilute urine with a lower than normal specific gravity. Serum sodium and osmolality may be increased by the excessive renal loss of free water, resulting in thirst and polydipsia. Patients who can drink water generally compensate for the urinary losses, but patients who are obtunded, bedridden, or otherwise limited in their ability to obtain water may develop life-threatening dehydration. Rarely, diabetes insipidus has a genetic basis, either because of autosomal dominant mutations of the arginine vasopressin (*AVP*) gene, or mutations of arginine vasopressin receptor type 2 gene (*AVPR2*), an X-linked condition that usually presents in young boys.
- **Syndrome of inappropriate ADH (SIADH) secretion.** ADH excess causes over-resorption of free water, resulting in hyponatremia. The most frequent causes of SIADH are the secretion of ectopic ADH by malignant neoplasms (particularly small-cell carcinoma of the

lung), drugs that increase ADH secretion, and a variety of central nervous system disorders, including infections and trauma. The clinical manifestations of SIADH are dominated by hyponatremia, cerebral edema, and resultant neurologic dysfunction. Although total body water is increased, blood volume remains normal, and peripheral edema does not generally develop.

HYPOTHALAMIC SUPRASellar TUMORS

Neoplasms in this location may induce hypofunction or hyperfunction of the anterior pituitary, diabetes insipidus, or combinations of these manifestations. The most commonly implicated tumors are glioma (sometimes arising in the optic chiasm; Chapter 28) and craniopharyngioma. Craniopharyngioma is thought to arise from vestigial remnants of *Rathke pouch*. These slow-growing tumors account for 1% to 5% of intracranial tumors. A small minority of these lesions occurs within the sella, but most are suprasellar, with or without intrasellar extension. A bimodal age distribution is observed, with one peak in childhood (5 to 15 years of age) and a second peak in adults 65 years of age or older. Patients usually come to attention because of headaches and visual disturbances, while children sometimes present with growth retardation due to pituitary hypofunction and GH deficiency.

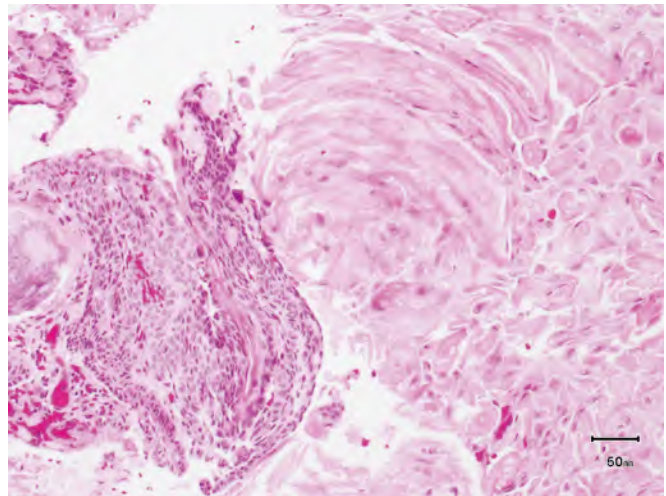


Fig. 24.7 Adamantinomatous craniopharyngioma, demonstrating characteristic compact, lamellar “wet” keratin (right half of photomicrograph) and cords of squamous epithelium with peripheral palisading on the left. (Courtesy Dr. Charles Eberhart, Department of Pathology, Johns Hopkins University, Baltimore.)

MORPHOLOGY

Craniopharyngiomas average 3 to 4 cm in diameter; they may be encapsulated and solid, but more commonly they are cystic and sometimes multiloculated. They often encroach on the optic chiasm or cranial nerves, and not infrequently, they bulge into the floor of the third ventricle and base of the brain. Two distinct histologic variants are recognized: adamantinomatous craniopharyngioma (most often observed in children) and papillary craniopharyngioma (most often observed in adults). The adamantinomatous type frequently contains radiologically demonstrable calcifications; the papillary variant calcifies only rarely.

Adamantinomatous craniopharyngioma consists of nests or cords of stratified squamous epithelium embedded in a spongy “reticulum” that becomes more prominent in the internal layers. “Palisading” of the squamous epithelium is frequently observed at the periphery. Compact, lamellar keratin (“wet keratin”) is a diagnostic feature of this tumor (Fig. 24.7). As mentioned earlier, **dystrophic calcification** is a frequent finding. Additional features include cyst formation, fibrosis, and chronic inflammation. The cysts of adamantinomatous craniopharyngiomas often contain a cholesterol-rich, thick brownish-yellow fluid that has been compared to “machine oil.” These tumors extend fingerlets of epithelium into adjacent brain, where they elicit a brisk glial reaction. This subtype of craniopharyngioma is characterized by recurrent mutations of the *CTNNB1* (β-catenin) gene, which leads to aberrant activation of the Wnt signaling pathway.

Papillary craniopharyngiomas contain both solid sheets of cells and papillae lined by well-differentiated squamous epithelium. These tumors usually are distinguished from the adamantinomatous type by the lack of lamellar keratin, calcification, cysts, peripheral palisading of squamous cells, and a spongy reticulum. Also unlike adamantinomatous craniopharyngioma, in greater than 90% of cases, this subtype has mutations that activate BRAF, a serine-threonine kinase that acts downstream of RAS (Chapter 7). Importantly, most papillary craniopharyngiomas respond well to treatment with BRAF inhibitors.

THYROID GLAND

The thyroid gland consists of two lateral lobes connected by a thin isthmus, usually located below and anterior to the larynx. It develops embryologically from an evagination of the pharyngeal epithelium

that descends from the foramen cecum at the base of the tongue to its normal position in the anterior neck. This pattern of descent explains the occasional presence of ectopic thyroid tissue at the base of the tongue (lingual thyroid) or at other sites high in the neck. The thyroid is divided by thin fibrous septa into lobules, each composed of 20 to 40 colloid-filled follicles lined by a cuboidal to low columnar epithelium.

In response to hypothalamic factors, TSH (*thyrotropin*) is released by thyrotrophs in the anterior pituitary into the circulation. The binding of TSH to its receptor on thyroid follicular epithelial cells activates the receptor, which then associates with a G_s protein (Fig. 24.8). Activation of the G-protein triggers downstream events that increase intracellular cAMP levels, which, in turn, stimulates thyroid growth and thyroid hormone synthesis and release via cAMP-dependent protein kinases.

Thyroid follicular epithelial cells convert thyroglobulin into *thyroxine* (T_4) and lesser amounts of *triiodothyronine* (T_3). T_4 and T_3 are released into the systemic circulation, where most of these hormones are reversibly bound to circulating plasma proteins, such as thyroxine-binding globulin (TBG), transthyretin, and albumin. The binding proteins act as a buffer that maintains the serum unbound (“free”) T_3 and T_4 concentrations within narrow limits, while ensuring that adequate amounts of the hormones are readily available to the tissues. In the periphery, the majority of free T_4 is deiodinated to T_3 ; the latter binds to thyroid hormone nuclear receptors in target cells with 10-fold greater affinity than does T_4 and has proportionately greater activity. Thyroid hormone has diverse cellular effects, including the stimulation of carbohydrate and lipid catabolism and protein synthesis in a wide range of cells. The net result is an increase in the basal metabolic rate. In addition, thyroid hormone has a critical role in brain development in the fetus and neonate (see later).

Mutations in the *THRB* gene, which encodes the TR-beta form of the thyroid hormone receptor, leads to varying degrees of tissue hyporesponsiveness to thyroid hormone, usually associate with elevated thyroid hormone levels because of ineffective feedback inhibition of the hypothalamus and pituitary. In contrast, mutations in the *THRA* gene (which encodes TR-alpha) manifest a different phenotype, including a mildly low T_4 level (with an elevated T_3/T_4

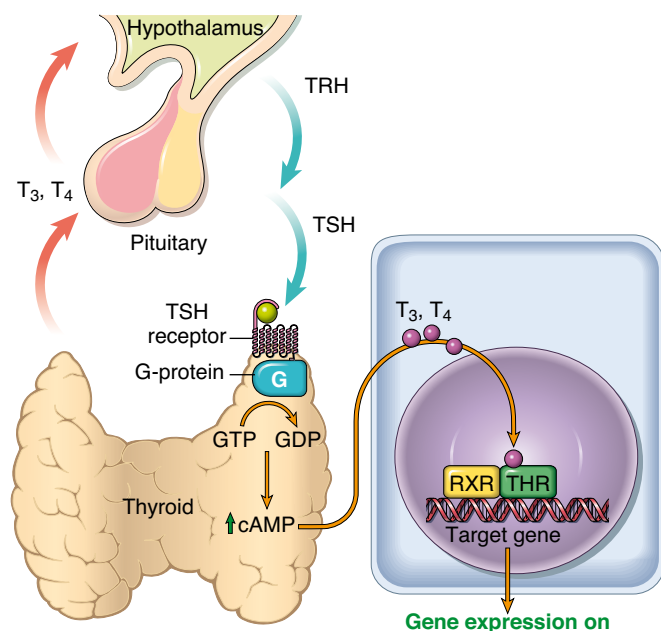


Fig. 24.8 Homeostasis in the hypothalamus-pituitary-thyroid axis and mechanism of action of thyroid hormones. Secretion of thyroid hormones (T_3 and T_4) is controlled by trophic factors secreted by both the hypothalamus and the anterior pituitary. Decreased levels of T_3 and T_4 stimulate the release of thyrotropin-releasing hormone (TRH) from the hypothalamus and thyroid-stimulating hormone (TSH) from the anterior pituitary, causing T_3 and T_4 levels to rise. Elevated T_3 and T_4 levels, in turn, give feedback to suppress the secretion of both TRH and TSH. TSH binds to the TSH receptor on the thyroid follicular epithelium, which causes activation of G-proteins, and cAMP-mediated synthesis and release of thyroid hormones (T_3 and T_4). In the periphery, T_3 and T_4 interact with the thyroid hormone receptor (THR) to form a hormone-receptor complex in the nucleus that binds to so-called thyroid response elements on target genes to initiate transcription. cAMP, Cyclic adenosine monophosphate; GDP, guanosine diphosphate; GTP, guanosine triphosphate.

ratio), short stature and other skeletal abnormalities, and neurocognitive defects.

The function of the thyroid gland can be inhibited by a variety of chemical agents, collectively referred to as *goitrogens*. Because they suppress T_3 and T_4 synthesis, the level of TSH increases, causing subsequent hyperplastic enlargement of the gland (*goiter*). The antithyroid agents propylthiouracil and methimazole inhibit the oxidation of iodide and thus block the production of thyroid hormones; propylthiouracil also inhibits the peripheral deiodination of circulating T_4 into T_3 , thus ameliorating symptoms of thyroid hormone excess (see later). Iodide, when given in large doses to individuals with thyroid hyperfunction, blocks the release of thyroid hormones.

The thyroid gland also contains a population of parafollicular cells, or C cells, which synthesize and secrete the hormone calcitonin. This hormone promotes the absorption of calcium by the skeletal system and inhibits the resorption of bone by osteoclasts.

Diseases of the thyroid include conditions associated with excessive release of thyroid hormones (hyperthyroidism or thyrotoxicosis), thyroid hormone deficiency (hypothyroidism), and mass lesions of the thyroid. We will first consider the clinical consequences of disturbed thyroid function, and then turn to the disorders that generate these problems.

HYPERTHYROIDISM

Thyrotoxicosis is a hypermetabolic state caused by elevated circulating levels of free T_3 and T_4 . Because it is caused most commonly by hyperfunction of the thyroid gland, it is often referred to as *hyperthyroidism*. However, in certain conditions the oversupply is related to either excessive release of preformed thyroid hormone (e.g., in transient thyroiditis) or to an extrathyroidal source, rather than hyperfunction of the gland (Table 24.3). Thus, strictly speaking, hyperthyroidism is only one (albeit the most common) cause of thyrotoxicosis, though in practice they are often used interchangeably. The terms *primary* and *secondary hyperthyroidism* are sometimes used to designate hyperthyroidism arising from an intrinsic thyroid abnormality and that arising from the pituitary, such as a TSH-secreting pituitary tumor. With this caveat, we follow the common practice of using the terms thyrotoxicosis and hyperthyroidism interchangeably.

The most common causes of thyrotoxicosis are associated with hyperfunction of the gland and include:

- Diffuse hyperplasia of the thyroid associated with Graves disease (approximately 85% of cases)
- Hyperfunctional multinodular goiter
- Hyperfunctional thyroid adenoma

Clinical Features

The manifestations of hyperthyroidism are protean and include changes referable to the hypermetabolic state induced by excess thyroid hormone and to overactivity of the sympathetic nervous system (i.e., an increase in the β -adrenergic “tone”).

- Excessive levels of thyroid hormone result in *an increase in the basal metabolic rate*. The skin of thyrotoxic patients tends to be soft, warm, and flushed because of increased blood flow and peripheral vasodilation, adaptations that serve to increase heat loss. Heat intolerance is common. Sweating is increased. Heightened catabolic metabolism results in weight loss despite increased appetite.
- *Cardiac manifestations* are among the earliest and most consistent features. Individuals with hyperthyroidism often have elevated cardiac contractility and output, a response to increased peripheral oxygen requirements. Tachycardia, palpitations, and cardiomegaly are common. Arrhythmias such as atrial fibrillation occur frequently, particularly in older patients. Congestive heart failure may

TABLE 24.3 Disorders Associated With Thyrotoxicosis

Associated With Increased Thyroid Gland Production of Hormone

Primary

Diffuse hyperplasia (Graves disease)
Hyperfunctioning (“toxic”) multinodular goiter
Hyperfunctioning (“toxic”) adenoma
Iodine-induced hyperthyroidism
Neonatal thyrotoxicosis associated with maternal Graves disease

Secondary

TSH-secreting pituitary neuroendocrine tumor (rare)^a

Not Associated With Increased Thyroid Gland Production of Hormone

Granulomatous (de Quervain) thyroiditis (painful)
Lymphocytic thyroiditis (painless)
Struma ovarii (ovarian teratoma with ectopic thyroid)
Factitious thyrotoxicosis (exogenous thyroxine intake)

^aAssociated with increased thyroid-stimulating hormone (TSH); all other causes of thyrotoxicosis associated with decreased TSH.

develop, especially in older adults with preexisting cardiac disease. Myocardial changes, including focal lymphocytic and eosinophilic infiltrates, mild fibrosis, myofibril fatty change, and an increase in size and number of mitochondria, have been described. Some individuals with thyrotoxicosis develop reversible left ventricular dysfunction, so-called *thyrotoxic* or *hyperthyroid cardiomyopathy*.

- *Overactivity of the sympathetic nervous system* produces tremor, hyperactivity, emotional lability, anxiety, inability to concentrate, and insomnia. Proximal muscle weakness and decreased muscle mass may occur (*thyroid myopathy*). In the gastrointestinal system, hypermotility and reduced transit times can lead to hyperdefecation. Some patients also develop fat malabsorption.
- *Ocular changes* often call attention to hyperthyroidism. A wide, staring gaze and lid lag are present because of sympathetic overstimulation of the superior tarsal muscle (also known as Müller muscle), which functions alongside the levator palpebrae superioris muscle to raise the upper eyelid (Fig. 24.9). However, true *thyroid ophthalmopathy* associated with proptosis occurs only in Graves disease (see later).
- The *skeletal system* is also affected. Thyroid hormone stimulates bone resorption, increasing cortical bone porosity and reducing trabecular bone volume. The net effect is osteoporosis and an increased risk of fractures. Other findings include atrophy of skeletal muscle, with fatty infiltration and focal interstitial lymphocytic infiltrates; hepatic fatty change (steatosis) leading to mild hepatomegaly; and generalized lymphoid hyperplasia and lymphadenopathy in patients with Graves disease.
- *Thyroid storm* refers to the abrupt onset of severe hyperthyroidism. This condition occurs most commonly in patients with Graves disease and often occurs in the setting of infection, surgery, cessation of antithyroid medication, or other form of stress. Patients are often febrile and present with tachycardia out of proportion to the fever. Thyroid storm is a medical emergency. A significant number of untreated patients die of cardiac arrhythmias.
- *Apathetic hyperthyroidism* refers to thyrotoxicosis occurring in older adults, in whom advanced age and various comorbidities blunt the typical features of thyroid hormone excess. The diagnosis in these individuals is often made during laboratory workup for unexplained weight loss or worsening cardiovascular disease.



Fig. 24.9 A person with hyperthyroidism. A wide-eyed, staring gaze, caused by overactivity of the sympathetic nervous system, is one feature of this disorder. In Graves disease, an important cause of hyperthyroidism, accumulation of loose connective tissue behind the orbit also adds to the protuberant appearance of the eyes.

Laboratory findings that support the diagnosis of primary hyperthyroidism include a low TSH value accompanied by an increase in free T_4 . The measurement of serum TSH concentration is the most useful single screening test for primary hyperthyroidism, because its levels are decreased even at the earliest stages, when the disease may still be subclinical. In occasional patients, hyperthyroidism results predominantly from increased circulating levels of T_3 but not T_4 (“ T_3 toxicosis”). In these cases, free T_4 levels may actually be decreased, and direct measurement of serum T_3 is useful. In rare cases of pituitary-associated (secondary) hyperthyroidism, TSH levels are either normal or raised.

Once the diagnosis of primary hyperthyroidism has been confirmed by a combination of TSH assays and free thyroid hormone levels, measurement of radioactive iodine uptake by the thyroid gland can help determine the etiology. For example, there may be diffusely increased uptake in the gland (Graves disease), increased uptake in a solitary nodule (toxic adenoma), or decreased uptake (thyroiditis). Alternatively, the presence of thyroid-stimulating immunoglobulins (TSI) and/or TSH receptor antibodies (TRAb) are generally diagnostic of Graves disease (see below).

The therapeutic options for primary hyperthyroidism include several medications, each with a different mechanism of action. Typically, these include a β -blocker to control symptoms induced by increased adrenergic tone and a thionamide (e.g., methimazole or propylthiouracil) to block new thyroid hormone synthesis. Temporarily, iodine solution can be used to suppress the release of thyroid hormone in the setting of severe hyperthyroidism or thyroid storm. In addition, certain agents can be used that inhibit peripheral conversion of T_4 to T_3 . Radioactive iodine, which is incorporated into thyroid tissues and results in ablation of thyroid function, or surgery may be used for definitive therapy. The treatment of TSH-secreting PitNETs can be complex, but may include surgery, somatostatin analogs, or pituitary irradiation.

HYPOTHYROIDISM

Hypothyroidism is a condition caused by a structural or functional derangement that interferes with the production of thyroid hormone.

Hypothyroidism is a common disorder. By some estimates, the prevalence of overt hypothyroidism is 0.3%, while that of subclinical hypothyroidism is greater than 4%. The prevalence increases with age, and it is nearly 10-fold more common in females than in males. Hypothyroidism can result from a defect anywhere in the hypothalamic-pituitary-thyroid axis. Like hyperthyroidism, it can be divided into *primary*, *secondary*, and *tertiary* forms, depending on whether it arises from an intrinsic thyroid abnormality (primary), or as a result of pituitary (secondary), or hypothalamic (tertiary) disease (Table 24.4). Secondary and tertiary hypothyroidism are sometimes grouped together as central hypothyroidism. Primary hypothyroidism accounts for most cases and may sometimes be accompanied by enlargement of the thyroid gland (goiter).

Primary hypothyroidism can be congenital, autoimmune, or iatrogenic.

- **Congenital hypothyroidism. Worldwide, congenital hypothyroidism (hypothyroidism from birth) is most often the result of endemic iodine deficiency in the diet** (see later). Other rare forms of congenital hypothyroidism include inborn errors of thyroid metabolism (*dysmorphogenetic goiter*), wherein one of the several steps leading to thyroid hormone synthesis is defective, such as (1) iodide transport into thyrocytes, (2) “organification” of iodine (covalent binding of iodine to tyrosine residues in thyroglobulin), and (3) further processing to form hormonally active T_3 and T_4 . In rare instances, there may be complete absence of thyroid parenchyma (*thyroid agenesis*), or the gland may be greatly reduced in size (*thyroid hypoplasia*) due to germline mutations in genes responsible for thyroid development (see Table 24.4).

TABLE 24.4 Causes of Hypothyroidism

Primary
Genetic defects in thyroid development (<i>PAX8</i> , <i>FOXE1</i> , TSH receptor gene mutations; rare)
Postablative
Surgery, radioiodine therapy, or external irradiation
Autoimmunity
Hashimoto thyroiditis ^a
Iodine deficiency ^a
Drugs (lithium, amiodarone, iodides, <i>p</i> -aminosalicylic acid) ^a
Congenital biosynthetic defect (dyshormonogenetic goiter; rare) ^a
Secondary (“Central Hypothyroidism”)
Pituitary failure (rare)
Hypothalamic failure (rare)

^aAssociated with enlargement of thyroid (“goitrous hypothyroidism”). Hashimoto thyroiditis and postablative hypothyroidism account for most cases of hypothyroidism in high-income countries. *FOXE1*, Forkhead box E1; *PAX8*, paired box 8; *THRB*, thyroid hormone receptor β .

- **Autoimmune hypothyroidism.** Autoimmune hypothyroidism is the most common cause of hypothyroidism in iodine-sufficient areas of the world. Most cases of autoimmune hypothyroidism are due to Hashimoto thyroiditis. Circulating autoantibodies, including antithyroid peroxidase (formerly antimicrosomal) and antithyroglobulin antibodies, are found in this disorder, and the thyroid may be enlarged (goitrous). Autoimmune hypothyroidism can occur in isolation or in conjunction with autoimmune polyendocrine syndrome types 1 and 2 (APS-1 and APS-2; discussed later).
- **Iatrogenic hypothyroidism.** This can be caused by either surgical or radiation-induced ablation of the thyroid. Resection of a large portion of the gland for the treatment of hyperthyroidism or a primary neoplasm can lead to hypothyroidism. The gland may also be ablated by radiation in the form of radioiodine administered for the treatment of hyperthyroidism or external radiation therapy to the neck. Drugs given to decrease thyroid secretion (e.g., thionamides like methimazole and propylthiouracil) can also cause acquired hypothyroidism, as can agents used to treat other conditions (e.g., lithium, *p*-aminosalicylic acid). Amiodarone use may be associated with either hypothyroidism or hyperthyroidism.

Central hypothyroidism is caused by deficiencies of TSH or, more uncommonly, thyrotropin-releasing hormone (TRH). Any of the causes of hypopituitarism (e.g., pituitary tumor, postpartum pituitary necrosis, trauma, and nonpituitary tumors) or hypothalamic damage (e.g., tumors, trauma, radiation therapy, or infiltrative diseases) can lead to central hypothyroidism.

Congenital Iodine Deficiency and Myxedema

Congenital iodine deficiency refers to hypothyroidism developing in infancy or early childhood and has been historically referred to as *cretinism*. In the past, this disorder occurred fairly commonly in regions of the world where dietary iodine deficiency is endemic, such as the Himalayas, inland China, Africa, and other mountainous areas. It is now fortunately much less prevalent as a result of the widespread supplementation of foods with iodine. On rare occasions, cretinism results from genetic defects that interfere with the biosynthesis of thyroid hormone (dyshormonogenetic goiter).

Clinical features of congenital iodine deficiency include severe intellectual disability, short stature, coarse facial features, a protruding tongue, and umbilical hernia. The severity of the mental impairment seems to be related to the time at which thyroid deficiency occurs in utero. Normally, maternal T_3 and T_4 cross the placenta and are critical

for fetal brain development. If there is maternal thyroid deficiency before the development of the fetal thyroid gland, intellectual disability is severe. In contrast, maternal thyroid hormone deficiency later in pregnancy, after the fetal thyroid has become functional, does not affect normal brain development.

The term myxedema is an older term for hypothyroidism developing in the older child or adult. Myxedema was first linked with thyroid dysfunction in 1873 by Sir William Gull in an article addressing the development of a “cretinoid state” in adults. Currently, myxedema is most often used in the setting of severe hypothyroidism, ultimately leading to myxedema coma.

The clinical findings of hypothyroidism vary with age of onset. In the adult, the condition appears insidiously and may take years before arousing clinical suspicion. Hypothyroidism is marked by a slowing of physical and mental activity. Very mild hypothyroidism may be asymptomatic (also called “subclinical hypothyroidism” when TSH levels are elevated but free T_4 levels remain normal). Early symptoms include generalized fatigue, apathy, and mental sluggishness, which may mimic depression. In more severe cases, speech and intellectual functions are slowed, and patients are listless, cold intolerant, and frequently overweight. Decreased sympathetic activity results in constipation and decreased sweating. The skin is cool and pale because of decreased blood flow. Reduced cardiac output probably contributes to shortness of breath and decreased exercise capacity. Thyroid hormones regulate the transcription of several sarcolemmal genes, such as calcium ATPases and the β -adrenergic receptor, and diminished expression of these genes may result in a decrease in cardiac output. In addition, changes in lipid metabolism produce an increase in total cholesterol and low-density lipoprotein (LDL) levels that is proatherogenic, contributing to increased cardiovascular mortality. Histologically, there is an accumulation of matrix substances, such as glycosaminoglycans and hyaluronic acid, in skin, subcutaneous tissue, and visceral sites. This results in nonpitting edema, a broadening and coarsening of facial features, enlargement of the tongue, and deepening of the voice.

Laboratory evaluation plays a vital role in the diagnosis of suspected hypothyroidism because of the nonspecific nature of symptoms. Patients with unexplained increases in body weight or hypercholesterolemia should be assessed for potential hypothyroidism. **Measurement of the serum TSH level is the most sensitive screening test for primary hypothyroidism.** The TSH level is increased in primary hypothyroidism as a result of a loss of feedback inhibition of TRH and TSH production by the hypothalamus and pituitary, respectively. In contrast, TSH levels are not increased in persons with hypothyroidism due to hypothalamic or pituitary disease, and are usually low or inappropriately normal. T_4 levels are decreased in individuals with overt hypothyroidism of any origin. As noted above, subclinical hypothyroidism is an early form of primary hypothyroidism where TSH levels are elevated but free T_4 and T_3 levels are normal.

THYROIDITIS

Thyroiditis encompasses a diverse group of disorders characterized by some form of thyroid inflammation. This discussion focuses on the three most common and clinically significant subtypes: (1) Hashimoto thyroiditis, (2) subacute granulomatous thyroiditis (also known as de Quervain or painful thyroiditis), and (3) chronic lymphocytic thyroiditis (also known as silent or painless thyroiditis). Hashimoto thyroiditis, also known as chronic autoimmune thyroiditis or chronic lymphocytic thyroiditis, is the most common form of hypothyroidism in iodine-sufficient areas. In contrast, subacute thyroiditis and painless thyroiditis are examples of transient thyroiditis, which are characterized by an initial thyrotoxic phase, followed by hypothyroidism and often eventual recovery to normal thyroid levels (euthyroidism).

Hashimoto Thyroiditis

Hashimoto thyroiditis is an autoimmune disease that results in destruction of the thyroid gland and gradual and progressive thyroid failure. The name is derived from the 1912 report by Hashimoto describing patients with goiter and lymphocytic infiltration of the thyroid (*struma lymphomatosa*). It is most prevalent between 45 and 65 years of age and is more common in females than in males, with a female predominance of 10:1 to 20:1. It can also occur in children and is a major cause of nonendemic goiter in the pediatric population.

Pathogenesis

Hashimoto thyroiditis is caused by a breakdown in self-tolerance to thyroid autoantigens. This is exemplified by the presence of circulating autoantibodies against thyroglobulin and thyroid peroxidase in the vast majority of patients. The inciting events have not been elucidated, but possibilities include abnormalities of regulatory T cells (Tregs) or exposure of normally sequestered thyroid antigens (Chapter 6). Similar to other autoimmune diseases, Hashimoto thyroiditis predisposition has a strong genetic component. Increased susceptibility is associated with polymorphisms in several immune regulation–associated genes, such as cytotoxic T lymphocyte–associated antigen-4 (*CTLA4*), protein tyrosine phosphatase non-receptor type 22 (*PTPN22*), and interleukin-2 receptor α chain (*IL2RA*), all of which encode regulators of T-cell responses.

Induction of thyroid autoimmunity is accompanied by a progressive depletion of thyroid epithelial cells and replacement of the thyroid parenchyma by lymphocytic infiltrates and fibrosis. Multiple immunologic mechanisms may contribute to thyroid cell death, including (Fig. 24.10):

- **CD8+ cytotoxic T cell–mediated cell death:** CD8+ cytotoxic T cells may destroy thyroid follicular cells.
- **Cytokine-mediated cell death:** Activation of CD4+ Th1 cells leads to the production of inflammatory cytokines such as interferon- γ in the thyroid gland, with resultant recruitment and activation of macrophages and damage to follicles.
- In the past, it was believed that a contributing mechanism involves binding of antithyroid antibodies (antithyroglobulin, and antithyroid peroxidase antibodies) followed by antibody-dependent cell-mediated cytotoxicity (Chapter 6). It is also possible that antithyroid antibodies damage follicular cells by complement-dependent mechanisms. However, it is unclear if these antibodies are the cause or consequence of thyroid injury.

MORPHOLOGY

The thyroid is usually diffusely enlarged and remains well demarcated from adjacent structures. The cut surface is pale, yellow-tan, firm, and somewhat nodular. There is extensive infiltration of the parenchyma by a **mononuclear inflammatory infiltrate** containing small lymphocytes, plasma cells, and lymphoid follicles with well-developed **germinal centers** (Fig. 24.11). The thyroid follicles are atrophic and are lined in many areas by epithelial cells with abundant eosinophilic, granular cytoplasm, termed **oncocytes** (formerly known as **Hürthle cells**), which represent a metaplastic response of the normally low cuboidal follicular epithelium to chronic injury. In fine-needle aspiration biopsy samples, the presence of oncocytes in conjunction with a heterogeneous population of lymphocytes is characteristic of Hashimoto thyroiditis. In “classic” Hashimoto thyroiditis, interstitial connective tissue is increased and may be abundant. Unlike Reidel thyroiditis (see later), the fibrosis does not extend beyond the periphery of the gland.

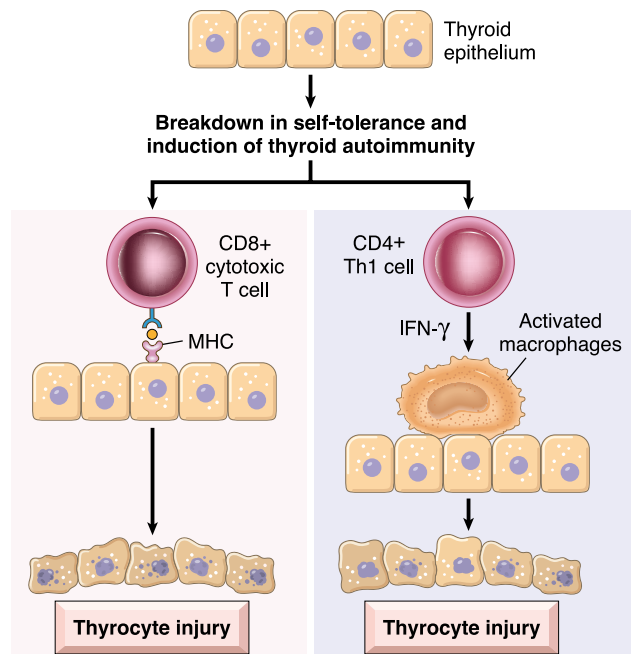


Fig. 24.10 Pathogenesis of Hashimoto thyroiditis. Breakdown of peripheral tolerance to thyroid autoantigens results in progressive autoimmune destruction of thyroid cells by infiltrating cytotoxic T cells, locally released cytokines, and other mechanisms (not shown). IFN- γ , Interferon gamma; MHC, major histocompatibility complex.

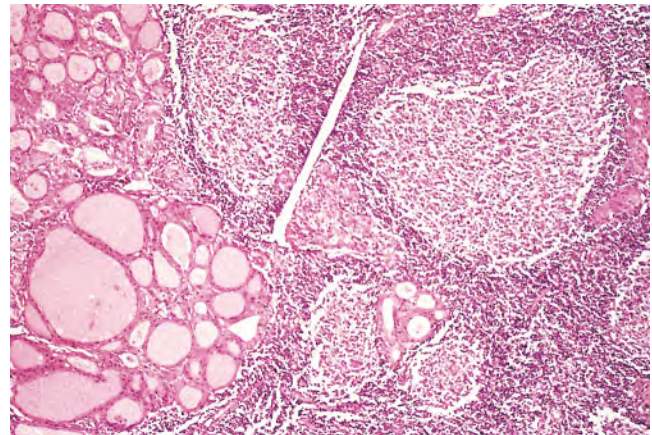


Fig. 24.11 Hashimoto thyroiditis. The thyroid parenchyma contains a dense lymphocytic infiltrate with germinal centers. Residual thyroid follicles lined by pink oncocytic cells are also seen.

Clinical Features

Hashimoto thyroiditis classically comes to attention as painless enlargement of the thyroid, usually associated with some degree of hypothyroidism. The enlargement of the gland is generally symmetric and diffuse, but in some cases it may be sufficiently localized to raise the suspicion of a neoplasm. In the typical case, hypothyroidism develops gradually. In some patients, however, it may be preceded by transient thyrotoxicosis caused by disruption of thyroid follicles and release of thyroid hormones (*hashitoxicosis*). As hypothyroidism develops, T_4 and T_3 levels fall, accompanied by a compensatory increase in TSH.

Individuals with Hashimoto thyroiditis are at increased risk for developing other autoimmune diseases, both endocrine (type 1 diabetes, autoimmune adrenalitis) and nonendocrine (systemic lupus erythematosus, myasthenia gravis, and Sjögren syndrome; Chapter 6).

They are also at increased risk for development of extranodal marginal zone B-cell lymphoma within the thyroid gland (Chapter 13), though this is rare. The relationship between Hashimoto disease and thyroid epithelial cancers remains controversial, with some studies suggesting a predisposition to papillary carcinoma.

Chronic Lymphocytic (Painless) Thyroiditis

Painless thyroiditis, also known as *silent thyroiditis*, is a presumed autoimmune disease. Most patients have circulating antithyroid peroxidase antibodies or a family history of other autoimmune disorders. This can occur during the postpartum period in up to 5% of females, when it is known as *postpartum thyroiditis*.

MORPHOLOGY

Except for possible mild symmetric enlargement, the thyroid appears grossly normal. Microscopic examination reveals lymphocytic infiltration with large germinal centers within the thyroid parenchyma and patchy disruption and collapse of thyroid follicles. Unlike Hashimoto thyroiditis, however, fibrosis and oncocytic metaplasia are not prominent.

Clinical Features

Painless thyroiditis usually comes to attention because of mild, transient hyperthyroidism, painless goiter, or both. Although it can occur at any age, it is most often seen in middle-aged adults and is more common in females. Patients often transition from thyrotoxicosis to hypothyroidism before recovery to normal thyroid function. As many as 30% of cases may evolve into permanent hypothyroidism, with pathologic features that resemble Hashimoto thyroiditis.

Subacute Granulomatous Thyroiditis

Subacute granulomatous thyroiditis (also called *de Quervain thyroiditis*) occurs much less frequently than does Hashimoto thyroiditis. The disorder is most common between 40 and 50 years of age and, like other forms of thyroiditis, affects females more often than males (approximately 4:1).

Pathogenesis

Subacute thyroiditis is believed to be triggered by a viral infection. Many patients have a history of an upper respiratory infection before the onset of thyroiditis. The disease may be seasonal, with some but not all studies suggesting a peak during the summer months, and clusters of cases have been reported in association with coxsackievirus, SARS-CoV-2, mumps, measles, adenovirus, and other viral infections. Although the pathogenesis of the disease is unclear, one model suggests that it results from virus-induced host tissue damage that stimulates a cytotoxic T-lymphocyte response to one or more thyroid antigens that damages thyroid follicle cells. In contrast to autoimmune thyroid disease, the immune response is virus-initiated and not self-perpetuating, so the process is limited.

MORPHOLOGY

The gland may be unilaterally or bilaterally enlarged and firm, with occasional fibrous adherence to surrounding structures. On cut section, the involved areas are firm and yellow-white and stand out from adjacent normal rubbery, brown thyroid gland. Histologic changes are patchy and depend on the stage of the disease. Early in the active inflammatory phase, scattered follicles may be disrupted and replaced by neutrophils forming microabscesses. Later, more characteristic features appear in the form of aggregates of lymphocytes, activated

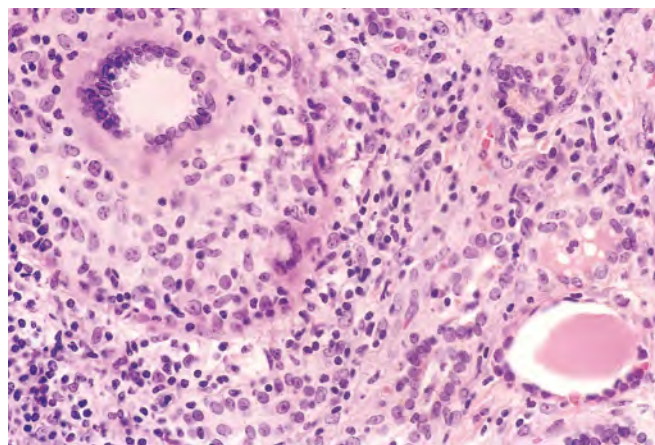


Fig. 24.12 Subacute granulomatous thyroiditis. The thyroid parenchyma contains a chronic inflammatory infiltrate with a multinucleated giant cell and adjacent follicles.

macrophages, and plasma cells associated with collapsed and damaged thyroid follicles. **Multinucleate giant cells** enclose pools of colloid (Fig. 24.12), hence the designation **granulomatous thyroiditis**. In later stages of the disease, a chronic inflammatory infiltrate and fibrosis may replace the foci of injury. Different histologic stages are sometimes found in the same gland, suggesting waves of destruction over a period of time.

Clinical Features

Subacute thyroiditis is the most common cause of thyroid pain. There is variable enlargement of the gland. Inflammation of the thyroid and thyrotoxicosis are transient, usually diminishing in 2 to 6 weeks, even if the patient is not treated. Nearly all patients have high serum T_4 and T_3 levels and low serum TSH levels during this phase. However, unlike in hyperthyroid states such as Graves disease, radioactive iodine uptake is diminished. As in silent thyroiditis, there may be a hypothyroid phase, followed by recovery to normal thyroid function, generally in 6 to 8 weeks.

Other, less common forms of thyroiditis include *Riedel thyroiditis*, a rare disorder characterized by extensive fibrosis involving the thyroid and contiguous neck structures. The presence of a hard and fixed thyroid mass clinically simulates a thyroid carcinoma. It may be associated with fibrosis in other sites in the body, such as the retroperitoneum, and appears to be a manifestation of IgG4-related disease, which is associated with fibrosis and tissue infiltration by plasma cells producing IgG4 (Chapter 6).

KEY CONCEPTS

Thyroiditis

- Hashimoto thyroiditis is the most common cause of hypothyroidism in regions where dietary iodine levels are sufficient.
- Hashimoto thyroiditis is an autoimmune thyroiditis characterized by progressive destruction of thyroid parenchyma, oncocytic change, and mononuclear (lymphoplasmacytic) infiltrates, with germinal centers and variable degrees of fibrosis.
- Silent thyroiditis often occurs after pregnancy (postpartum thyroiditis), typically is painless, and is characterized by lymphocytic inflammation in the thyroid. It is another type of autoimmune thyroiditis.
- Subacute (granulomatous or de Quervain) thyroiditis is a self-limited disease, probably secondary to a viral infection, and is characterized by painful granulomatous inflammation of the thyroid.

GRAVES DISEASE

Graves disease is the most common cause of endogenous hyperthyroidism. In 1835, Graves reported his observations of a disease characterized by “violent and long continued palpitations in females” associated with enlargement of the thyroid gland. The disease is characterized by primary hyperthyroidism, and can be associated with a triad of clinical findings:

- *Hyperthyroidism* associated with diffuse enlargement of the gland
- Infiltrative *ophthalmopathy* and resultant exophthalmos
- Localized, infiltrative *dermopathy*, sometimes called *pretibial myxedema*, which is present in a minority of patients

Graves disease has a peak incidence between 20 and 50 years of age. Females are affected as much as 5 to 10 times more frequently than males.

Pathogenesis

Graves disease is an autoimmune disorder characterized by the production of autoantibodies against multiple thyroid proteins, most importantly the TSH receptor. A variety of antibodies that either stimulate or block the TSH receptor are detected in the circulation. The most common antibody subtype, known as *thyroid-stimulating immunoglobulin* (TSI), is observed in approximately 90% of patients. In contrast to antibodies against thyroglobulin and thyroid peroxidase, TSI is almost never observed in other autoimmune diseases of the thyroid. TSI binds to the TSH receptor and mimics its actions, stimulating adenyl cyclase and increasing the release of thyroid hormones. As stated earlier, TSH receptor blocking antibodies may also be present, and in a minority of patients, these lead to hypothyroidism.

Perhaps unsurprisingly, factors that predispose to Graves disease overlap with risk factors for Hashimoto thyroiditis, the other major form of autoimmune thyroid disease. The concordance rate in monozygotic twins is 30% to 40%, compared with less than 5% among dizygotic twins, and as in Hashimoto thyroiditis, genetic susceptibility is associated with polymorphisms in immune-function genes like *CTLA4*, *PTPN22*, and *IL2RA*. In addition, genome-wide association studies (GWAS) have implicated variants in the TSH receptor (*TSHR*) gene in susceptibility to Graves disease.

Autoimmunity also plays a role in the development of the *infiltrative ophthalmopathy* that is characteristic of Graves disease. Activated CD4+ helper T cells secrete cytokines that stimulate fibroblast proliferation and

synthesis of extracellular matrix proteins (glycosaminoglycans), leading to progressive infiltration of the retro-orbital space and ophthalmopathy. The protrusion of the eyeball (*exophthalmos*) is caused by an increase in the volume of retro-orbital connective tissues and extraocular muscles, which occurs for several reasons: (1) marked infiltration of connective tissue by mononuclear cells, predominantly T cells; (2) inflammation, edema, and swelling of extraocular muscles; (3) accumulation of extracellular matrix components, specifically hydrophilic glycosaminoglycans such as hyaluronic acid and chondroitin sulfate; and (4) increased numbers of adipocytes (fatty infiltration). These changes not only displace the eyeball but also may interfere with the function of the extraocular muscles.

MORPHOLOGY

The thyroid gland is usually symmetrically enlarged due to **diffuse hypertrophy and hyperplasia** of thyroid follicular epithelial cells (Fig. 24.13A). Increases in weight to over 80 g are not uncommon. On cut section, the parenchyma has a soft, brown-red appearance. Histologically, in untreated cases, the follicular epithelial cells are tall and more crowded than usual. This crowding often results in the formation of small papillae that project into the follicle lumen and encroach on the colloid, sometimes filling the follicles (Fig. 24.13B). Such papillae lack fibrovascular cores, in contrast to those of papillary carcinoma (see later). The colloid within the follicle lumen is pale, with scalloped margins. Lymphoid infiltrates, consisting predominantly of T cells, along with scattered B cells and mature plasma cells, are present throughout the interstitium. Germinal centers are common.

Preoperative therapy alters the morphology of the thyroid in Graves disease. Administration of iodine causes involution of the epithelium and the accumulation of colloid by blocking thyroglobulin secretion. Treatment with the antithyroid drug propylthiouracil exaggerates the epithelial hypertrophy and hyperplasia by stimulating TSH secretion.

Changes in extrathyroidal tissue include lymphoid hyperplasia, especially enlargement of the thymus in younger patients. The heart may be hypertrophied, and ischemic changes may be present, particularly in patients with pre-existing coronary artery disease. In patients with ophthalmopathy, the tissues of the orbit are edematous because of the presence of hydrophilic mucopolysaccharides. In addition, there is infiltration by lymphocytes and fibrosis. Orbital muscles are edematous initially but may undergo fibrosis late in the course of the disease. The dermopathy, if present, is characterized by thickening of the dermis due to deposition of glycosaminoglycans and lymphocyte infiltration.

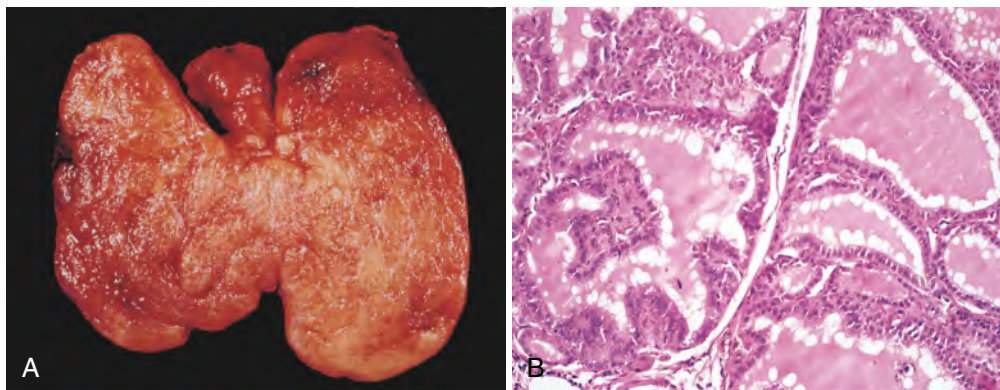


Fig. 24.13 Graves disease. (A) There is diffuse symmetric enlargement of the gland and a beefy deep red parenchyma. Compare with gross photograph of multinodular goiter in Fig. 24.15. (B) Diffusely hyperplastic thyroid in a case of Graves disease. The follicles are lined by tall, columnar epithelium. The crowded, enlarged epithelial cells project into the lumens of the follicles. These cells actively resorb the colloid in the centers of the follicles, resulting in the scalloped appearance of the edges of the colloid. (A, From Lloyd RV, Douglas BR, Young WF Jr, eds. *Atlas of Nontumor Pathology: Endocrine Diseases*. American Registry of Pathology; 2002.)

Clinical Features

The findings in Graves disease include changes associated with thyrotoxicosis and others associated with Graves disease, such as diffuse hyperplasia of the thyroid, ophthalmopathy, and dermopathy. The thyroid enlargement may be accompanied by increased flow of blood through the hyperactive gland, often producing an audible “bruit.” Sympathetic overactivity produces a characteristic wide, staring gaze and lid lag, while the ophthalmopathy causes exophthalmos. The exophthalmos may persist or progress despite successful treatment of the thyrotoxicosis, sometimes resulting in corneal injury. The extraocular muscles are often weak. The infiltrative dermopathy (*pretibial myxedema*), which presents in a minority of patients, most commonly occurs as scaly thickening and induration in the skin overlying the shins. Sometimes individuals spontaneously develop thyroid hypofunction. Patients are at increased risk for other autoimmune diseases, such as systemic lupus erythematosus, pernicious anemia, type 1 diabetes, and Addison disease.

Laboratory findings in Graves disease include elevated free T_4 and T_3 levels and depressed TSH levels. Because of ongoing stimulation by thyroid-stimulating immunoglobulins, radioiodine scans show diffusely increased uptake of iodine in the thyroid gland. TSIs and TSH-receptor blocking antibodies (TRABs) can be measured and are elevated.

Graves disease is treated with β -blockers, which dampen symptoms related to increased sympathetic nervous system activity (e.g., tachycardia, palpitations, tremulousness, and anxiety), and by measures that decrease thyroid hormone synthesis, such as the administration of thionamides (e.g., methimazole and propylthiouracil), radioiodine ablation, and thyroidectomy. Surgery is used mostly in patients who have large goiters that are compressing surrounding structures, or in patients who require definitive therapy but cannot be given radioiodine.

KEY CONCEPTS

Graves Disease

- Graves disease, the most common cause of endogenous hyperthyroidism, is characterized by the triad of thyrotoxicosis, ophthalmopathy, and dermopathy.
- Graves disease is an autoimmune disorder caused by activation of thyroid epithelial cells by autoantibodies to the TSH receptor that mimic TSH action (thyroid-stimulating immunoglobulins).
- The thyroid in Graves disease shows diffuse hypertrophy and hyperplasia of follicles and lymphoid infiltrates; glycosaminoglycan deposition and lymphoid infiltrates are responsible for the ophthalmopathy and dermopathy.
- Laboratory features include elevations in serum free T_3 and T_4 and decreased serum TSH.

DIFFUSE AND MULTINODULAR GOITER

Goiter can be associated with hyperthyroidism, hypothyroidism, or euthyroidism. For example, reduced thyroid hormone production leads to a compensatory rise in the serum TSH level, which, in turn, causes hypertrophy and hyperplasia of thyroid follicular cells and, ultimately, enlargement of the thyroid gland. This increase in functional mass of the gland usually overcomes the hormone deficiency, ensuring a euthyroid metabolic state in most individuals. If the underlying disorder is sufficiently severe (e.g., a congenital biosynthetic defect or endemic iodine deficiency, discussed later), the compensatory responses may be inadequate, resulting in *goitrous hypothyroidism*. The degree of thyroid enlargement is proportional to the level and duration of thyroid hormone deficiency.

Goiters can be broadly divided into two types: *diffuse nontoxic* and *multinodular*.

Diffuse Nontoxic (Simple) Goiter

Diffuse nontoxic (simple) goiter causes enlargement of the entire gland without producing nodularity. The follicles are enlarged and filled with colloid. This disorder occurs in both an endemic and a sporadic distribution.

- *Endemic goiter* occurs in geographic areas where the soil, water, and food supply contain low levels of iodine. Such conditions are particularly common in mountainous areas of the world, including the Andes and Himalayas, where iodine deficiency is widespread. The lack of iodine leads to decreased synthesis of thyroid hormone and a compensatory increase in TSH, leading to follicular cell hypertrophy and hyperplasia and goitrous enlargement. With increasing dietary iodine supplementation, the frequency and severity of endemic goiter have declined significantly.

Variations in the prevalence of endemic goiter in regions with similar levels of iodine deficiency point to the existence of other causative influences, particularly dietary substances, referred to as *goitrogens*. The ingestion of substances that interfere with thyroid hormone synthesis, such as vegetables belonging to the Brassicaceae (Cruciferae) family (e.g., cabbage, cauliflower, Brussels sprouts, turnips, and cassava), has been documented to be goitrogenic, although large amounts of ingested vegetables are required. Native populations subsisting on cassava root are particularly at risk. Cassava contains a thiocyanate that inhibits iodide transport within the thyroid, worsening any concurrent iodine deficiency.

- *Sporadic goiter* occurs less frequently than does endemic goiter. Sporadic goiter can be caused by several conditions, including the ingestion of substances that interfere with thyroid hormone synthesis. In other instances, goiter may result from hereditary enzymatic defects that interfere with thyroid hormone synthesis, all transmitted as autosomal recessive conditions (dys-hormonogenetic goiter; see earlier). In most cases, however, the cause of sporadic goiter is not apparent.

MORPHOLOGY

Two phases can be identified in the evolution of diffuse nontoxic goiter: the **hyperplastic phase** and the phase of **colloid involution**. In the hyperplastic phase, the thyroid gland is diffusely and symmetrically enlarged, although the increase is usually modest, and the gland rarely exceeds 100 to 150 g. The follicles are lined by crowded columnar cells, which may pile up and form projections similar to those seen in Graves disease. The accumulation is not uniform throughout the gland, and some follicles are hugely distended, whereas others remain small. If dietary iodine subsequently increases or if the demand for thyroid hormone decreases, the stimulated follicle epithelium involutes to form an enlarged, colloid-rich gland. In these cases, the cut surface of the thyroid is usually brown, somewhat glassy, and translucent. Histologically the follicle epithelium is flattened and cuboidal, and colloid is abundant during periods of involution.

Clinical Features

As stated earlier, most persons with simple goiters are clinically euthyroid. Therefore, the clinical manifestations are primarily related to *mass effects* from the enlarged thyroid gland (Fig. 24.14). In children, dys-hormonogenetic goiter, caused by a congenital biosynthetic defect, may induce congenital iodine deficiency.

Multinodular Goiter

Recurrent episodes of hyperplasia and involution combine to produce a more irregular enlargement of the thyroid, termed *multinodular goiter*.

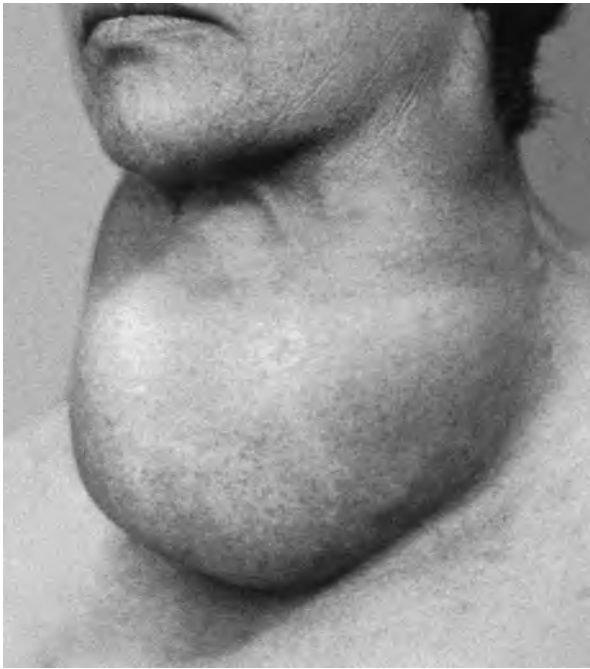


Fig. 24.14 A 52-year-old female with a huge colloid goiter who developed compressive symptoms. (From Lloyd RV, Douglas BR, Young WF Jr, eds. *Atlas of Nontumor Pathology: Endocrine Diseases*. American Registry of Pathology; 2002.)

Multinodular goiters produce the most extreme thyroid enlargements and are more frequently mistaken for neoplasms than any other form of thyroid disease. Although generally benign, thyroid cancers can reside within multinodular goiters. Because multinodular goiters may arise from simple goiters, they occur in both sporadic and endemic forms, presumably having the same origins but affecting older individuals because they are late complications.

Pathogenesis

It is believed that multinodular goiters arise because of variations among follicular cells in their response to external stimuli, such as trophic hormones. If some cells in a follicle have a growth advantage, perhaps because of acquired genetic abnormalities similar to those that give rise to adenomas, they may begin to proliferate autonomously, producing a nodule. Consistent with this model, both polyclonal and monoclonal nodules coexist within the same multinodular

goiter, the latter presumably having arisen because of the acquisition of a genetic abnormality favoring growth. Not surprisingly, activating mutations affecting proteins of the TSH signaling pathway have been identified in a subset of autonomous thyroid nodules (TSH signaling pathway mutations and their implications in adenomas are discussed later).

MORPHOLOGY

Multinodular goiters are multilobulated, asymmetrically enlarged glands that can reach weights of more than 2000 g. The pattern of enlargement is unpredictable and may involve one lobe far more than the other, producing pressure on midline structures, such as the trachea and esophagus. In other instances, the goiter grows behind the sternum and clavicles to produce the **intrathoracic** or **substernal goiter**. Occasionally, most of it is hidden behind the trachea and esophagus; in other instances, one nodule may stand out, mimicking a solitary tumor. On cut section, irregular nodules containing variable amounts of brown, gelatinous colloid are present (Fig. 24.15A). Older lesions have areas of hemorrhage, fibrosis, calcification, and cystic change. The microscopic appearance includes colloid-rich follicles lined by flattened, inactive epithelium and areas of **follicle hyperplasia**, accompanied by degenerative changes related to physical stress. A prominent capsule between the nodules and residual compressed thyroid parenchyma is not present (Fig. 24.15B). The histologic term for this constellation of features is **follicular nodular disease**.

Clinical Features

The dominant clinical features of multinodular goiter are those caused by mass effects. In addition to their cosmetic effects, goiters may cause airway obstruction, dysphagia, and compression of large vessels in the neck and upper thorax (*superior vena cava syndrome*). Most patients are euthyroid but others have hyperthyroidism (*toxic multinodular goiter*), also known as *Plummer disease*. The incidence of malignancy in long-standing multinodular goiters is low but the possibility of malignancy should not be overlooked, particularly in nodules that exhibit suspicious ultrasound features or in goiters that demonstrate sudden changes in size or symptoms (e.g., hoarseness). Dominant nodules in a multinodular goiter can present as a “solitary thyroid nodule,” mimicking a thyroid neoplasm. A radioiodine scan demonstrates uneven iodine uptake (including the both “hot” autonomous nodules and “cold” nonfunctioning nodules). A fine-needle aspiration biopsy of cold nodules can be helpful for identifying thyroid neoplasms (see later).

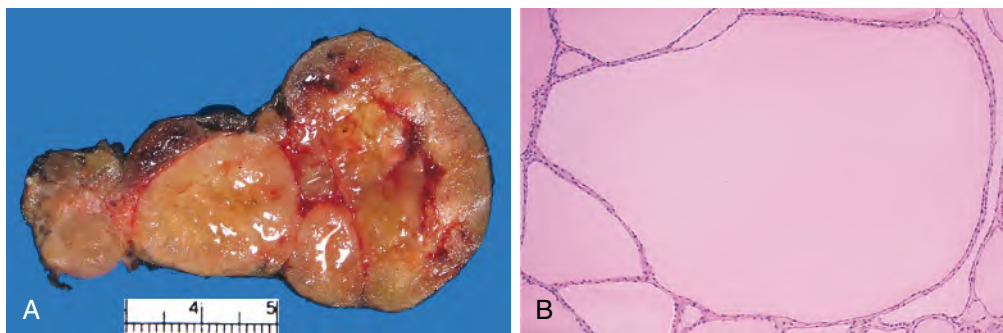


Fig. 24.15 Follicular nodular disease. (A) Gross morphology demonstrating a gland expanded by tan, glistening, gelatinous (colloid-rich) nodules. (B) The enlarged thyroid follicles are lined with inactive, flattened epithelial cells and filled with abundant stored pink colloid. (Courtesy Dr. William Westra, Department of Pathology, Johns Hopkins University, Baltimore.)

NEOPLASMS OF THE THYROID

The solitary thyroid nodule is a palpably discrete swelling within an otherwise apparently normal thyroid gland. Autopsy studies suggest up to 50% of individuals may have thyroid nodules. The incidence of thyroid nodules increases throughout life.

From a clinical standpoint, the major concern in persons who present with thyroid nodules is the possibility of a malignant neoplasm. Fortunately, most solitary nodules of the thyroid prove to be localized, nonneoplastic lesions (e.g., a benign nodule in multinodular goiter, simple cysts, or foci of thyroiditis) or benign neoplasms such as follicular adenoma. In fact, benign neoplasms outnumber thyroid carcinomas by a ratio of about 10:1. Fortunately, most thyroid cancers are readily treated; more than 90% of affected patients are alive 20 years after being diagnosed.

Several criteria provide clues to the nature of a thyroid nodule:

- *Solitary nodules, nodules in younger patients, and nodules in males* are more likely to be neoplastic.
- A history of *radiation treatment* to the head and neck region is associated with an increased incidence of thyroid malignancy.
- Functional nodules that take up radioactive iodine in imaging studies (*hot nodules*) are much more likely to be benign.

Ultimately, microscopic evaluation of a given thyroid nodule provides the most definitive information about its nature. Ultrasonographic evaluation is often used to determine which thyroid nodules should undergo fine-needle aspiration (FNA), based on suspicious features such as hypoechoic echotexture and irregular borders. Cytologic examination of FNA material can distinguish benign from malignant in many, but not all, cases. Molecular analysis can aid in the diagnosis of nodules with indeterminate cytology. The following sections consider the major thyroid neoplasms.

Thyroid Adenomas

Adenomas of the thyroid are typically discrete, solitary neoplasms, derived from follicle epithelium, and hence they are known as follicular adenomas. Clinically, follicular adenomas can be difficult to distinguish from a dominant nodule of follicular nodular disease or from less common follicular carcinoma. In general, follicular adenomas are not forerunners to carcinomas; nevertheless, shared genetic alterations raise the possibility that at least a subset of follicular carcinoma arises in preexisting adenomas (see later). Although most adenomas are nonfunctional, a small minority produces thyroid hormones and causes clinically apparent thyrotoxicosis. Hormone production in functional adenomas (called *toxic adenoma*) is independent of TSH stimulation.

Pathogenesis

Somatic mutations that lead to constitutive activation of the TSH receptor signaling pathway are found in toxic adenomas and toxic multinodular goiter. Gain-of-function mutations—most often in the gene encoding the TSH receptor (*TSHR*), and less commonly, the α -subunit of G_s (*GNAS*)—cause follicular cells to secrete thyroid hormone independent of TSH stimulation (“thyroid autonomy”). This leads to hyperthyroidism and produces a functional “hot” nodule on imaging. *TSHR* and *GNAS* mutations have been identified in 50% to 80% of toxic adenomas, and a recent study suggested a two-hit model in which mutation of the histone methyltransferase *EZH1* is also required to produce a follicular adenoma. Notably, *TSHR*, *GNAS*, and *EZH1* mutations are rarely seen in follicular carcinomas; thus, toxic adenomas (and toxic multinodular goiters) do not seem to be precancerous lesions.

MORPHOLOGY

The typical thyroid follicular adenoma is a solitary, spherical or ovoid lesion that is often demarcated by a well-defined capsule (Fig. 24.16A). By contrast, as the name implies, follicular nodular disease (clinically, multinodular goiter) contains numerous nodules, even in patients presenting with a solitary dominant nodule. In freshly resected specimens, the adenoma bulges from the cut surface and compresses the adjacent thyroid. The color ranges from gray-white to red-brown, depending on the cellularity of the adenoma and its colloid content. Areas of hemorrhage, fibrosis, calcification, and cystic change, similar to those encountered in follicular nodular disease, are common in follicular adenomas, particularly within larger lesions.

Microscopically, the constituent cells often form uniform-appearing follicles that contain colloid (Fig. 24.16B). The follicular growth pattern is usually distinct from the adjacent nonneoplastic thyroid. The neoplastic cells show little variation in cell size, shape, or nuclear morphology, and mitotic figures are rare. Occasionally the neoplastic cells acquire brightly eosinophilic granular cytoplasm (**oncocyctic change**) (Fig. 24.17). The hallmark of many follicular adenomas is the presence of an intact, well-formed capsule encircling the tumor. A capsule is not present in all cases, but circumscription without invasion is a requisite of diagnosis. **Careful evaluation of the lesion periphery is therefore critical in distinguishing follicular adenoma from follicular carcinoma**, which demonstrates capsular, vascular, or intrathyroidal invasion (see later). Mitotic activity, necrosis, or high cellularity also raises the index of suspicion for carcinoma (see later).

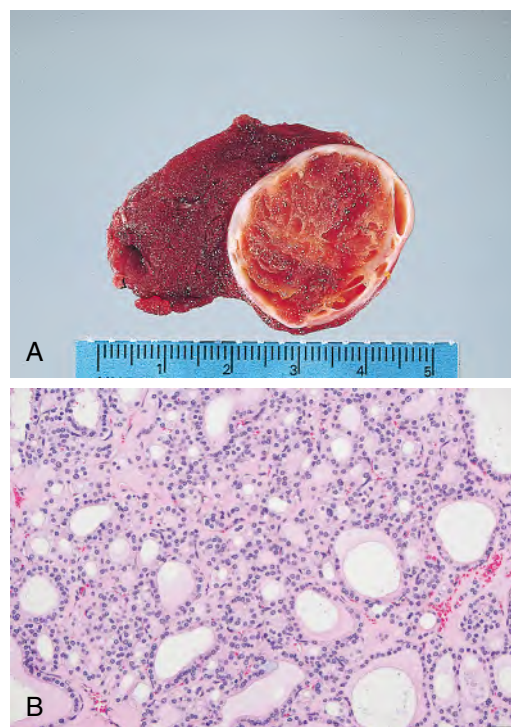


Fig. 24.16 Follicular adenoma of the thyroid. (A) A solitary, well-circumscribed nodule is seen. (B) The photomicrograph shows well-differentiated follicles resembling normal thyroid parenchyma.

Clinical Features

Many follicular adenomas present as solitary painless masses that are discovered on routine physical examination or incidentally in the setting of various radiology procedures. Larger masses may produce local symptoms, such as difficulty in swallowing. Nonfunctioning adenomas

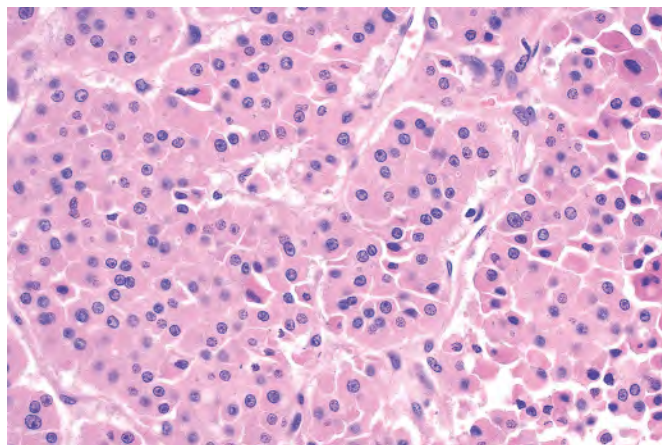


Fig. 24.17 Oncocytic adenoma. A high-power view showing that the tumor is composed of cells with abundant eosinophilic cytoplasm and small regular nuclei. (Courtesy Dr. Mary Sunday, Duke University, Durham, NC.)

take up less radioactive iodine than does normal thyroid parenchyma and on radionuclide scanning appear as cold nodules. Other techniques used to evaluate suspected adenomas are ultrasonography and fine-needle aspiration. **Because of the need for evaluating capsular integrity, the definitive diagnosis of adenoma can be made only after careful histologic examination of the resected specimen.** Suspected adenomas of the thyroid may have to be removed surgically to exclude malignancy, but recently molecular diagnostic studies of indeterminate thyroid nodules have been helpful in identifying which nodules require surgery for further assessment. Follicular adenomas do not recur or metastasize and have an excellent prognosis.

Thyroid Carcinomas

It is estimated that there were than 52,000 new cases of thyroid cancer in the United States in 2019, and about 2000 individuals succumbed to the disease. The rate of newly diagnosed thyroid cancer has been rising dramatically (currently, at the rate of 3% increase in incidence each year in the United States). Countries such as South Korea saw a veritable “epidemic” of thyroid cancer when ultrasound screening programs were put into effect. Despite the rising rates of newly diagnosed cancer, mortality from this disease is one of the lowest among solid tumors, with a 5-year survival greater than 98%. The major reason for this good outcome is that many of these cancers are found incidentally during neck palpation or by ultrasonography, and consist of small, localized lesions (typically papillary carcinomas) that have an excellent prognosis.

The major subtypes of thyroid carcinoma and their relative frequencies are:

- *Papillary thyroid carcinoma* (PTC) (~85% of cases)
- *Follicular or oncocytic thyroid carcinoma* (~5% – 15% of cases)
- *High grade differentiated and poorly differentiated carcinoma* (~5% of cases)
- *Anaplastic (undifferentiated) carcinoma* (~1% of cases)
- *Medullary thyroid carcinoma* (a neoplasm of C cells) (<5% of cases)

Noninvasive follicular thyroid neoplasm with papillary-like nuclear features (NIFTP) is a new name for encapsulated noninvasive follicular variant papillary thyroid cancer. These tumors are noninvasive and do not require aggressive treatment (though surgery is required for definitive diagnosis).

Most thyroid carcinomas are derived from thyroid follicular epithelium and of these the vast majority are well-differentiated lesions. Most high grade, poorly differentiated, and anaplastic (undifferentiated) carcinomas arise from well-differentiated PTC or follicular carcinomas through acquisition of additional mutations. These aggressive neoplasms and medullary thyroid carcinomas are the major causes of mortality from thyroid cancer.

We begin with a discussion of the molecular pathogenesis of thyroid neoplasms.

Pathogenesis

Driver Mutations. Distinct sets of driver mutations in cancer genes are found in the four major histologic variants of thyroid cancer. Among follicular neoplasms, there are both unique and shared genetic alterations between the histologic subtypes (Fig. 24.18), most of which involve components of the receptor tyrosine kinase (RTK) pathway, which is frequently mutated in many different cancers (Chapter 7).

- *Conventional papillary thyroid carcinomas.* Conventional PTCs have two defining genetic abnormalities: point mutations in *BRAF* and translocations that result in gene fusions of *RET*, *NTRK*, or *BRAF* (see Fig. 24.18).
- Up to 80% of conventional PTCs harbor gain-of-function mutations *BRAF* that lead to a valine-to-glutamate change in codon 600 (*BRAF*^{V600E}). *BRAF* encodes a serine/threonine kinase that

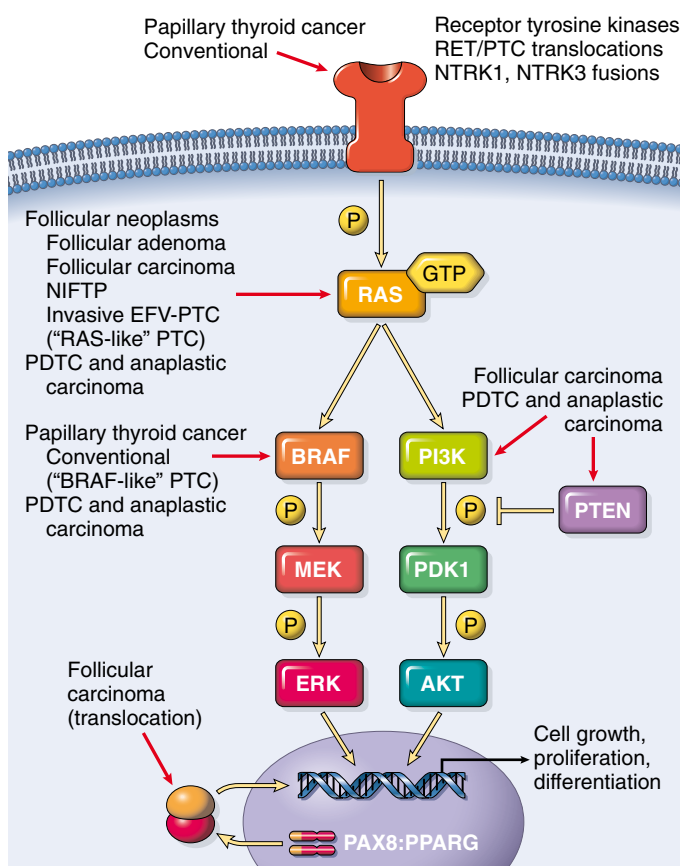


Fig. 24.18 Genetic alterations in follicle cell-derived malignancies of the thyroid. The most common mutations involve components of the MAP-kinase and PI3K/AKT signaling pathways. *EFV-PTC*, Encapsulated follicular variant of papillary thyroid carcinoma; *NIFTP*, noninvasive follicular thyroid neoplasm with papillary-like nuclear features; *PDTC*, poorly differentiated thyroid carcinoma.

lies downstream of receptor tyrosine kinases and RAS in growth factor signaling pathways. The presence of *BRAF* mutation in conventional PTC correlates with reduced expression of thyroid differentiation markers (such as thyroglobulin and thyroid peroxidase) and may be associated with a higher risk of extrathyroidal extension and recurrence. As discussed elsewhere, the *BRAF*^{V600E} mutation is found in many other cancers including melanoma and hairy cell leukemia (Chapter 7).

- The *RET* gene is located on chromosome 10q11 and encodes an RTK that is normally not expressed in thyroid follicular cells. As with many other RTKs implicated in cancer, PTC rearrangements involving *RET* create fusion genes that encode constitutively active forms of RET tyrosine kinase. There are more than 20 fusion partners of *RET*; *PTC1* (also known as *CCDC6*) being the most common. Similarly, paracentric inversions or translocations of *NTRK1* or *NTRK3*, also encoding receptor tyrosine kinases, are present in a subset of PTCs, particularly those that arise in childhood. These rearrangements also produce constitutively active NTRK fusion proteins.
- **Follicular neoplasms.** In contrast to papillary carcinomas, follicular neoplasms are often associated with gain-of-function mutations in *RAS*. Neoplasms with *RAS* mutations retain expression of thyroid differentiation markers (e.g., thyroglobulin, thyroid peroxidase), which may contribute to a growth pattern that resembles normal thyroid. Overall, up to 50% of follicular adenomas and follicular carcinomas harbor *RAS* mutations (see Fig. 24.18). Similarly, noninvasive follicular thyroid neoplasm with papillary-like nuclear features (discussed subsequently) demonstrate *RAS* point mutations.

A unique (2;3)(q13;p25) translocation has been described in a subset of follicular neoplasms. This translocation creates a fusion gene composed of portions of *PAX8*, a paired homeobox gene that is important in thyroid development, and the peroxisome proliferator-activated receptor gene (*PPARG*), which encodes a nuclear hormone receptor implicated in terminal differentiation of thyrocytes. Finally, up to 10% of follicular carcinomas exhibit gain-of-function mutations of *PIK3CA* (the gene that encodes phosphoinositide 3-kinase) or loss-of-function mutations of *PTEN*, a tumor suppressor gene and negative regulator of the PI3K pathway (see Fig. 24.18).

- **High grade, poorly differentiated, and anaplastic (undifferentiated) carcinomas.** These highly aggressive and lethal tumors can arise *de novo*, or, much more commonly, by transformation of a papillary or follicular thyroid carcinoma. In addition to having driver mutations seen in well-differentiated thyroid cancers, other common genetic “hits”—point mutations of *TP53* and the promoter region of *TERT* (which encodes the catalytic component of the enzyme telomerase)—are essentially restricted to aggressive carcinomas and likely have central roles in their genesis and behavior.
- **Medullary thyroid carcinomas.** Familial medullary thyroid carcinomas occur in multiple endocrine neoplasia, type 2 (MEN-2, see later) and are associated with germline *RET* mutations that lead to constitutive activation of the receptor. Somatic *RET* mutations are also seen in approximately one-half of nonfamilial (sporadic) medullary thyroid cancers. Chromosomal rearrangements involving *RET*, such as the *RET/PTC* translocations reported in papillary cancers, are not seen in medullary carcinomas.

Environmental Factors. The major risk factor predisposing to thyroid cancer is exposure to *ionizing radiation*, particularly during the first 2 decades of life. There was a marked increase in the incidence of conventional PTCs among children exposed to ionizing radiation after

the Chernobyl nuclear disaster in 1986. Radiation-induced PTCs have a higher frequency of chromosome rearrangements that produce gene fusions, possibly because of the ability of ionizing radiation to induce double-stranded DNA breaks. Deficiency of dietary iodine (and by extension, goiter) is associated with a higher frequency of follicular neoplasms.

Papillary Carcinoma

Papillary carcinoma is the most common form of thyroid cancer, accounting for nearly 85% of cases in the United States. It occurs throughout life but most often between 25 and 50 years of age, and accounts for most thyroid carcinomas associated with previous exposure to ionizing radiation.

MORPHOLOGY

Conventional papillary carcinomas may be solitary or multifocal. Some tumors are well circumscribed, while others infiltrate the adjacent parenchyma and have ill-defined margins. The tumors may contain areas of fibrosis and calcification and are often cystic. The cut surface sometimes reveals papillary foci that point to the diagnosis. The microscopic hallmarks of papillary carcinomas include (Fig. 24.19):

- Branching **papillae** having a fibrovascular stalk covered by a single to multiple layers of cuboidal epithelial cells. The papillae tend to be branching and complex with thin fibrovascular cores.
- Nuclei with finely dispersed chromatin and an optically clear or empty appearance, giving rise to the designation **ground-glass** or **Orphan Annie eye nuclei**. In addition, nuclear membrane irregularities result in wrinkling, nuclear grooves (coffee-bean shapes), or frank invaginations of the cytoplasm into the nucleus (nuclear **pseudoinclusions**).
- Concentrically calcified structures termed **psammoma bodies** are often present, usually within the cores of papillae or adjacent stroma. These structures are almost never found in follicular or medullary carcinomas, and they are a strong indication that the lesion is a papillary carcinoma when present in fine-needle aspiration material.
- Foci of **lymphatic invasion** by tumor are often present, but involvement of blood vessels is uncommon, particularly in smaller lesions. Metastases to adjacent cervical lymph nodes occur in up to one-half of cases.

The histologic subtypes of papillary carcinoma are beyond the scope of this book, with the exception of the follicular variant (discussed later).

Clinical Features

Most conventional papillary carcinomas present as asymptomatic thyroid nodules, but the first manifestation may be an enlarged cervical lymph node. Most carcinomas are single nodules that move freely with the thyroid gland during swallowing and are not distinguishable on examination from benign nodules. Hoarseness, dysphagia, cough, or dyspnea suggests advanced disease. In a minority of patients, particularly with follicular thyroid cancer, hematogenous metastases are present at the time of diagnosis, most commonly in the lung.

A variety of diagnostic tests have been used to help separate benign from malignant thyroid nodules, particularly fine-needle aspiration. Papillary carcinomas are cold masses on scintigraphy, and “hot” nodules are almost never malignant. However, this test is generally not performed unless the TSH level is low, since “hot” nodules produce excess thyroid hormone. Improvements in cytologic analysis accompanied by molecular testing have made fine-needle aspiration cytology a reliable test for distinguishing between benign and malignant nodules. In addition, recent

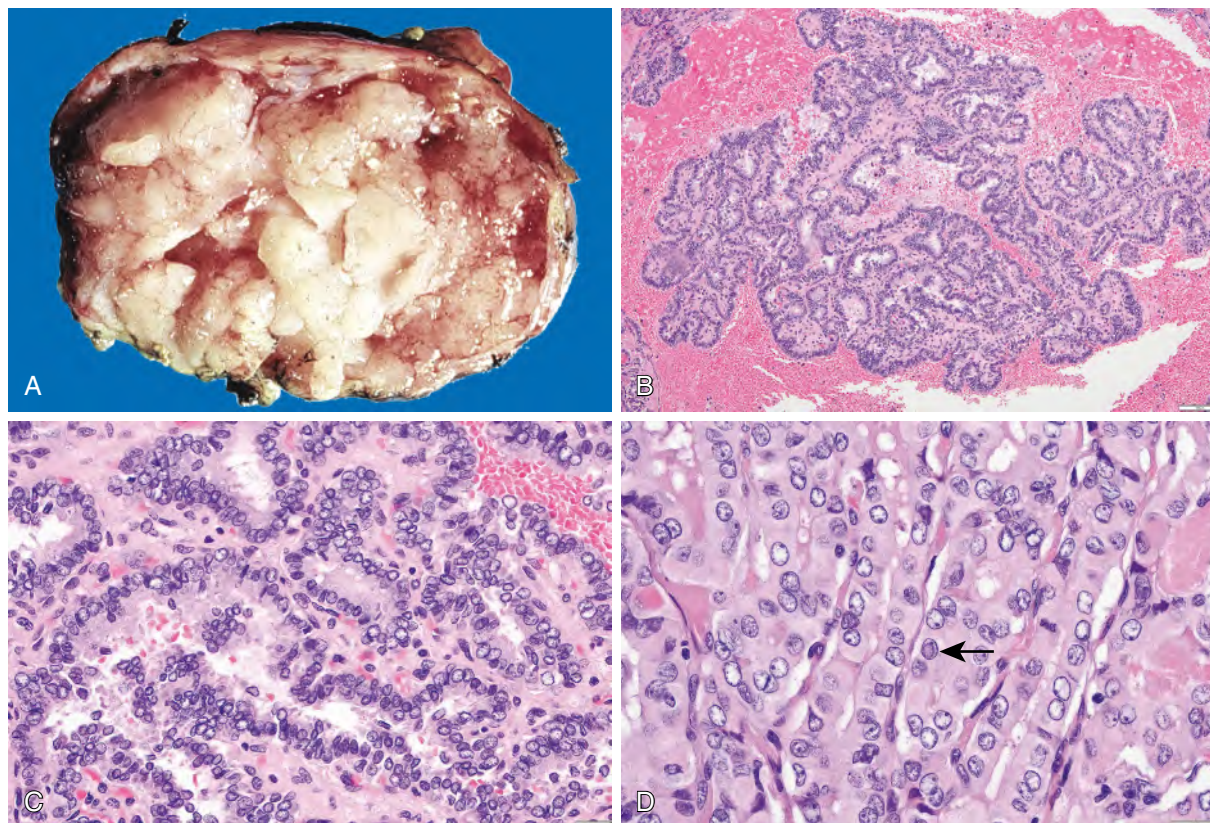


Fig. 24.19 Papillary carcinoma of the thyroid. (A) The macroscopic appearance of a papillary carcinoma with grossly discernible papillary structures. (B) This particular example contains well-formed papillae. (C) High power shows cells with characteristic enlarged, overlapping, and grooved nuclei with pale or empty-appearing chromatin, sometimes called “Orphan Annie eye” nuclei. (D) An intranuclear cytoplasmic pseudoinclusion is present (arrow).

advances in molecular diagnostics have also helped distinguish between these categories when the cytology is indeterminate.

Papillary thyroid carcinomas have an excellent prognosis, with a 10-year survival rate in excess of 95%. Based on recurrence risk profile, some patients have excellent prognoses with lobectomy alone. Those with higher risk of recurrence require total thyroidectomy followed by radioactive iodine ablation. Thyroid-sparing surgery has become increasingly important in order to preserve endocrine thyroid function and minimize vocal cord palsy (due to recurrent laryngeal nerve injury).

Between 5% and 20% of patients with PTC have local or regional recurrences, and 10% to 15% have distant metastases. The

prognosis is dependent on several factors including age (in general, being less favorable among patients older than 55 years), presence of extrathyroidal extension, and presence of distant metastases (stage).

Follicular Carcinoma

Follicular carcinoma accounts for 5% to 15% of primary thyroid cancers; it is more frequent in areas with dietary iodine deficiency, where it constitutes 25% to 40% of thyroid cancers. It presents more often in older patients than does papillary carcinoma; the peak incidence is between 40 and 60 years of age.

MORPHOLOGY

Follicular carcinomas are usually solitary nodules that may be circumscribed or widely infiltrative (Fig. 24.20A). Sharply demarcated lesions may be exceedingly difficult to distinguish from follicular adenomas by gross examination. Larger lesions may penetrate the capsule and infiltrate into the adjacent neck. They are gray to tan to pink on cut section and may be translucent due to the presence of large, colloid-filled follicles. Degenerative changes, such as central fibrosis and foci of coarse calcification, may be present.

Microscopically, most follicular carcinomas are composed of uniform cells forming small follicles containing colloid, quite reminiscent of normal thyroid (Fig. 24.20B). Occasional tumors are dominated by cells with abundant granular, eosinophilic cytoplasm (**oncocytic [formerly known as Hürthle cell] thyroid**

carcinoma). Whatever the pattern, the nuclei lack the features typical of papillary carcinoma, and psammoma bodies are not present. **There is no reliable cytologic difference between follicular adenomas and minimally invasive follicular carcinomas, making them indistinguishable on fine-needle aspiration biopsy.** Making this distinction requires extensive histologic sampling of the tumor capsule to exclude capsular and/or vascular invasion (Fig. 24.21). Vascular invasion involves venous vessels within and just beyond the capsule. Unlike in papillary carcinomas, lymphatic spread is uncommon in follicular carcinomas. By contrast, the diagnosis of carcinoma is obvious in widely invasive follicular carcinomas, which infiltrate the thyroid parenchyma and extrathyroidal soft tissues.

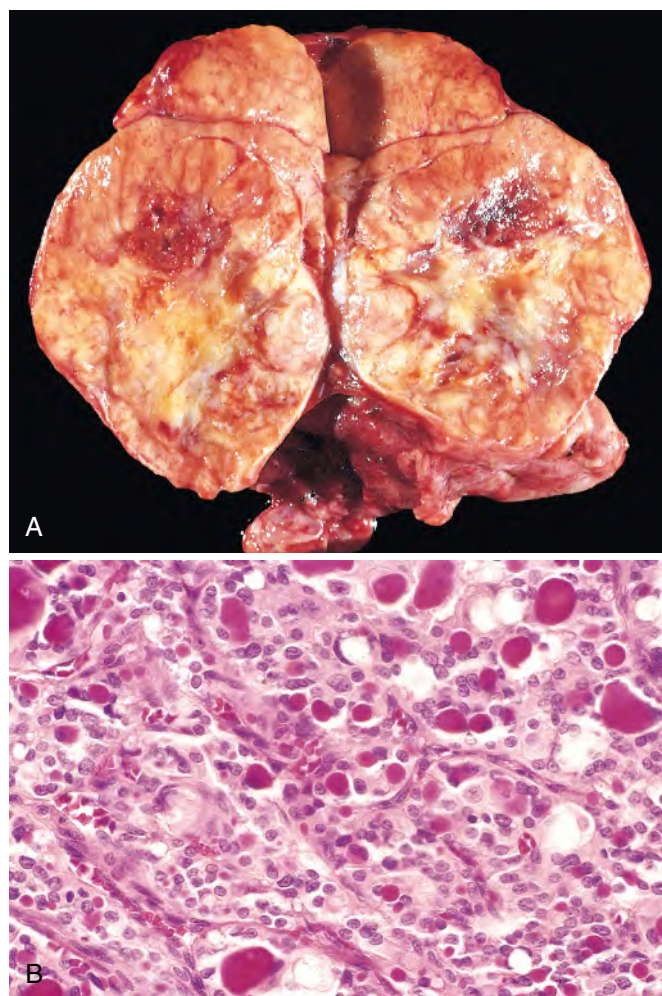


Fig. 24.20 Follicular carcinoma. (A) Cut surface of a follicular carcinoma with substantial replacement of the lobe of the thyroid. The tumor has a light-tan appearance and contains small foci of hemorrhage. (B) A few of the glandular lumens contain recognizable colloid.

Clinical Features

Follicular carcinomas present as slowly enlarging painless nodules. Most frequently they are cold nodules on scintigrams. Very rarely, better-differentiated lesions may be hyperfunctional, take up radioactive iodine, and appear warm on scintigraphy. Because follicular carcinomas have little propensity for invading lymphatics, regional lymph nodes are typically not involved, while vascular (hematogenous) dissemination is common, with metastases to bone, lungs, liver, brain, and elsewhere.

The prognosis depends largely on the extent of invasion and stage at presentation. Widely invasive follicular carcinoma often presents with systemic metastases, and as many as one-half of affected patients succumb to their disease within 10 years. This is in sharp contrast to minimally invasive follicular carcinomas, which have a 10-year survival rate of greater than 90%. Aside from some minimally invasive follicular carcinomas which have a low recurrence risk, most are treated with total thyroidectomy followed by radioactive iodine administration, which can be used to identify metastases and to ablate such lesions. In addition, because any residual follicular carcinoma may respond to TSH stimulation, patients are usually treated with thyroid hormone after surgery to suppress endogenous TSH levels. Serum thyroglobulin levels are used for monitoring tumor recurrence, because this thyroid protein

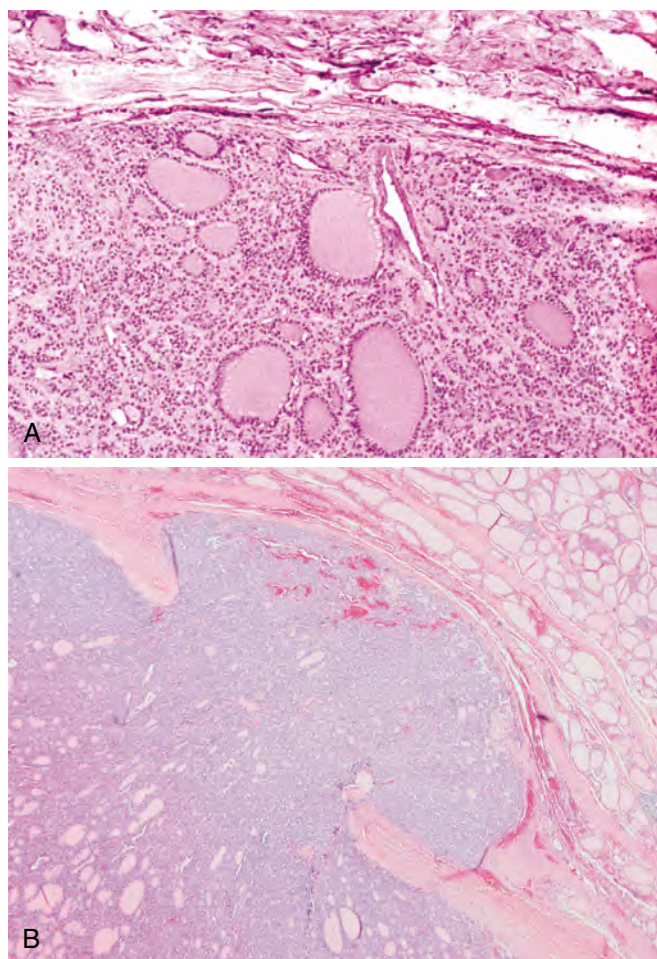


Fig. 24.21 Capsular integrity in follicular neoplasms. In adenoma (A), a fibrous capsule, usually thin but occasionally more prominent, circumferentially surrounds the neoplastic follicles, and no capsular invasion is seen; compressed normal thyroid parenchyma is usually present external to the capsule (*top of panel*). In contrast, follicular carcinomas demonstrate capsular invasion (B) that may be minimal, as in this case, or widespread. The presence of vascular invasion is another feature of follicular carcinomas.

should be barely detectable in a patient who is free of disease if they have had a total thyroidectomy.

Noninvasive Follicular Thyroid Neoplasm With Papillary-like Nuclear Features and Invasive Encapsulated Follicular Variant of PTC

Some neoplasms with entirely follicular growth show nuclear features reminiscent of PTC (enlargement, chromatin clearing, membrane irregularities). When showing no invasion, they were historically termed “noninvasive encapsulated follicular variant of PTC.” However, in retrospect, these lesions showed very low (essentially zero) risk of recurrence or metastasis, and the name attached to them was changed to “noninvasive follicular thyroid neoplasm with papillary-like nuclear features” or NIFTP if strict diagnostic criteria were met. Those showing invasion (capsular, vascular, or intrathyroidal) are termed “invasive encapsulated follicular variant of PTC.” These neoplasms tend to be driven by mutations in *RAS*, similar to follicular adenoma and carcinoma, and they tend to have hematogenous metastasis secondary to vascular invasion, also similar to follicular carcinoma. How the nomenclature may shift in the future is to be determined.

High Grade Differentiated, Poorly Differentiated, and Anaplastic (Undifferentiated) Carcinoma

These carcinomas are aggressive neoplasms of thyroid follicular epithelium, cumulatively accounting for less than 5% of thyroid tumors, but with a high mortality rate (approaching 100% for anaplastic lesions). Patients with these tumors are older than those with other types of thyroid cancer, with a mean age of 65 years. Approximately one-quarter of patients have a past history of a well-differentiated follicular neoplasm, and another one-quarter harbor a concurrent well-differentiated neoplasm in the resection specimen.

MORPHOLOGY

Microscopically, high grade differentiated and poorly differentiated carcinomas have either tumor necrosis or increased mitotic activity. Poorly differentiated carcinomas are arranged in solid, insular (nested), or trabecular growth patterns. Anaplastic carcinomas no longer resemble thyroid follicular cells, and they demonstrate variable morphology, including epithelioid (sometimes resembling squamous cell carcinoma), spindled (resembling sarcoma), pleomorphic, or giant-cell rich. Foci of papillary or follicular carcinoma may be present in some tumors, consistent with an origin from a well-differentiated carcinoma. The neoplastic cells of anaplastic carcinoma often lose markers of thyroid follicular cell differentiation, but retain genetic driver mutations.

Clinical Features

High grade differentiated and poorly differentiated carcinomas may present similarly to papillary or follicular carcinomas, and the diagnosis may only be made on histologic examination of the resected tumor. By contrast, anaplastic carcinomas usually present as rapidly enlarging bulky neck masses. In most cases, the disease has spread beyond the thyroid into adjacent neck structures or has metastasized to the lungs at the time of presentation. Symptoms related to compression and invasion of adjacent structures, such as dyspnea, dysphagia, hoarseness, and cough, are common. Anaplastic carcinomas of the thyroid are one of the most aggressive cancers known; in most cases death occurs in less than 1 year. High grade differentiated and poorly differentiated carcinomas fare somewhat better with surgery, external beam radiotherapy, and radioactive iodine.

Medullary Carcinoma

Medullary carcinomas of the thyroid are neuroendocrine neoplasms derived from parafollicular cells (C cells). They account for approximately 5% of thyroid neoplasms. Medullary carcinomas, like normal C cells, secrete *calcitonin*, the measurement of which plays an important role in the diagnosis and postoperative follow-up of patients. Calcitonin is a regulator of calcium metabolism. It is normally produced in response to hypercalcemia, and it reduces serum calcium by inhibiting osteoclast activity and renal tubular reabsorption of calcium. In some instances, the tumor cells elaborate other polypeptide hormones, such as serotonin, corticotropin, and vasoactive intestinal peptide (VIP). About 70% of tumors arise sporadically. The remainder occur in the setting of MEN2 syndrome (caused by germline *RET* mutations). Activating point mutations in the *RET* proto-oncogene play an important role in the development of both familial and sporadic medullary carcinomas. Syndromic cases tend to occur in younger patients, even during the first decade of life. In contrast, sporadic medullary carcinomas are neoplasms of adulthood, with a peak incidence in the 40s and 50s.

MORPHOLOGY

Sporadic medullary thyroid carcinomas present as a solitary nodule (Fig. 24.22A). In contrast, **bilaterality and multicentricity are common in syndromic cases**. Larger lesions often contain areas of necrosis and hemorrhage and may extend through the capsule of the thyroid. The tumors are firm, pale gray to tan, and infiltrative. There may be foci of hemorrhage and necrosis in larger lesions.

Microscopically, medullary carcinomas are composed of polygonal to spindle-shaped cells, which may form nests, trabeculae, and even follicles. **Amyloid deposits** derived from calcitonin polypeptides are present in the stroma in many cases (Fig. 24.22B). Calcitonin is readily demonstrable within the tumor cells and the stromal amyloid by immunohistochemical methods. As with all neuroendocrine tumors, electron microscopy reveals variable numbers of membrane-bound electron-dense granules within the cytoplasm of the neoplastic cells (Fig. 24.23). One of the features of syndromic medullary carcinomas is the presence of **C-cell hyperplasia** in the surrounding thyroid parenchyma, a feature that is usually absent in sporadic lesions, and that is believed to be a precursor lesion in syndromic cases. Thus, the presence of multiple prominent clusters of C cells (often in the lobe contralateral to the main carcinoma) should raise the possibility of an inherited predisposition, even if a family history is not present.

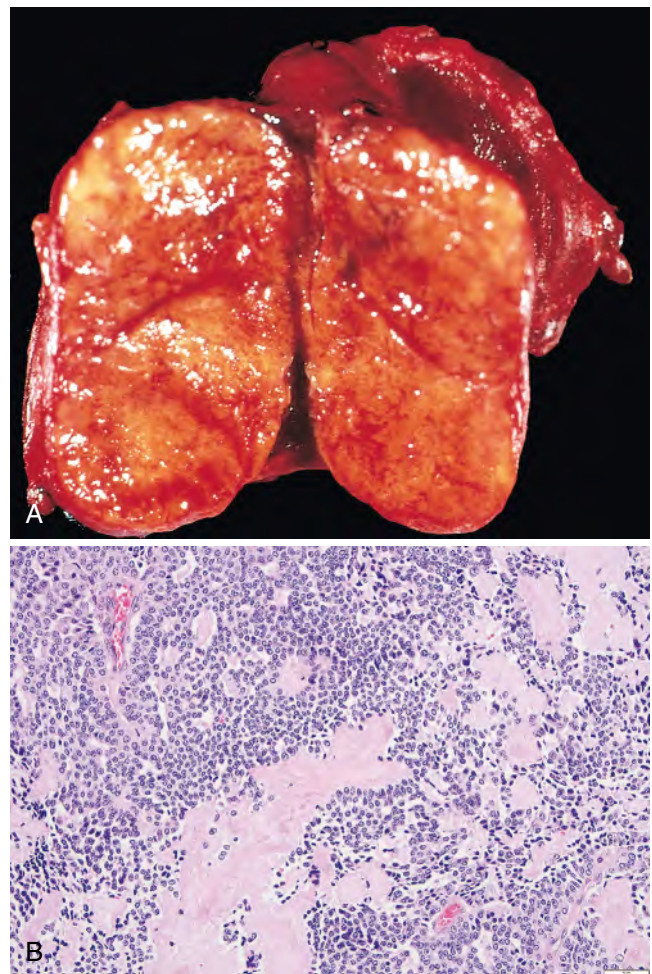


Fig. 24.22 Medullary carcinoma of the thyroid. (A) These tumors typically do not have connective tissue capsules. (B) Histology demonstrates a nested neoplasm with round nuclei, stippled chromatin, and intercellular deposition of pink-colored amyloid derived from calcitonin secreted by the neoplastic cells.

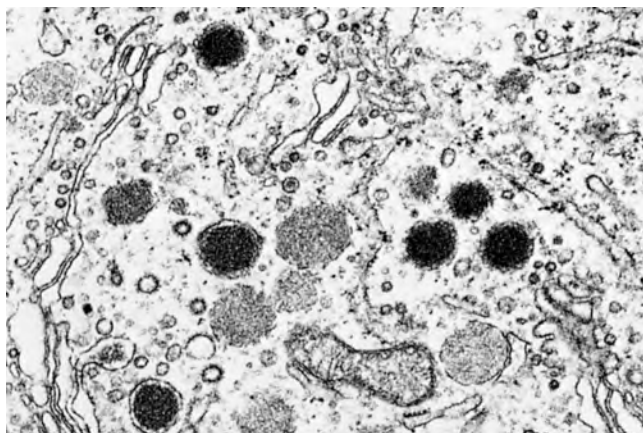


Fig. 24.23 Electron micrograph of medullary thyroid carcinoma. These cells contain membrane-bound secretory granules that are the sites of storage of calcitonin and other peptides.

Clinical Features

Sporadic cases of medullary carcinoma come to medical attention most often as a mass in the neck, sometimes associated with dysphagia or hoarseness. In some instances, the initial manifestations are those of a paraneoplastic syndrome caused by the secretion of a peptide hormone (e.g., diarrhea due to the secretion of VIP, or Cushing syndrome due to corticotropin). Hypocalcemia is not a prominent feature, despite the presence of raised calcitonin levels. In addition to circulating calcitonin, secretion of carcinoembryonic antigen (CEA) by the neoplastic cells is a useful biomarker, especially for presurgical assessment of tumor load and in uncommon calcitonin-negative tumors.

Patients with familial syndromes may come to attention because of symptoms localized to the thyroid or because of endocrine neoplasms in other organs (e.g., adrenal or parathyroid glands). Asymptomatic MEN2 individuals carrying germline *RET* mutations are offered early prophylactic thyroidectomy to prevent the otherwise inevitable development of medullary carcinoma, the major risk factor for poor outcome in these individuals; the specific timing of planned surgery often depends on the particular *RET* mutation. Sometimes the only histologic finding in the resected thyroid of asymptomatic carriers is the presence of C-cell hyperplasia or small (<1 cm) medullary microcarcinomas. Several small-molecule inhibitors of RET tyrosine kinase have recently been developed and are now being used to treat patients with medullary carcinoma.

KEY CONCEPTS

Thyroid Neoplasms

- Most thyroid neoplasms manifest as thyroid nodules.
- Multiple genetic pathways are involved in thyroid carcinogenesis. Conventional papillary carcinomas harbor either *BRAF* V600E point mutations or fusion genes (*RET*, *NTRK*, *BRAF*). Follicular neoplasms of all subtypes are characterized by oncogenic *RAS* mutations and *PAX8:PPARG* fusions. Progression to aggressive neoplasms is associated with *TP53* and *TERT* promoter mutations.
- Follicular adenomas and carcinomas are both composed of well-differentiated follicular epithelial cells; the latter are distinguished by evidence of capsular and vascular invasion.
- Conventional papillary carcinomas are recognized based on architecture and nuclear features (chromatin clearing, membrane wrinkling, pseudoinclusions).

Psammoma bodies are a characteristic feature of papillary carcinomas; although these neoplasms often metastasize by way of lymphatics, the prognosis is excellent.

- High grade differentiated, poorly differentiated, and anaplastic carcinomas are thought to arise from more differentiated thyroid carcinomas. They are highly aggressive, often lethal cancers.
- Medullary carcinomas are neoplasms arising from parafollicular C cells and can occur in either sporadic (70%) or syndromic (30%) settings. Multicentricity and C cell hyperplasia are features of syndromic cases. Amyloid deposits are a characteristic histologic finding.

CONGENITAL ANOMALIES

Thyroglossal duct cyst is the most common clinically significant congenital anomaly of the thyroid. A sinus tract may persist as a vestige of the tubular development of the thyroid gland. Parts of this tube may be obliterated, leaving small segments that form cysts. These present at any age and may not become evident until adult life. Secretions may collect within the cysts to form either spherical masses or fusiform swellings, rarely over 2 to 3 cm in diameter, that present in the midline of the neck anterior to the trachea. The cyst may be lined by stratified squamous or pseudostratified respiratory-type epithelium. Characteristically, the adjacent stroma contains a lymphocytic infiltrate and, in many cases, clusters of thyroid follicles. Superimposed infection may convert these lesions into abscess cavities, and rarely, they give rise to cancers.

PARATHYROID GLANDS

The parathyroid glands develop from pharyngeal pouches that also give rise to the thymus. They are most commonly located in close proximity to the upper and lower poles of each thyroid lobe but may be found anywhere along the pathway of descent of the pharyngeal pouches, including the carotid sheath, the thymus, and elsewhere in the anterior mediastinum. The four parathyroid glands are composed of two cell types: chief cells and oxyphil cells. *Chief cells* predominate; they are polygonal, 12 to 20 μ m in diameter, and have central, round, uniform nuclei and light to dark pink cytoplasm. Chief cells have secretory granules containing *parathyroid hormone* (PTH). *Oxyphil cells* and transitional oxyphils are found throughout the normal parathyroid, either singly or in small clusters. They are slightly larger than the chief cells, have eosinophilic cytoplasm, and are tightly packed with mitochondria. Glycogen granules are also present in these cells, but secretory granules are sparse or absent. Oxyphil cells have a large number of calcium sensing receptors and are thought to produce various factors like parathyroid hormone-related protein (PTHr) and calcitriol. In early infancy and childhood, the parathyroid glands are composed almost entirely of solid sheets of chief cells. The amount of stromal fat increases up to 25 years of age, reaching a maximum of approximately 30% of the gland, and then plateaus.

The function of the parathyroid glands is to regulate calcium homeostasis. The activity of the parathyroid glands is controlled by the level of free (ionized) calcium in the blood. Normally, decreased levels of free calcium stimulate the synthesis and secretion of PTH. PTH regulates blood calcium levels by:

- Increasing renal tubular reabsorption of calcium, thereby conserving free calcium
- Increasing conversion of vitamin D to its active dihydroxy form in the kidneys, which, in turn, augments gastrointestinal calcium absorption
- Increasing urinary phosphate excretion, thereby lowering serum phosphate levels and further increasing calcium (since phosphate binds to ionized calcium)

- Increasing osteoclastic activity (i.e., bone resorption, thus releasing ionized calcium), by promoting the differentiation of osteoclast progenitor cells into mature osteoclasts

The net result of these activities is to elevate the level of free calcium, which, in turn, inhibits further PTH secretion in a classic feedback loop.

Similar to the other endocrine organs, abnormalities of the parathyroid glands include both hyperfunction and hypofunction. Tumors of the parathyroid glands, in contrast to thyroid tumors, usually come to attention because of excessive secretion of PTH rather than mass effects.

HYPERPARATHYROIDISM

Hyperparathyroidism is caused by elevated PTH and is classified into primary, secondary, and tertiary types.

- Primary hyperparathyroidism:** an autonomous overproduction of PTH, usually resulting from an adenoma or hyperplasia of parathyroid tissue
- Secondary hyperparathyroidism:** compensatory hypersecretion of PTH, usually in response to chronic renal failure or vitamin D deficiency
- Tertiary hyperparathyroidism:** persistent hypersecretion of PTH, representing autonomous PTH production that usually develops after long-standing secondary hyperparathyroidism

Primary Hyperparathyroidism

Primary hyperparathyroidism is one of the most common endocrine disorders and an important cause of hypercalcemia. The frequency of the various parathyroid lesions underlying hyperfunction is:

- Adenoma: 85% to 95%
- Primary hyperplasia (diffuse or nodular): 5% to 10%
- Parathyroid carcinoma: ~1%

Primary hyperparathyroidism is usually a disease of adults and is more common in females than males by a ratio of nearly 4:1. The annual incidence is estimated to be about 25 cases per 100,000 in the United States and Europe; as many as 80% of patients with this condition are identified in the outpatient setting when hypercalcemia is discovered incidentally on a serum electrolyte panel. Most cases occur in adults in their 50s or older.

The most common cause of primary hyperparathyroidism is a solitary sporadic parathyroid adenoma (Fig. 24.24). Most, if not all,

sporadic parathyroid adenomas are monoclonal, consistent with a neoplastic origin. There are two molecular defects that have an established role in the development of sporadic adenomas:

- Cyclin D1 (CCND1) gene inversions** leading to overexpression of cyclin D1 protein, a major regulator of the cell cycle. A pericentromeric inversion on chromosome 11 results in relocation of *CCND1* (normally on 11q), so that it is positioned adjacent to the *PTH* gene (on 11p). As a consequence of these changes, a 5'-regulatory element flanking the *PTH* gene directs overexpression of cyclin D1, causing the cells to proliferate. In addition, in tumors lacking *CCND1* rearrangements, cyclin D1 is overexpressed in up to 40% of parathyroid adenomas, suggesting that mechanisms other than *CCND1* gene inversion can lead to its overexpression.
- MEN1* mutations:** Approximately 15% of sporadic parathyroid tumors have somatic mutations of the *MEN1* gene, a tumor suppressor gene on chromosome 11q13, with loss of heterozygosity (LOH) of the second allele. Germline mutations of *MEN1* are also found in patients with familial parathyroid adenomas (see later).
- In addition, *CDC73*, which encodes a protein likely involved in transcriptional regulation called parafibromin, is mutated in ~70% of sporadic parathyroid carcinomas, and occasionally in adenomas. Germline mutations of *CDC73* lead to a rare syndrome known as *hyperparathyroidism-jaw tumor (HPT-JT) syndrome*, which includes parathyroid carcinomas and ossifying jaw tumors as part of the disease spectrum.

Familial syndromes are a distant second to sporadic parathyroid adenomas as causes of primary hyperparathyroidism. The genetic syndromes associated with *familial parathyroid adenomas* include MEN-1, MEN-2, and MEN-4 (discussed in detail later). *Familial hypocalciuric hypercalcemia (FHH)* is a rare autosomal dominant disorder most often caused by heterozygous loss-of-function mutations in the parathyroid calcium-sensing receptor (*CASR*) gene that results in decreased sensitivity to extracellular calcium. It is marked by elevated calcium levels in the setting of a nonsuppressed PTH; however, such patients do not in general exhibit the downstream negative effects of hyperparathyroidism.

MORPHOLOGY

The morphologic changes seen in primary hyperparathyroidism involve the parathyroid glands and organs that are affected by elevated levels of PTH and calcium. Parathyroid **adenomas** are almost always solitary and, like normal parathyroid glands, may be in close proximity to the thyroid or in an ectopic site (e.g., the mediastinum). The typical adenoma averages 0.5 to 5 g and consists of a well-circumscribed, soft, tan to reddish-brown nodule. In contrast to primary hyperplasia, the glands outside the adenoma are usually normal in size or shrunk because of feedback inhibition by elevated levels of serum calcium. Microscopically, parathyroid adenomas are mostly composed of sheets of uniform, polygonal chief cells with small, centrally placed nuclei (Fig. 24.25). At least a few nests of larger oxyphil cells are present as well; uncommonly, adenomas are composed entirely of oxyphils (**oxyphil adenomas**) or clear cells (**water-clear adenomas**). A rim of compressed, nonneoplastic parathyroid tissue may be visible at the edge of the adenoma. Mitotic figures are rare, but it is not uncommon to find bizarre and pleomorphic nuclei even within adenomas (so-called **endocrine atypia**); this is not a criterion for malignancy. In contrast to the normal parathyroid parenchyma, adipose tissue is inconspicuous.

Primary hyperplasia may occur sporadically or as a component of MEN syndrome. Although classically all four glands are involved, there is frequently asymmetry with apparent sparing of one or two glands, making the distinction between hyperplasia and adenoma difficult. The combined weight of all glands rarely exceeds 1 g. Microscopically, the most common pattern seen is that of

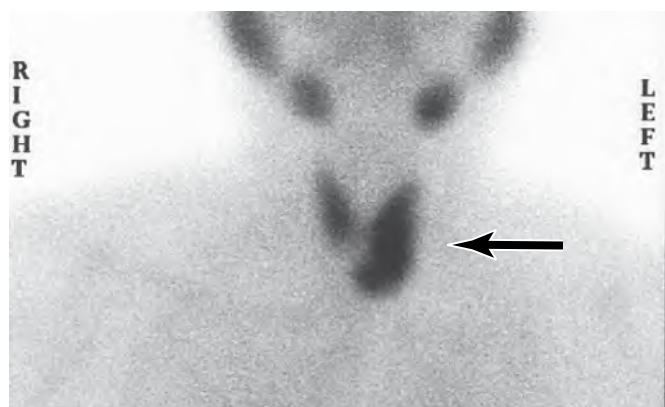


Fig. 24.24 Parathyroid adenoma imaging. Technetium-99m-sestamibi radionuclide scan demonstrates an area of increased uptake corresponding to the left inferior parathyroid gland (arrow), which contained a parathyroid adenoma. Preoperative scintigraphy is useful in localizing and distinguishing adenomas from parathyroid hyperplasia, where more than one gland demonstrates increased uptake.

chief cell hyperplasia, which may involve the glands in a diffuse or multinodular pattern. Less commonly, the constituent cells contain abundant glycogen, giving them a clear appearance in histologic sections (“water-clear cell hyperplasia”). In many instances, there are islands of oxyphils, and poorly developed, delicate fibrous strands may envelop the nodules. As in the case of adenomas, stromal fat is inconspicuous within hyperplastic glands.

Parathyroid carcinomas are defined by an invasive growth pattern and/or metastasis. These tumors involve one parathyroid gland and consist of gray-white, irregular masses that sometimes exceed 10 g in weight. The cells are usually uniform and resemble normal parathyroid cells and are arrayed in nodular or trabecular patterns. **Diagnosis of carcinoma based on cytologic detail is unreliable, and invasion of surrounding tissues and metastasis are the only reliable criteria.** Local recurrence occurs in one-third of cases, and more distant dissemination occurs in another third.

Atypical parathyroid neoplasms are a newly-defined entity consisting of neoplasms with features raising the possibility of, but not diagnostic of, carcinoma (fibrosis, cystic change, macronucleoli). Many demonstrate parafibromin loss by immunohistochemistry, reflecting underlying mutation and/or deletion of *CDC73*. Patients with parafibromin-deficient parathyroid neoplasms or parathyroid carcinomas should be screened for germline *CDC73* gene mutation, because up to 30% of patients with parathyroid carcinoma may have an underlying HPT-JT syndrome.

Morphologic changes of hyperparathyroidism in the skeletal system (Chapter 26) and the urinary tract deserve special mention. Symptomatic, untreated primary hyperparathyroidism manifests with three interrelated skeletal abnormalities: osteoporosis, brown tumors, and osteitis fibrosa cystica. The osteoporosis results in decreased bone mass, with preferential involvement of the phalanges, vertebrae, and proximal femur. For unknown reasons, the increased osteoclast activity in hyperparathyroidism affects cortical bone (subperiosteal and endosteal surfaces) more severely than medullary bone. In medullary bone, osteoclasts tunnel centrally along the length of the trabeculae, creating the appearance of railroad tracks and producing what is known as **dissecting osteitis** (Fig. 24.26). The marrow spaces around the affected surfaces are replaced by fibrovascular tissue.

The bone loss predisposes to microfractures and secondary hemorrhages that elicit an influx of macrophages and an ingrowth of reparative fibrous tissue, creating a mass of reactive tissue known as a **brown tumor** (see Fig. 26.13, Chapter 26). The brown color is the result of the vascularity, hemorrhage, and hemosiderin deposition, and it is not uncommon for the lesions to undergo cystic degeneration. The combination of increased osteoclast activity, peritrabecular fibrosis, and cystic brown tumors is the hallmark of severe hyperparathyroidism and is known as **generalized osteitis fibrosa cystica (von Recklinghausen disease of bone)**.

PTH-induced hypercalcemia favors formation of **urinary tract stones** (nephrolithiasis) as well as calcification of the renal interstitium and tubules (nephrocalcinosis). Metastatic calcification secondary to severe hypercalcemia may also be seen in other sites, including the stomach, lungs, myocardium, and blood vessels.

Clinical Features

Primary hyperparathyroidism may be (1) asymptomatic and identified on routine blood chemistry profile, or (2) associated with the classic clinical manifestations of primary hyperparathyroidism.

- **Asymptomatic hyperparathyroidism.** Because serum calcium levels are now assessed in blood tests, most patients with primary hyperparathyroidism are diagnosed incidentally. In fact, primary hyperparathyroidism is the most common cause of asymptomatic hypercalcemia. Hence, many of the classic

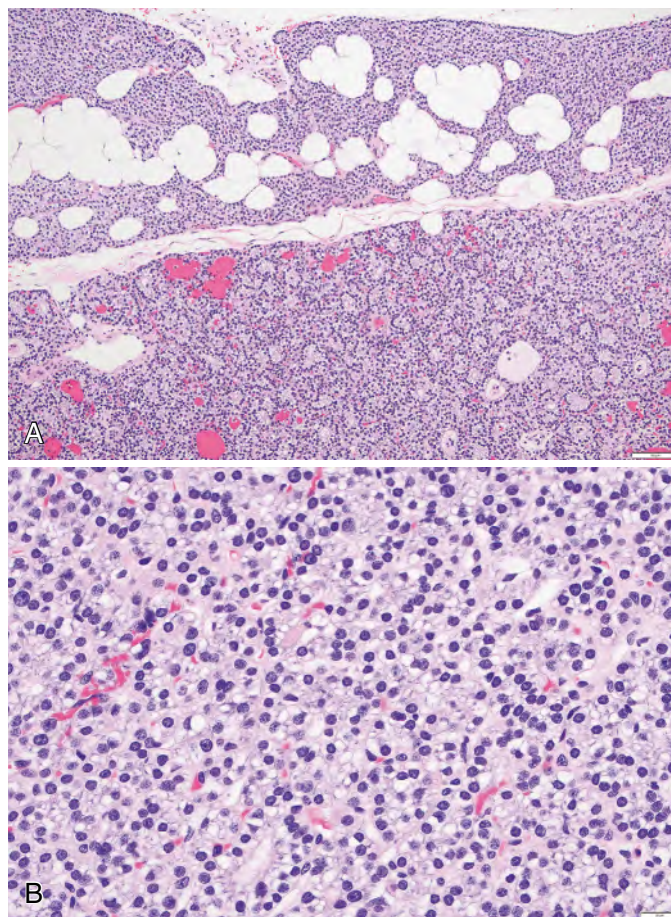


Fig. 24.25 Parathyroid adenoma. (A) Solitary chief cell parathyroid adenoma (low-power photomicrograph) revealing delineation from the residual fat-containing gland above. (B) High-power detail of a chief cell parathyroid adenoma. Cells are monotonous with dark round nuclei and pale purple to vacuolated cytoplasm.

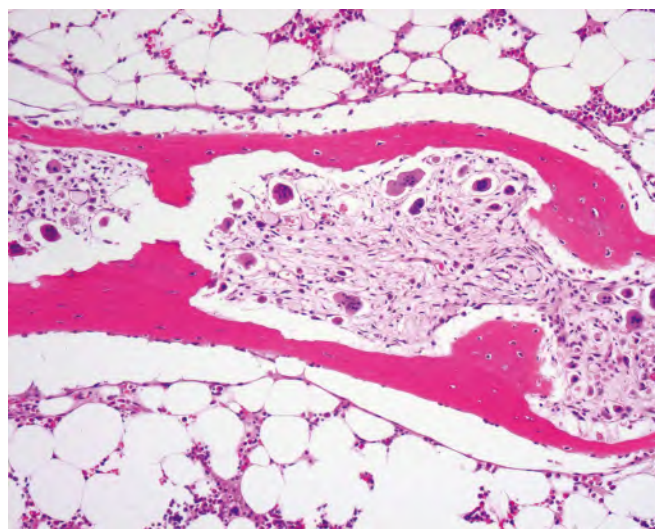


Fig. 24.26 Hyperparathyroidism with osteoclasts boring into the center of the trabeculum (dissecting osteitis). (From Horvai A. *Bone and Soft Tissue Pathology: A Volume in the High Yield Pathology Series*. Elsevier; 2012.)

manifestations and associated pathologic changes, particularly those referable to bone and renal disease, such as brown tumors, osteitis fibrosa cystica, and urinary stones, are now seen infrequently in clinical practice. Among other causes of hypercalcemia (Table 24.5), malignancy stands out as the most frequent cause of symptomatic hypercalcemia in adults and must be excluded by appropriate clinical and laboratory investigations. As discussed in Chapter 7, hypercalcemia can occur with solid tumors, such as lung, breast, head and neck, and renal cancers, and with hematologic malignancies, particularly multiple myeloma. The most common mechanism (in ~80% of cases) through which osteolytic solid tumors induce hypercalcemia is by secretion of PTH-related peptide (PTHrP), which (like PTH) induces osteoclastic bone resorption and hypercalcemia; the remaining 20% induce hypercalcemia through metastases to the bone and subsequent cytokine-induced bone resorption. In individuals with primary hyperparathyroidism, serum PTH levels are elevated or inappropriately normal for the level of serum calcium, whereas PTH levels are low to undetectable in hypercalcemia caused by nonparathyroid diseases (see Table 24.5). Immunoassays specific for PTH and PTHrP are useful in distinguishing primary hyperparathyroidism and malignancy-associated hypercalcemia. Other laboratory alterations referable to PTH excess include hypophosphatemia and increased urinary excretion of both calcium and phosphate. Secondary renal disease may in turn lead to phosphate retention, “normalizing” the serum phosphate levels.

- **Symptomatic primary hyperparathyroidism.** The signs and symptoms of hyperparathyroidism reflect the combined effects of increased PTH secretion and hypercalcemia. Classically, severe primary hyperparathyroidism has been associated with “painful bones, renal stones, abdominal groans, and psychic moans.” The constellation of symptoms includes
 - *Bone disease* and bone pain secondary to fractures of bones weakened by osteoporosis or osteitis fibrosa cystica
 - *Nephrolithiasis* (renal stones), with attendant pain and obstructive uropathy. Chronic renal insufficiency and abnormalities in renal function lead to polyuria and secondary polydipsia
 - Gastrointestinal disturbances, including constipation, nausea, peptic ulcers, pancreatitis, and gallstones
 - Central nervous system alterations, including depression, lethargy, and eventually seizures
 - Neuromuscular abnormalities, including weakness and fatigue
 - Cardiac manifestations, including aortic or mitral valve calcifications (or both)

TABLE 24.5 Causes of Hypercalcemia

Increased PTH	Decreased PTH
Hyperparathyroidism	Hypercalcemia of malignancy ^a
Primary (adenoma > hyperplasia) ^a	Vitamin D toxicity
Tertiary ^b	Immobilization
	Thiazide diuretics
Familial hypocalciuric hypercalcemia	Granulomatous disease (e.g., sarcoidosis)

PTH, Parathyroid hormone.

^aPrimary hyperparathyroidism is the most common cause of hypercalcemia overall. Malignancy is the most common cause of symptomatic hypercalcemia. Primary hyperparathyroidism and malignancy account for nearly 90% of cases of hypercalcemia.

^bSecondary and tertiary hyperparathyroidism are most commonly associated with progressive renal failure.

Secondary Hyperparathyroidism

Secondary hyperparathyroidism is caused by any condition that gives rise to chronic hypocalcemia, which, in turn, leads to compensatory overactivity of the parathyroid glands. Renal failure is a common cause of secondary hyperparathyroidism, which may also be caused by several other diseases, including inadequate dietary intake of calcium, steatorrhea, and vitamin D deficiency. The mechanisms by which chronic renal failure induces secondary hyperparathyroidism are complex and not fully understood. Chronic renal insufficiency is associated with decreased phosphate excretion, which, in turn, results in hyperphosphatemia. The elevated serum phosphate levels directly depress serum calcium levels and thereby stimulate parathyroid gland activity. In addition, loss of kidney function reduces the availability of α_1 -hydroxylase, which is necessary for the synthesis of the active form of vitamin D, leading in turn to reduced intestinal absorption of calcium (Chapter 9). Because vitamin D has suppressive effects on parathyroid growth and PTH secretion, its deficiency compounds the hyperparathyroidism in renal failure.

MORPHOLOGY

The parathyroid glands in secondary hyperparathyroidism are hypercellular. As in primary hyperparathyroidism, the degree of glandular enlargement may be asymmetric. Microscopically, the hyperplastic glands contain an increased number of chief cells or water-clear cells in a diffuse or multinodular distribution. Fat cells are decreased in number. **Metastatic calcification** may be seen in many tissues, including the lungs, heart, stomach, and blood vessels. Histologically, distinction between hyperplasia and adenoma may not be possible in light of overlapping features.

Clinical Features

The clinical features of secondary hyperparathyroidism are usually dominated by the inciting chronic renal failure. Control of the hyperparathyroidism allows the bony changes (referred to as *renal osteodystrophy*) to regress significantly. The vascular calcification associated with secondary hyperparathyroidism may occasionally result in ischemic damage to skin and other organs, a process referred to as *calciophylaxis*. Patients with secondary hyperparathyroidism often respond to dietary vitamin D supplementation as well as phosphate binders, which decrease the prevailing hyperphosphatemia.

In a minority of patients with secondary hyperparathyroidism, parathyroid activity becomes autonomous and excessive, with resultant hypercalcemia, a process termed *tertiary hyperparathyroidism*. Parathyroidectomy may be necessary to control the hyperparathyroidism in such patients.

KEY CONCEPTS

Hyperparathyroidism

- Primary hyperparathyroidism is the most common cause of asymptomatic hypercalcemia.
- In most cases, primary hyperparathyroidism is caused by a sporadic parathyroid adenoma and, less commonly, by parathyroid hyperplasia.
- Parathyroid adenomas are solitary, while hyperplasia is typically a multiglandular process.
- Skeletal manifestations of hyperparathyroidism include bone resorption, osteitis fibrosa cystica, and brown tumors. Renal changes include nephrolithiasis (stones) and nephrocalcinosis.
- The clinical manifestations of hyperparathyroidism have been classically summarized as “painful bones, renal stones, abdominal groans, and psychic moans.”

- Secondary hyperparathyroidism most often is caused by renal failure or vitamin D deficiency, which lowers serum calcium levels, resulting in reactive hyperplasia of parathyroid glands.
- Malignancy is the most important cause of symptomatic hypercalcemia, which results from osteolytic metastases or release of PTH-related protein from nonparathyroid tumors.

HYPOPARATHYROIDISM

Hypoparathyroidism is far less common than hyperparathyroidism. Acquired hypoparathyroidism is almost always an inadvertent consequence of surgery, but there are also several genetic causes.

- *Surgically induced hypoparathyroidism* occurs with inadvertent removal of all parathyroid glands during thyroidectomy, excision of parathyroid glands in the mistaken belief that they are lymph nodes during radical neck dissection for malignant disease, or removal of too large a proportion of parathyroid tissue in the treatment of primary hyperparathyroidism.
- *Autoimmune hypoparathyroidism* is often associated with chronic mucocutaneous candidiasis and primary adrenal insufficiency; this syndrome is known as autoimmune polyglandular syndrome (APS-1) and is caused by mutations in the *autoimmune regulator (AIRE)* gene. The syndrome typically presents in childhood with the onset of candidiasis, followed several years later by hypoparathyroidism and then adrenal insufficiency during adolescence. APS-1 is discussed later in this chapter.
- *Autosomal dominant hypoparathyroidism* is caused by gain-of-function mutations in the *calcium-sensing receptor (CASR)* gene. Inappropriate CASR activity due to heightened calcium sensing suppresses PTH, resulting in hypocalcemia and hypercalciuria. Recall that heterozygous loss-of-function CASR mutations are a rare cause of familial hypocalciuric hypercalcemia, whereas homozygous inactivating mutations cause severe neonatal primary hyperparathyroidism.
- *Familial isolated hypoparathyroidism (FIH)* is a rare condition with autosomal dominant or autosomal recessive patterns of inheritance. Autosomal dominant FIH is caused by a mutation in the gene encoding PTH that impairs PTH processing to the mature active hormone. Autosomal recessive FIH is caused by loss-of-function mutations in the transcription factor gene *GCM2*, which is essential for development of the parathyroid.
- *Congenital absence of parathyroid glands* can occur in conjunction with other malformations, such as thymic aplasia and cardiovascular defects, or as a component of the 22q11 deletion syndrome (Chapter 5). As discussed in Chapter 6, when thymic defects are present, the condition is called *DiGeorge syndrome*.

Clinical Features

The major manifestations of hypoparathyroidism are related to the severity and chronicity of the hypocalcemia.

- The hallmark of hypocalcemia is *tetany*, which is characterized by neuromuscular irritability resulting from decreased blood calcium levels. The symptoms range from circumoral numbness or paresthesias (tingling) of the distal extremities and carpopedal spasm, to life-threatening laryngospasm and generalized seizures. The classic findings on physical examination are *Chvostek sign* and *Trousseau sign*. Chvostek sign is elicited in subclinical disease by tapping along the course of the facial nerve, which induces contractions of the muscles of the eye, mouth, or nose. Trousseau sign refers to carpal spasms produced by occlusion of the circulation to the distal arm with a blood pressure cuff for several minutes.

- *Mental status changes* include emotional instability, anxiety and depression, confusion, hallucinations, and frank psychosis.
- *Intracranial manifestations* include calcifications of the basal ganglia, parkinsonian-like movement disorders, and increased intracranial pressure with resultant papilledema. The paradoxical association of hypocalcemia with calcifications may be because of an increase in phosphate levels, leading to deposition of calcium phosphate in vulnerable tissues.
- *Ocular disease* takes the form of calcification of the lens and cataract formation.
- *Cardiovascular manifestations* include a conduction defect that produces a characteristic prolongation of the QT interval in the electrocardiogram.
- *Dental abnormalities* occur when hypocalcemia is present during early development. These findings are highly characteristic of hypoparathyroidism and include dental hypoplasia, failure of eruption, defective enamel and root formation, and abraded, carious teeth.

Pseudohypoparathyroidism

Pseudohypoparathyroidism occurs because of end-organ resistance to the actions of PTH. Indeed, serum PTH levels are normal or elevated. In one form of pseudohypoparathyroidism, there is end-organ resistance to TSH and FSH/LH as well as PTH. All these hormones signal via G-protein-coupled receptors, and the disorder results from genetic defects in components of this pathway that are shared across endocrine tissues. PTH resistance is the most obvious clinical manifestation. It presents as hypocalcemia, hyperphosphatemia, and elevated circulating PTH. TSH resistance is generally mild, while LH/FSH resistance manifests as hypergonadotropic hypogonadism in females.

ENDOCRINE PANCREAS

The endocrine pancreas consists of about 1 million clusters of cells, the *islets of Langerhans*, which contain four major and two minor cell types. The four main types are β , α , δ , and PP (pancreatic polypeptide) cells. They can be differentiated by the ultrastructural characteristics of their granules and their hormone content (Fig. 24.27).

- The β cells produce insulin, which regulates glucose utilization in tissues and reduces blood glucose levels, as will be detailed in the discussion of diabetes.
- The α cells secrete glucagon, which stimulates glycogenolysis in the liver and thus increases blood sugar.
- The δ cells secrete somatostatin, which suppresses both insulin and glucagon release.
- The PP cells secrete pancreatic polypeptide, which exerts several gastrointestinal effects, such as stimulation of secretion of gastric and intestinal enzymes and inhibition of intestinal motility. These cells are present in islets and scattered throughout the exocrine pancreas.
- There are also other rare cells that secrete ghrelin, serotonin, and gastrin, and perhaps some with unknown content. *Enterochromaffin cells* synthesize serotonin and may be the source of pancreatic tumors that cause the carcinoid syndrome (Chapter 19).

The following discussion focuses on the two main disorders of islet cells: diabetes mellitus and pancreatic endocrine tumors.

DIABETES MELLITUS

Diabetes mellitus, now commonly called diabetes, is a group of metabolic disorders sharing the common feature of hyperglycemia caused by defects in insulin secretion, insulin action, or, most commonly, both. The chronic hyperglycemia and attendant metabolic

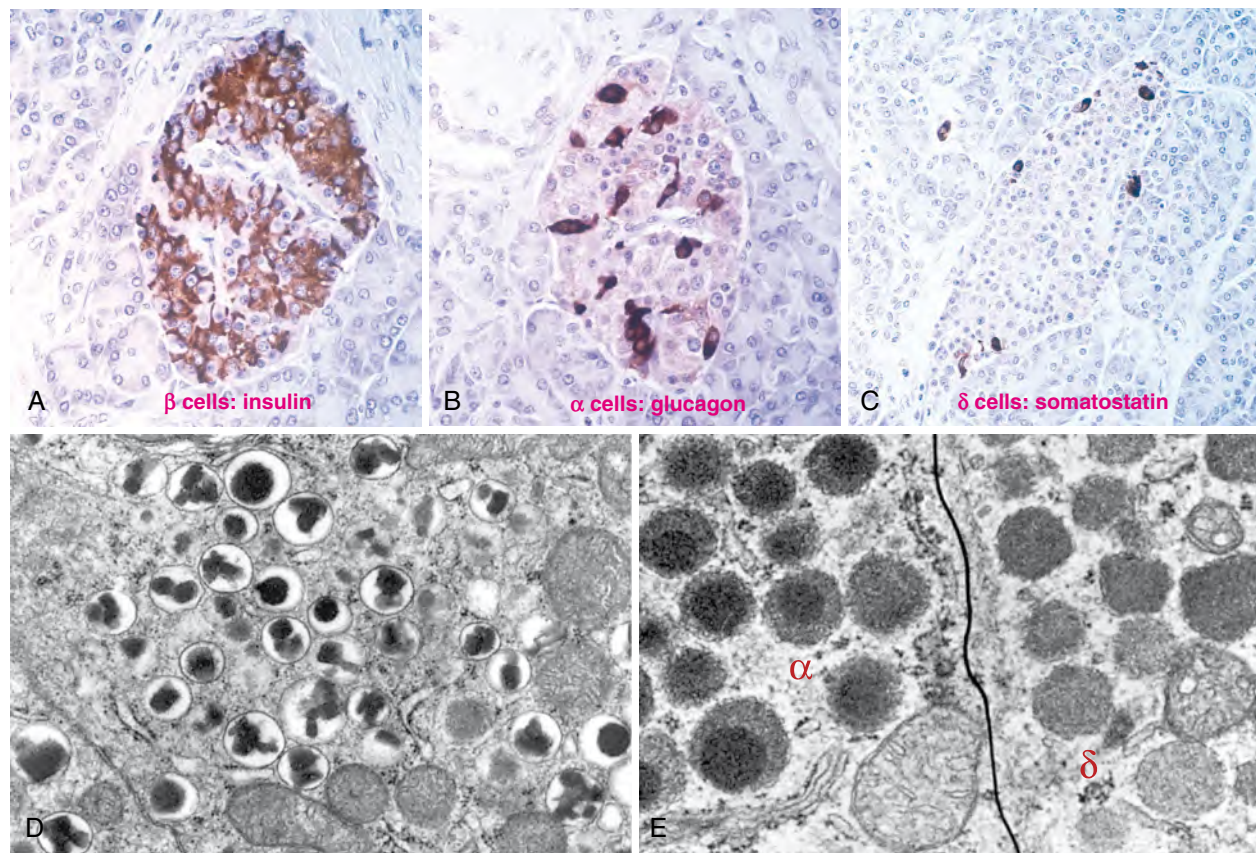


Fig. 24.27 Hormone production in pancreatic islet cells. Immunoperoxidase staining shows insulin in β cells (A), glucagon in α cells (B), and somatostatin in δ cells (C). (D) Electron micrograph of a β cell shows the characteristic membrane-bound granules, each containing a dense, often rectangular core with a distinct halo. (E) Portions of an α cell (left) and a δ cell (right) also contain granules, but with closely apported membranes. The α -cell granule shows a dense, round center. (Electron micrographs courtesy Dr. Arthur Like, University of Massachusetts Medical School, Worcester.)

deregulation may be associated with secondary damage in multiple organ systems, especially the kidneys, eyes, nerves, and blood vessels. **In the United States, diabetes is the leading cause of end-stage renal disease, adult-onset blindness, and nontraumatic lower extremity amputations.**

Diabetes and related disorders of glucose metabolism are common. According to the American Diabetes Association (ADA), diabetes affects more than 30 million children and adults, or more than 11% of the population, in the United States, of which about 1.9 million have the form of diabetes called type 1 and the remainder have type 2. Astonishingly, nearly one-quarter of these individuals are currently unaware that they have hyperglycemia. Approximately 1.4 million new cases of adult diabetes are diagnosed each year in the United States. Furthermore, a staggering 96 million adults in this country have impaired glucose tolerance or “prediabetes,” which is defined as elevated blood sugar that does not reach the criterion used for an outright diagnosis of type 2 diabetes (T2D; see later), and individuals with prediabetes are at risk for developing frank T2D. Compared to non-Hispanic people of European descent, Native Americans, African Americans, and Hispanics are about 1.5 to 2 times more likely to develop diabetes in their lifetimes. The World Health Organization estimates that as many as 422 million people have diabetes worldwide. Increasingly sedentary lifestyles and dietary habits have contributed to the simultaneous escalation of T2D and obesity, which some have called the *diabetes epidemic*. The rates of T2D in adolescents has increased dramatically in recent years.

The mortality rate from diabetes varies across countries, with diabetes representing the seventh leading cause of death in the United States in 2019, per the American Diabetes Association. The total yearly cost related to diabetes in the United States is estimated to be an astounding \$327 billion, including \$237 billion in direct medical costs and \$90 billion in indirect costs stemming from the reduced productivity of individuals with diabetes.

Diagnosis

Blood glucose is normally maintained in a very narrow range of 70 to 120 mg/dL. According to the ADA and WHO, diagnostic criteria for diabetes include the following:

1. A fasting plasma glucose ≥ 126 mg/dL
2. A random plasma glucose ≥ 200 mg/dL (in a patient with classic hyperglycemic signs, as discussed later)
3. A 2-hour plasma glucose ≥ 200 mg/dL during an oral glucose tolerance test (OGTT) with a loading dose of 75 g
4. A glycated hemoglobin (HbA1c) level $\geq 6.5\%$ (glycated hemoglobin is discussed later)

All tests, except the random blood glucose test in a patient with classic hyperglycemic signs, need to be repeated and confirmed on a separate day. If there is discordance between two assays (e.g., fasting glucose and HbA1c level), the result with the greatest degree of abnormality is considered the “readout.” Of note, many acute stresses, such as severe

infections, burns, or trauma, can lead to transient hyperglycemia due to secretion of hormones such as catecholamines and cortisol that oppose the action of insulin. The diagnosis of diabetes requires persistence of hyperglycemia following resolution of the acute illness.

Prediabetes, a state of dysglycemia that often precedes development of frank T2D, is defined by one or more of the following:

1. A fasting plasma glucose between 100 and 125 mg/dL (“impaired fasting glucose”),
2. A 2-hour plasma glucose between 140 and 199 mg/dL following a 75-g OGTT (“impaired glucose tolerance”), and/or
3. An HbA1c level between 5.7% and 6.4%.

As many as one-quarter of individuals with impaired glucose tolerance will develop overt diabetes over 5 years, with additional factors such as obesity and family history compounding the risk. In addition, individuals with prediabetes are also at significant risk for cardiovascular complications.

Classification

Although all forms of diabetes have hyperglycemia as a common feature, the underlying abnormalities involved in its development vary widely. The previous classification schemes of diabetes were based on clinical features; in contrast, the current classification reflects a greater understanding of the pathogenesis of each variant (Table 24.6). Most cases of diabetes fall into one of two broad classes:

- **Type 1 diabetes (T1D)** is an autoimmune disease characterized by pancreatic β -cell destruction and an absolute deficiency of insulin. It accounts for approximately 5% to 10% of diabetes and is the most common subtype diagnosed in patients younger than 20 years of age.
- **Type 2 diabetes (T2D)** is caused by a combination of peripheral resistance to insulin action and a secretory response by pancreatic β cells that is inadequate to overcome insulin resistance (“relative insulin deficiency”). Approximately 90% to 95% of diabetes patients have T2D, and most such individuals are overweight. Although classically considered “adult-onset,” the prevalence of T2D in children and adolescents has been increasing at an alarming pace due to the increasing rates of obesity in children and young adults.

The important similarities and differences between T1D and T2D are summarized in Table 24.7. A variety of monogenic and secondary causes are responsible for the remaining cases (discussed later). Before discussing the pathogenesis of the two major types, we will first briefly review normal insulin secretion and the mechanism of insulin action, since these are critical to understanding the pathogenesis of diabetes.

Glucose Homeostasis

Glucose homeostasis is tightly regulated by three interrelated processes: glucose production in the liver; glucose uptake and utilization by peripheral tissues, chiefly skeletal muscle; and actions of insulin and counterregulatory hormones, including glucagon, on glucose uptake and metabolism. Insulin and glucagon have opposing effects on glucose homeostasis. During fasting states, low insulin and high glucagon levels facilitate hepatic gluconeogenesis and glycogenolysis (glycogen breakdown) while decreasing glycogen synthesis, thereby preventing hypoglycemia. Thus, fasting plasma glucose levels are determined primarily by hepatic glucose output. Following a meal, insulin levels rise, and glucagon levels fall in response to the large glucose load. Insulin promotes glucose uptake and utilization in tissues (discussed later). The skeletal muscle is the major insulin-responsive site for postprandial glucose utilization, and it is critical for preventing hyperglycemia and maintaining glucose homeostasis. Although less dependent on insulin, brain and adipose tissues also extract a significant amount of glucose from the circulation.

TABLE 24.6 Classification of Diabetes

Type 1 Diabetes (β -Cell Destruction, Usually Leading to Absolute Insulin Deficiency)

Immune-mediated
Idiopathic (autoantibody-negative)

Type 2 Diabetes (Combination of Insulin Resistance and β -Cell Dysfunction)

Other Types

Genetic Defects of β -Cell Function

Maturity-onset diabetes of the young (MODY) caused by mutations in:
Hepatocyte nuclear factor 4 α (*HNF4A*) (MODY1)
Glucokinase (*GCK*) (MODY2)
Hepatocyte nuclear factor 1 α (*HNF1A*) (MODY3)
Pancreatic and duodenal homeobox 1 (*PDX1*) (MODY4)
Hepatocyte nuclear factor 1 β (*HNF1B*) (MODY5)
Neurogenic differentiation factor 1 (*NEUROD1*) (MODY6)
Neonatal diabetes (activating mutations in *KCNJ11* and *ABCC8*, encoding Kir6.2 and SUR1, respectively)
Maternally inherited diabetes and deafness (MIDD) due to mitochondrial DNA mutations (m.3243A→G)
Defects in proinsulin conversion
Insulin gene mutations

Genetic Defects in Insulin Action

Type A insulin resistance
Lipodystrophic diabetes

Exocrine Pancreatic Defects (“Pancreatogenic” or Type 3C Diabetes)

Chronic pancreatitis
Pancreatectomy/trauma
Pancreatic cancer
Cystic fibrosis
Hemochromatosis
Fibrocalculous pancreatopathy

Endocrinopathies

Acromegaly
Cushing syndrome
Hyperthyroidism
Pheochromocytoma
Glucagonoma

Infections

Cytomegalovirus
Coxsackie B virus
Congenital rubella

Drugs

Glucocorticoids
Thyroid hormone
Interferon- α
Protease inhibitors
 β -adrenergic agonists
Thiazides
Nicotinic acid
Phenytoin (Dilantin)
Vacor

Genetic Syndromes Associated With Diabetes

Down syndrome
Klinefelter syndrome
Turner syndrome
Prader-Willi syndrome

Gestational Diabetes Mellitus

Modified from American Diabetes Association. Diagnosis and classification of diabetes mellitus. *Diabetes Care*. 2014;37(Suppl 1):S81-S90.

TABLE 24.7 Comparative Features of Type 1 and Type 2 Diabetes

Type 1 Diabetes	Type 2 Diabetes
Clinical	
Onset: usually childhood and adolescence	Onset: usually adult; increasing incidence in childhood and adolescence
Normal weight or weight loss preceding diagnosis	Majority are obese (80%)
Progressive decrease in insulin levels	Increased blood insulin (early); normal or moderate decrease in insulin (late)
Circulating islet autoantibodies (anti-GAD65, anti-IA2, anti-Znt8)	No islet autoantibodies
Diabetic ketoacidosis in absence of insulin therapy	Nonketotic hyperosmolar coma more common
Genetics	
Major linkage to MHC class II genes; also linked to polymorphisms in <i>CTLA4</i> and <i>PTPN22</i> , and insulin gene VNTRs	No HLA linkage; linkage to candidate diabetogenic and obesity-related genes (e.g., <i>TCF7L2</i> , <i>PPARG</i> , <i>FTO</i>)
Pathogenesis	
Dysfunction in T-cell selection and regulation leading to break down in self-tolerance to islet autoantigens	Insulin resistance in peripheral tissues, failure of compensation by β cells
Pathology	
Insulinitis (inflammatory infiltrate of T cells and macrophages)	No insulinitis; amyloid deposition in islets
β -cell depletion, islet atrophy	Mild β -cell depletion

HLA, Human leukocyte antigen; MHC, major histocompatibility complex; VNTRs, variable number of tandem repeats.

Regulation of Insulin Release

Insulin is produced in the β cells of the pancreatic islets (see Fig. 24.27) as a precursor protein and is proteolytically cleaved in the Golgi complex to generate the mature hormone and a peptide byproduct, *C-peptide*. Both insulin and *C-peptide* are then stored in secretory granules and secreted in equimolar quantities after physiologic stimulation; thus, *C-peptide* levels serve as a surrogate for β -cell function, decreasing with loss of β -cell mass in T1D and increasing with insulin resistance–associated hyperinsulinemia.

The most important stimulus for insulin synthesis and release is glucose. An increase in blood glucose levels results in glucose uptake into pancreatic β cells, facilitated by an insulin-independent glucose transporter, GLUT2 (Fig. 24.28). Metabolism of glucose generates ATP, which leads to the influx of Ca^{2+} through plasma membrane calcium channels. The resultant increase in intracellular Ca^{2+} stimulates secretion of insulin, presumably from hormone stored in β -cell granules. This is the phase of immediate insulin release, sometimes called the first phase of β -cell insulin secretion. If the secretory stimulus persists, a delayed and protracted response follows that involves active synthesis of insulin, the second phase.

Oral intake of food leads to secretion of multiple hormones that play a role in glucose homeostasis and satiety. Of these, the most important class of hormones responsible for promoting insulin secretion from pancreatic β cells following feeding is the *incretins*, which act by binding G-protein–coupled receptors that are expressed on pancreatic β cells. The two most important incretins are

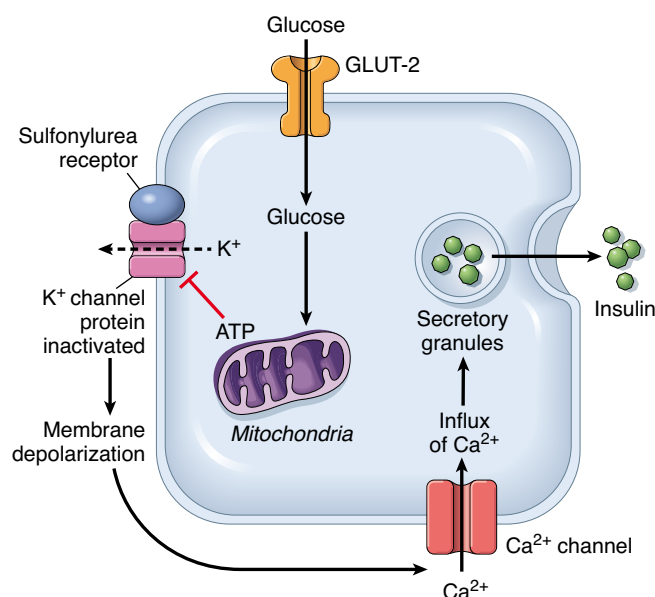


Fig. 24.28 Insulin synthesis and secretion. The influx of glucose into β cells through the GLUT-2 receptors initiates a cascade of signaling events that culminates in calcium (Ca^{2+})-induced release of stored insulin (see text for details). ATP, Adenosine triphosphate; K^+ , potassium ions.

glucose-dependent insulinotropic polypeptide (GIP) and glucagon-like peptide-1 (GLP-1), both secreted by cells in the intestines following oral food intake. The elevation in GIP and GLP-1 levels is known as the “incretin effect,” and the production of these factors in the intestines explains why an oral glucose load leads to greater insulin secretion than equivalent intravenous administration. In addition to increasing insulin secretion from β cells, these hormones reduce glucagon secretion from pancreatic α cells and delay gastric emptying, which promotes satiety. Once released, circulating GIP and GLP-1 are degraded in the circulation by a class of enzymes known as dipeptidyl peptidases (DPPs), especially DPP-4. The “incretin effect” is significantly blunted in patients with T2D, and efforts to restore incretin function can improve glycemic control and promote weight loss (through restoration of satiety). These insights have led to the recent development of two classes of drugs for treating T2D: GLP-1 receptor agonists, which are synthetic GLP-1 mimetics that bind to and activate the GLP-1 receptor on islet and extrapancreatic cells; and DPP-4 inhibitors, which enhance levels of endogenous incretins by delaying their degradation. In addition, tirzepatide is a dual GIP and GLP-1 receptor agonist that was recently approved to treat patients with diabetes and has particularly profound effects on weight loss. GLP-1 also increases energy expenditure, so the weight-loss-inducing effects are likely multifactorial, but seem to involve a significant decrease in appetite. GLP-1 receptor agonists are also now approved for the treatment of obesity and appear to have a number of beneficial effects on renal and cardiac function, albeit through uncertain mechanisms.

Insulin Action and Insulin-Signaling Pathways

Insulin is the most potent anabolic hormone known, with multiple synthetic and growth-promoting effects (Fig. 24.29). The principal metabolic function of insulin is to increase the rate of glucose transport into certain cells in the body, thus providing a major source of energy and metabolic intermediates derived from glucose that are used in the biosynthesis of cellular building blocks such as lipids, nucleotides, and amino acids. The most important targets of insulin action are striated muscle

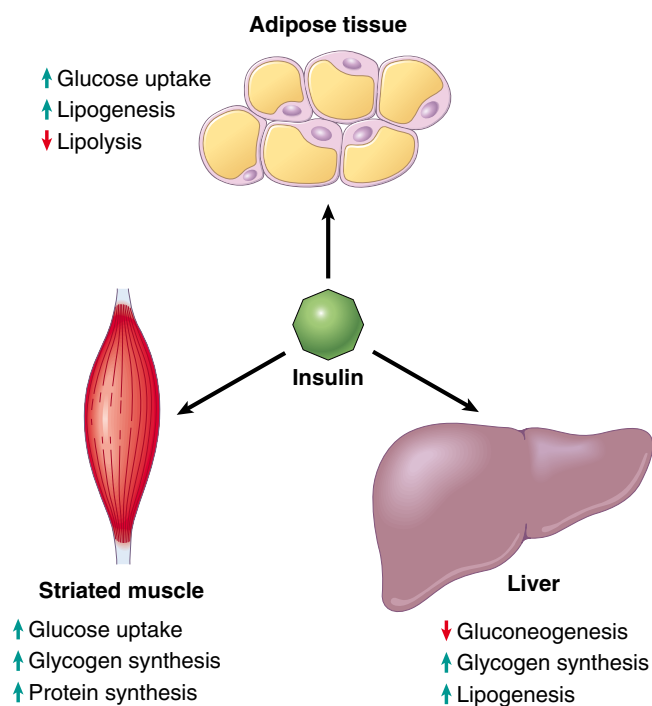


Fig. 24.29 Metabolic actions of insulin in striated muscle, adipose tissue, and liver.

cells (including cardiomyocytes) and, to a lesser extent, adipocytes, which together normally represent about two-thirds of the body's weight. Adipose tissue is composed of white adipose tissue, which stores energy and accumulates in obesity, and brown adipose tissue, which utilizes energy to promote heat generation. Beige adipose tissue is composed of brown-like adipocytes that are inducible in white adipose tissue depots. The importance of brown and beige adipose tissue in humans is an active area of investigation.

In muscle cells, glucose is either stored as glycogen or oxidized to generate ATP. In adipose tissue, glucose is primarily used as a substrate for synthesis of lipids, which are stored as triglycerides. Besides promoting lipid synthesis, insulin also inhibits triglyceride hydrolysis and lipid release by adipocytes. Similarly, insulin promotes amino acid uptake and protein synthesis, while inhibiting protein degradation. Thus, the anabolic effects of insulin are attributable to increased synthesis and reduced degradation of glycogen, lipids, and proteins. In addition, insulin has several mitogenic activities, including initiation of DNA synthesis in certain cells and stimulation of their growth and differentiation.

The metabolic effects of insulin are exerted through its binding to the insulin receptor, which, in turn, sets into motion a series of signaling events through an array of mediators, the more pertinent of which are summarized in Fig. 24.30. The insulin receptor is a tetrameric protein composed of two α -subunits and two β -subunits. The β -subunit cytosolic domain possesses tyrosine kinase activity. Insulin binding to the α -subunit extracellular domain activates the β -subunit tyrosine kinase, which autophosphorylates itself and phosphorylates several intracellular docking or bridging proteins, including so-called insulin receptor substrate (IRS) proteins. These molecules in turn activate downstream factors such as PI3-kinase and AKT, a serine/threonine kinase that serves as a central signaling hub that mediates many insulin-dependent activities, including increased glucose uptake, reduced glucose synthesis, and increased glycogen and protein synthesis.

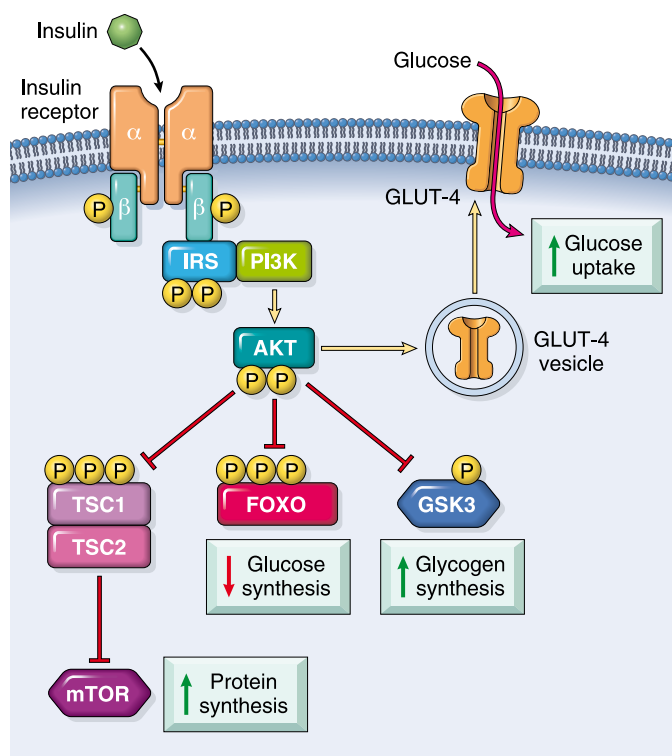


Fig. 24.30 Insulin action on a target cell. Insulin binding to the insulin receptor initiates a cascade of phosphorylation events that result in activation of PI-3-kinase/AKT signaling. AKT is a serine-threonine kinase that mediates its effector functions via phosphorylation-dependent events. For example, AKT phosphorylates and inhibits the function of the tuberous sclerosis complex (TSC) proteins, leading to activation of the downstream mammalian TOR (mTOR) complex, which enhances protein synthesis. AKT also inhibits the function of Forkhead box O (FOXO) protein, which, in turn, reduces glucose synthesis, while inhibition of glycogen synthase kinase 3 (GSK3) enhances glycogen production. Finally, AKT enhances glucose uptake by translocation of GLUT-4 vesicles to the cell membrane. IRS, Insulin receptor substrate; PI3K, phosphoinositide 3-kinase. (Modified from Brendan Manning, Harvard TH Chan School of Public Health.)

Pathogenesis of Type 1 Diabetes

T1D is an autoimmune disease in which islet destruction is caused primarily by immune effector cells reacting against endogenous β -cell antigens. T1D most commonly develops in childhood, becomes manifest at puberty, and progresses with age. Because the disease can develop at any age, including late adulthood, the old moniker “juvenile-onset diabetes” is no longer used. Similarly, “insulin-dependent diabetes mellitus” has been removed from the current classification of diabetes because many other forms of diabetes eventually require treatment with insulin. Nevertheless, most patients with T1D require insulin for survival; without insulin, they may develop serious metabolic complications such as ketoacidosis and coma. Latent autoimmune diabetes in adults (LADA) is a slowly progressive form of T1D that occurs in adults and can be distinguished from T2D by the presence of one or more diabetes autoantibodies; most patients eventually require insulin but may not at the time of diagnosis.

As with most autoimmune diseases, the pathogenesis of T1D involves an interplay of genetic and environmental factors.

Genetic Susceptibility

Epidemiologic studies, such as those demonstrating higher concordance rates in monozygotic versus dizygotic twins, have convincingly

established a genetic basis for T1D. More recently, GWASs have identified multiple genetic susceptibility loci for T1D, as well as for T2D (see later). **Of these, the most important locus is the HLA gene cluster, which according to some estimates contributes as much as 50% of the genetic susceptibility for T1D.** Between 90% and 95% of individuals of European descent with this disease have an *HLA-DR3* or *HLA-DR4* haplotype, in contrast to about 40% of unaffected subjects; moreover, 40% to 50% of patients with T1D are *DR3/DR4* compound heterozygotes, in contrast to 5% of normal subjects. Individuals who have either *DR3* or *DR4* concurrently with *DQ8* haplotype demonstrate one of the highest inherited risks for T1D in sibling studies. Predictably, the polymorphisms in the HLA molecules that are associated with risk are located in or adjacent to the peptide-binding pockets, consistent with the notion that disease-associated alleles code for HLA molecules that have the capacity to display particular antigens. However, as discussed in Chapter 6, it is still not known how particular HLA alleles contribute to the pathogenesis of T1D (and other autoimmune diseases).

Several *non-HLA* genes also confer susceptibility to T1D. The first disease-associated non-MHC gene variant to be identified consisted of a variable number of tandem repeats in the promoter region of the insulin gene. The mechanism underlying this association is unknown. It is possible that these polymorphisms influence insulin expression by thymic antigen-presenting cells, thus affecting the negative selection of insulin-reactive T cells (Chapter 6). The association between polymorphisms in *CTLA4*, *PTPN22* and autoimmune thyroiditis was mentioned earlier; not surprisingly, these genes have also been linked with susceptibility to T1D. More recently, patients have been identified with *STAT3* mutations that lead to T1D and other autoimmune disorders. The relationship of T1D to altered T-cell selection and regulation is also underscored by the striking prevalence of this disease in individuals with rare germline defects in genes that code for immune regulators, such as *AIRE*, mutations of which cause APS-1 (discussed later).

Environmental Factors

As in other autoimmune diseases, genetic susceptibility contributes to only a part of diabetes risk, and the concordance rate in monozygotic

twins is only about 50%, so environmental factors must play a role. The nature of these environmental influences remains an enigma. Although antecedent viral infections have been suggested as triggers, neither the type of virus nor how it promotes islet-specific autoimmunity is established. Some studies suggest that viruses might share epitopes with islet antigens, and the immune response to the virus results in cross-reactivity and destruction of islet tissues, a phenomenon known as *molecular mimicry*. On the other hand, certain infections are also thought to be protective against T1D.

Mechanisms of β -Cell Destruction

While the clinical onset of T1D is often abrupt, there is a lengthy lag period between initiation of the autoimmune process and the appearance of symptomatic disease, during which there is progressive loss of insulin reserves. Three distinct stages of T1D are now recognized (Fig. 24.31). In stage 1 (*normoglycemia, presymptomatic T1D*), individuals have developed two or more islet autoantibodies but are still normoglycemic. In stage 2 (*dysglycemia, presymptomatic T1D*), there is increasingly severe impairment of glucose tolerance due to progressive loss of β -cell mass, but symptoms are absent. Nonetheless, the 5-year risk of developing symptomatic T1D increases from less than 50% in stage 1 to 75% in stage 2. Finally, in stage 3 (*dysglycemia, symptomatic T1D*), classic manifestations of the disease (polyuria, polydipsia, polyphagia, ketoacidosis; see later) appear, typically after more than 90% of the β cells have been destroyed.

The fundamental immune abnormality in T1D is a failure of self-tolerance in T cells specific for islet antigens. This failure of tolerance may be a result of some combination of defective clonal deletion of self-reactive T cells in the thymus, as well as defects in the functions of regulatory T cells or abnormal resistance of effector T cells to suppression by regulatory cells. Thus, autoreactive T cells not only survive, but are poised to respond to self-antigens. The initial activation of these cells is thought to occur in the peripancreatic lymph nodes, perhaps in response to antigens that are released from damaged islets. The activated T cells then travel to the pancreas, where they cause β -cell injury. Multiple T-cell populations have been implicated in

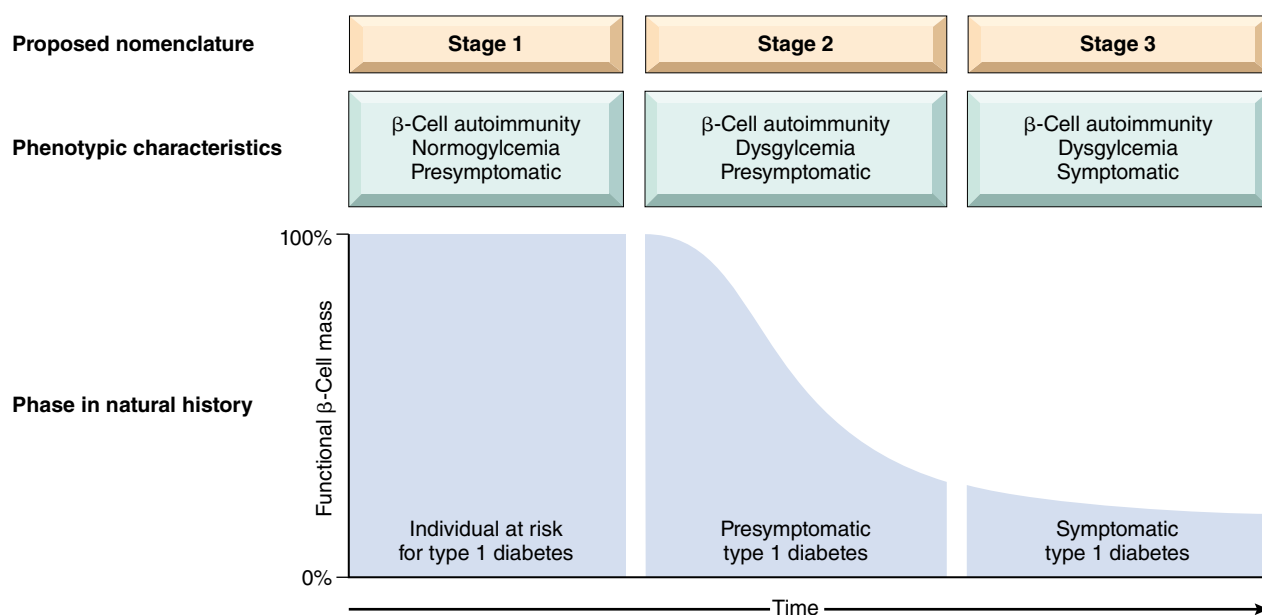


Fig. 24.31 The three stages of type 1 diabetes. (Modified from Insel RA, Dunne JL, Atkinson MA, et al. Staging presymptomatic type 1 diabetes: a scientific statement of JDRF, the Endocrine Society, and the American Diabetes Association. *Diabetes Care*. 2015;38(10):1964-1974.)

this damage, including Th1 cells (which may secrete cytokines, including IFN- γ and TNF, that injure β cells), and CD8+ CTLs (which kill β cells directly). The islet autoantigens that are the targets of immune attack may include insulin, the β -cell enzyme glutamic acid decarboxylase (GAD), and others. Consistent with the idea that failure of self-tolerance is fundamental to the pathogenesis of T1D, cancer patients treated with immune checkpoint blockade therapy, which disrupts tolerance mechanisms, sometimes develop the disease.

A role for antibodies in T1D is suspected because autoantibodies against islet antigens are found in most patients with T1D, including at the presymptomatic stages of disease, as described earlier. However, it is not clear if the autoantibodies cause injury or are merely a consequence of islet injury.

Pathogenesis of Type 2 Diabetes

T2D is a complex disease that involves the interplay of genetic and environmental factors and a proinflammatory state. Unlike T1D, there is no evidence of an autoimmune basis.

Genetic Factors

Genetic susceptibility contributes to the pathogenesis, as evidenced by the disease concordance rate of greater than 90% in monozygotic twins, a rate even higher than in T1D. Furthermore, first-degree relatives have a 5- to 10-fold higher risk of developing T2D than those without a family history, when matched for age and weight. GWASs performed over the past decade have identified at least 30 loci that individually confer a minimal to modest increase in the lifetime risk for T2D. Many of these genes are involved in adipose tissue function (through effects on bodily fat distribution [visceral vs. subcutaneous]), islet β -cell function, and obesity. It is believed that together, these genetic polymorphisms conspire to provide the genetic basis for T2D risk.

Environmental Factors

The most important acquired risk factor for T2D is obesity, particularly central or visceral obesity. Greater than 80% of individuals with T2D are obese, and the incidence of diabetes worldwide has risen in proportion to obesity. Obesity contributes to the cardinal metabolic abnormalities of diabetes (see later) and to insulin resistance early in disease. In fact, even modest weight loss through dietary modifications can reduce insulin resistance and improve glucose tolerance. A sedentary lifestyle (typified by lack of exercise) is another risk factor for diabetes, independent of obesity. Weight loss and exercise usually increase insulin sensitivity additively and are often first-line interventions in patients with milder T2D. The combination of obesity, hyperglycemia, increased serum cholesterol and triglycerides, and hypertension is called *metabolic syndrome*.

Sleep disorders (such as obstructive sleep apnea) and circadian disruption are additional risk factors for T2D. Circadian disruption is defined as misalignment between the endogenous circadian rhythm and the cycle or rhythm created by individual behaviors. Those at risk for circadian disruption include shift workers and those with sleep disorders or other conditions that restrict nighttime sleep and daytime wakefulness. Studies have shown that circadian disruption impairs glucose homeostasis by affecting both insulin secretion and insulin action. In addition, GWAS have shown an association between circadian-controlled genes and T2D.

Metabolic Defects in Type 2 Diabetes

The development of T2D involves two key abnormalities:

- **Insulin resistance:** Decreased response of peripheral tissues, especially skeletal muscle, adipose tissue, and liver, to insulin

- **β -cell dysfunction:** Inadequate insulin secretion in the face of insulin resistance and hyperglycemia

Insulin resistance generally predates the development of hyperglycemia and is usually accompanied by compensatory β -cell hyperfunction and hyperinsulinemia in the early stages of the evolution of T2D (Fig. 24.32). Over time, the inability of β cells to adapt to increasing secretory needs for maintaining a euglycemic state results in chronic hyperglycemia and the resulting long-standing complications of diabetes.

Insulin Resistance. Insulin resistance is the failure of target tissues to respond normally to insulin. The liver, skeletal muscle, and adipose tissue are the major tissues where insulin resistance gives rise to abnormal glucose tolerance. Insulin resistance results in:

- Failure to inhibit endogenous glucose production (gluconeogenesis) in the liver, which contributes to high fasting blood glucose levels
- Failure of glucose uptake and glycogen synthesis to occur in skeletal muscle following a meal, which contributes to high postprandial blood glucose level
- Failure to inhibit activation of “hormone-sensitive” lipases in adipose tissue, leading to excess triglyceride breakdown in adipocytes and high levels of circulating free fatty acids (FFAs).

A variety of functional defects in the insulin-signaling pathway underlie insulin resistance. For example, reduced tyrosine phosphorylation of the insulin receptor and IRS proteins is observed in peripheral tissues, which compromises insulin signaling and reduces the level of the glucose transporter GLUT-4 on the cell surface (see Fig. 24.30). In fact, one of the mechanisms by which exercise improves insulin sensitivity is by increasing the translocation of GLUT-4 to the plasma membrane of skeletal muscle cells.

Obesity. Multiple factors contribute to insulin resistance, of which obesity is probably the most important. The risk for diabetes rises as the body mass index (BMI) increases. BMI is an imperfect measurement for obesity, and not only the absolute amount of fat but also its distribution determines insulin sensitivity, as central obesity (abdominal fat) is more likely to be linked with insulin resistance than is peripheral (subcutaneous) obesity. Individuals, particularly from Asia and the Middle East, who develop diabetes without overt obesity have primarily visceral adiposity, and it is this increase in visceral fat that appears to engender T2D risk for them. By contrast, individuals who develop primarily subcutaneous adiposity may be relatively protected from T2D. Studies of these so-called “metabolically healthy” obese individuals is an area of active research.

Obesity can adversely impact insulin sensitivity in numerous ways (see Fig. 24.32):

- **Free fatty acids (FFAs).** Cross-sectional studies have demonstrated an inverse correlation between fasting plasma FFAs and insulin sensitivity. Central adipose tissue is more “lipolytic” than peripheral sites, which might explain the particularly deleterious consequences of this pattern of fat distribution. Excess FFAs overwhelm the intracellular fatty acid oxidation pathways, leading to the accumulation of cytoplasmic intermediates like diacylglycerol (DAG), phospholipids, and sphingolipids, including ceramides. These “toxic” lipid metabolites can attenuate signaling through the insulin receptor and activate inflammatory pathways in the islets, which further promote β -cell abnormalities. In liver cells, insulin normally inhibits gluconeogenesis by blocking the activity of phosphoenolpyruvate carboxykinase, the first enzymatic step in this process. Attenuated insulin signaling allows phosphoenolpyruvate carboxykinase to “ramp up” gluconeogenesis. Excess FFAs also compete with

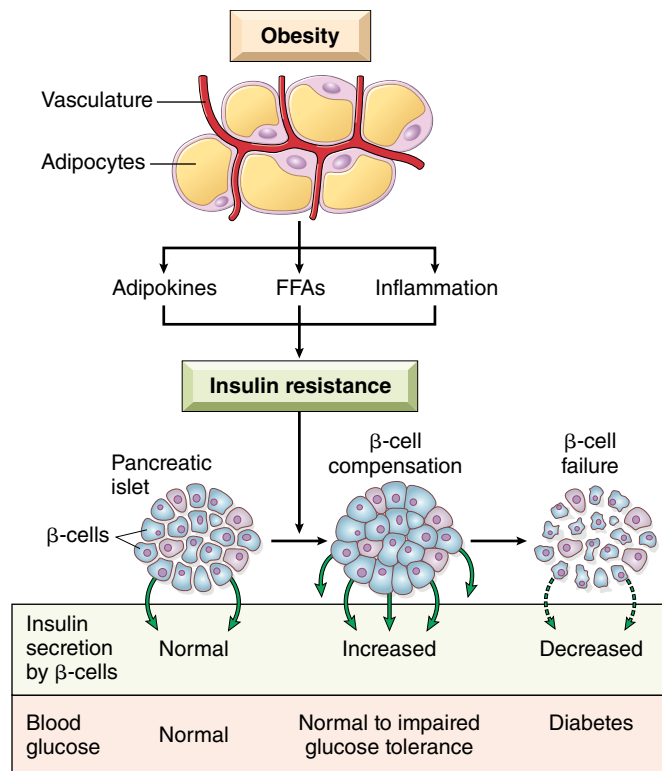


Fig. 24.32 Development of type 2 diabetes. Insulin resistance associated with obesity and β-cell failure result together in diabetes. FFAs, Free fatty acids. (Modified from Kasuga M. Insulin resistance and pancreatic β-cell failure. *J Clin Invest.* 2006;116(7):1756-1760.)

glucose as substrates for oxidation, further exacerbating the reduced glucose utilization.

- **Adipokines.** Adipose tissue is not merely a storage depot for fat but is also an endocrine organ that releases hormones in response to changes in metabolism (Chapter 9). A variety of proteins secreted into the circulation by adipose tissue have been identified that are collectively termed *adipokines* (or adipose cytokines). Some of these promote hyperglycemia, while others (such as adiponectin) decrease blood glucose, in part by increasing insulin sensitivity in peripheral tissues. Adiponectin levels are reduced in obesity, thus contributing to insulin resistance.
- **Inflammation.** Unlike T1D, T2D is not mediated by an autoimmune process. Rather, over the past several years, inflammation has emerged as an important factor in the pathogenesis of T2D. Proinflammatory cytokines secreted in response to excess nutrients such as FFAs and glucose create an inflammatory milieu that results in both insulin resistance and β-cell dysfunction. Excess FFAs within macrophages and β cells can activate the inflammasome, a multiprotein cytoplasmic complex that leads to secretion of the cytokine interleukin IL-1β (Chapter 3). IL-1β, in turn, mediates the secretion of additional proinflammatory cytokines from macrophages, islet cells, and other cells. IL-1 and other cytokines act on the major sites of insulin action to promote insulin resistance. Thus, excess FFAs can impede insulin signaling directly within peripheral tissues, as well as indirectly through the release of proinflammatory cytokines.
- **Liver steatosis.** High circulating levels of FFAs may result in the accumulation of excess fat (steatosis) in hepatocytes. This form of metabolic dysfunction-associated steatotic liver disease (MASLD), ranges in severity from hepatic steatosis without evidence of liver

injury to metabolic dysfunction-associated steatohepatitis with evidence of inflammation and hepatocyte injury with or without fibrosis (Chapter 18). MASLD is common in those with metabolic syndrome and T2D.

β-Cell Dysfunction. While insulin resistance by itself can lead to impaired glucose tolerance, **β-cell dysfunction is a requirement for the development of overt diabetes.** In contrast to the severe genetic defects in β-cell function that occur in monogenic forms of diabetes (see later), β-cell function actually increases early in the disease process in most patients with “sporadic” T2D as a compensatory measure to counter insulin resistance and maintain euglycemia. Eventually, however, β cells seemingly exhaust their capacity to adapt to the long-term demands posed by insulin resistance, and the hyperinsulinemic state gives way to a state of relative insulin deficiency, that is, insulin levels are inadequate for the level of blood glucose.

Several mechanisms have been implicated in promoting β-cell dysfunction in T2D, including:

- Excess FFAs that compromise β-cell function and attenuate insulin release (*lipotoxicity*).
- The impact of chronic hyperglycemia (*glucotoxicity*).
- An abnormal *incretin effect*, leading to reduced secretion of GIP and GLP-1, hormones that promote insulin release (see earlier).
- Amyloid deposition within islets. This is a characteristic finding in individuals with long-standing T2D, being present in more than 90% of diabetic islets examined, but it is unclear whether it is a cause or an effect of β-cell “burnout.”
- Finally, the impact of genetics cannot be discounted, as many of the polymorphisms associated with an increased lifetime risk for T2D occur in genes that control insulin secretion.

Monogenic Forms of Diabetes

Although genetically defined causes of diabetes are relatively uncommon, they have been intensively studied in the hope of gaining insights into the disease. As Table 24.6 illustrates, monogenic forms of diabetes are classified separately from types 1 and 2 diabetes. Monogenic forms of diabetes often result from either a primary defect in β-cell function or a defect in insulin receptor signaling (described later).

Genetic Defects in β-Cell Function

Approximately 1% to 2% of patients with diabetes harbor a primary defect in β-cell function that affects either β-cell mass and/or insulin production. This form of monogenic diabetes is caused by a heterogeneous group of genetic defects. The largest subgroup of patients in this category was designated as having “maturity-onset diabetes of the young” (MODY) because of its superficial resemblance to T2D and its occurrence in younger patients. MODY results from germline loss-of-function mutations in multiple genes (Table 24.6). Of these syndromes, mutations of *glucokinase* (*GCK*) are particularly important to consider, since these patients develop hyperglycemia (due to an altered “set point”) but do not appear to develop typical diabetes complications. Glucokinase is a rate-limiting step in oxidative glucose metabolism, which, in turn, is coupled to insulin secretion within islet β cells (Fig. 24.28). In contrast, the other forms of MODY lead to diabetes with its associated complications. Interestingly, insulin gene mutations have also been identified as a cause of monogenic diabetes.

Genetic Defects That Impair Tissue Response to Insulin

Rare insulin receptor mutations that affect receptor synthesis, insulin binding, or RTK activity can cause severe insulin resistance, accompanied by hyperinsulinemia and diabetes (*type A insulin resistance*). Such patients often show a velvety hyperpigmentation of the skin known as *acanthosis*

nigricans. Females with type A insulin resistance also frequently have polycystic ovaries and elevated androgen levels.

Diabetes and Pregnancy

Pregnancy can be complicated by diabetes in one of two settings: when females with preexisting diabetes become pregnant (“pregestational” or overt diabetes) or females who were previously euglycemic develop impaired glucose tolerance and diabetes for the first time during pregnancy (“gestational” diabetes). Approximately 5% to 9% of pregnancies occurring in the United States are complicated by hyperglycemia, and the incidence of both pregestational and gestational diabetes is increasing in the general population. Pregnancy is a “diabetogenic” state in which the prevailing hormonal milieu favors insulin resistance. In a previously euglycemic female who is otherwise susceptible due to concurrent genetic and environmental factors, the consequence may be gestational diabetes. In addition, females with pregestational diabetes have an increased risk of stillbirth and congenital malformations in the fetus. Poorly controlled diabetes that arises later in pregnancy, regardless of prior history, can lead to excessive birth weight in the newborn (*macrosomia*) and may have long-term sequelae for the child later in life, including an increased risk of obesity and diabetes. Gestational diabetes typically resolves following delivery; however, most affected females develop overt diabetes over the next 10 to 20 years.

Clinical Features of Diabetes

It is difficult to sketch with brevity the diverse clinical presentations of diabetes. We will discuss the most common initial presentation or mode of diagnosis for each of the two major subtypes, followed by a discussion of acute and then chronic (long-term) complications of the disease.

T1D may arise at any age. In the initial 1 or 2 years following the onset of overt T1D, exogenous insulin requirements may be minimal because of residual endogenous insulin secretion (referred to as the *honeymoon period*). Eventually, however, β -cell function declines to a tipping point, and insulin requirements increase dramatically. Although β -cell destruction is a gradual process, the transition from impaired glucose tolerance (stage 2, see earlier) to overt diabetes (stage 3) may be abrupt and is often brought on by a superimposed stress, such as infection, because of associated increase in insulin requirements.

In contrast to T1D, T2D is typically seen in obese patients older than 40 years; however, it is now being diagnosed in children and adolescents with increasing frequency due to increases in obesity and sedentary lifestyle. In some cases, medical attention is sought because of unexplained fatigue, dizziness, or blurred vision. Most frequently, however, the diagnosis of T2D is made after routine blood testing in asymptomatic persons. In fact, in light of the large number of asymptomatic individuals with undiagnosed hyperglycemia in the United States, routine blood glucose testing is recommended for everyone older than 45 years of age, and in younger individuals with obesity, family history, or the presence of the metabolic syndrome.

The Classic Triad of Type 1 Diabetes

The onset of clinical T1D is usually marked by the triad of polyuria, polydipsia, and polyphagia, and, when severe, diabetic ketoacidosis, all resulting from metabolic derangements. Because insulin is a major anabolic hormone, its deficiency results in a catabolic state that affects glucose, fat, and protein metabolism. Unopposed secretion of counter-regulatory hormones (such as glucagon) also plays a role in these metabolic derangements. The assimilation of glucose into muscle and adipose tissue is sharply diminished or abolished. Not only does storage of glycogen in liver and muscle cease, but reserves are depleted by

glycogenolysis. The resultant *hyperglycemia* leads to filtration of so much glucose in the kidney that the renal tubular threshold for reabsorption is exceeded. This leads to glycosuria, which induces an osmotic diuresis and thus *polyuria*, causing a profound loss of water and electrolytes (Fig. 24.33). The renal water loss combined with the hyperosmolarity due to increased levels of glucose in the blood depletes intracellular water, triggering the osmoreceptors of the thirst centers of the brain. Thus, intense thirst (*polydipsia*) appears. With a deficiency of insulin, the scales swing from insulin-promoted anabolism to catabolism of proteins and fats. Proteolysis follows, releasing gluconeogenic amino acids that are removed by the liver and used as building blocks for glucose. The catabolism of proteins and fats tends to induce a negative energy balance, which in turn leads to increasing appetite (*polyphagia*), thus completing the classic triad of polyuria, polydipsia, and polyphagia. Despite the increased appetite, catabolic effects prevail, resulting in weight loss and muscle weakness. The combination of polyphagia and weight loss is paradoxical and should always raise the suspicion of diabetes.

Acute Metabolic Complications of Diabetes

Diabetic ketoacidosis is a severe acute metabolic complication mainly seen in T1D. The most frequent precipitating factor is often a failure to take insulin, although other stressors such as infections, other illnesses, trauma, and certain drugs may also serve as triggers. It may also occur in T2D but generally under conditions of very severe stress such as caused by serious infections and trauma. Many of these factors are associated with the release of the catecholamine epinephrine, which blocks residual insulin action and stimulates the secretion of glucagon. The insulin deficiency coupled with glucagon excess decreases peripheral utilization of glucose while increasing gluconeogenesis, severely exacerbating hyperglycemia (the plasma glucose levels are usually in the range of 250 to 600 mg/dL). The hyperglycemia causes an osmotic diuresis and dehydration characteristic of the ketoacidotic state.

A second major effect of insulin deficiency is increased synthesis of ketone bodies. Insulin deficiency stimulates hormone-sensitive lipase, with a resultant breakdown of adipose stores and an increase in levels of FFAs. When these FFAs reach the liver, they are esterified to fatty acyl coenzyme A. Oxidation of fatty acyl coenzyme A molecules within the hepatic mitochondria produces *ketone bodies* (acetoacetic acid and β -hydroxybutyric acid). The rate at which ketone bodies are formed may exceed the rate at which they can be utilized by peripheral tissues, leading to *ketonemia* and *ketonuria*. If the urinary excretion of ketones is compromised by dehydration, the result is a systemic *metabolic ketoacidosis*. Release of ketogenic amino acids by protein catabolism aggravates the ketotic state.

The clinical manifestations of diabetic ketoacidosis include fatigue, nausea and vomiting, severe abdominal pain, a characteristic fruity odor, and deep, labored breathing (also known as *Kussmaul breathing*). Persistence of the ketotic state eventually leads to depressed consciousness and coma. Reversal of ketoacidosis requires administration of insulin, correction of metabolic acidosis, and treatment of any underlying precipitating factors, such as infection.

The lower frequency of ketoacidosis in T2D is believed to be due to higher portal vein insulin levels in these patients, which prevents unrestricted hepatic fatty acid oxidation and keeps the formation of ketone bodies in check. Instead, patients with T2D may develop a condition known as *hyperosmolar hyperglycemic state* due to severe dehydration resulting from sustained osmotic diuresis (particularly in patients who do not drink enough water to compensate for urinary losses from chronic hyperglycemia). Typically, this occurs in an older patient who has diabetes and is disabled by a stroke or an infection and thus unable

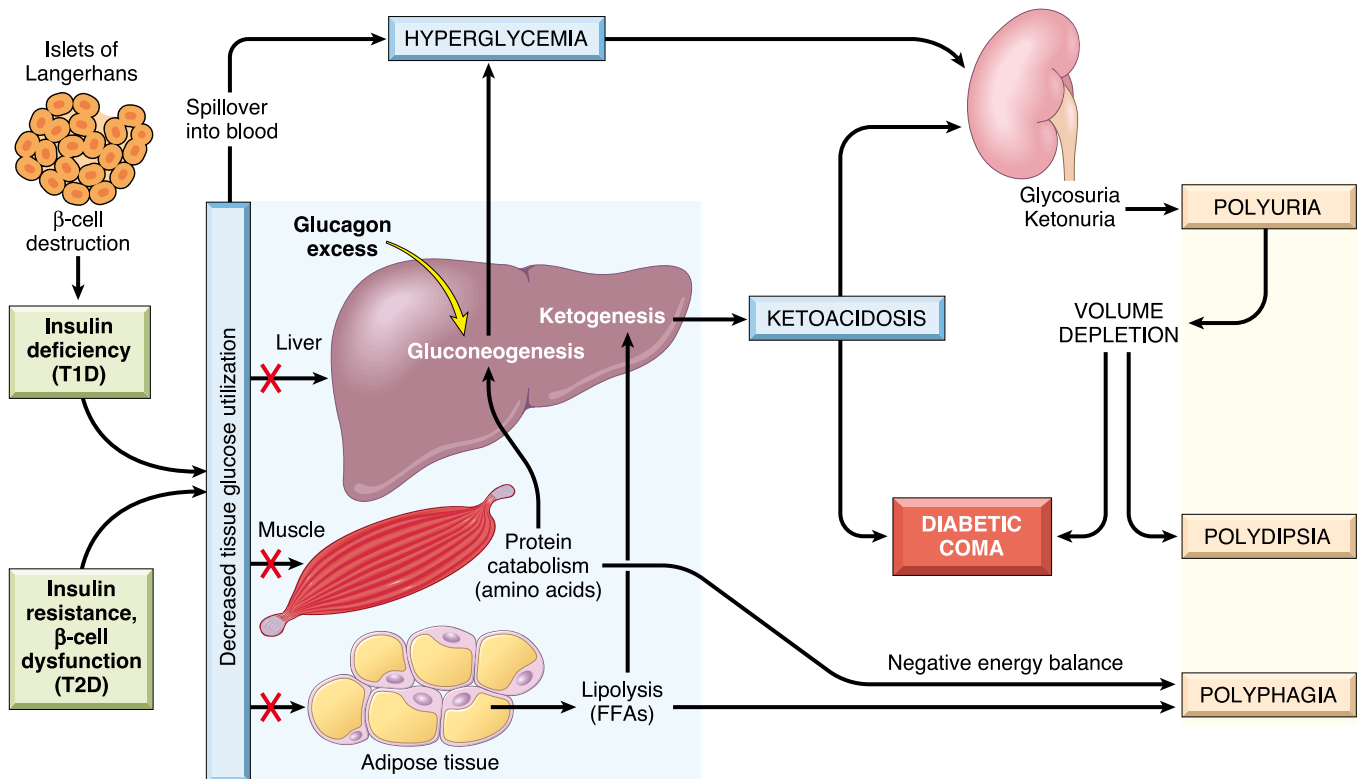


Fig. 24.33 Sequence of metabolic derangements underlying the clinical manifestations of diabetes. An absolute insulin deficiency leads to a catabolic state, culminating in ketoacidosis and severe volume depletion. These derangements bring about sufficient central nervous system compromise to cause coma and eventual death if left untreated. FFAs, Free fatty acids.

to maintain adequate water intake. Furthermore, the absence of ketoacidosis and its symptoms (nausea, vomiting, Kussmaul breathing) delays seeking of medical attention until severe dehydration and impairment of mental status occur. The degree of hyperglycemia is usually even more severe than in diabetic ketoacidosis, often in the range of 600 to 1200 mg/dL.

Hypoglycemia is a common and serious acute metabolic complication for those patients treated with insulin (in either T1D or T2D) or sulfonylureas (in T2D). Causes include missing a meal, excessive physical exertion, excessive insulin administration, or “misdosing.” The signs and symptoms of hypoglycemia include dizziness, confusion, sweating, palpitations, and tachycardia; if hypoglycemia persists, loss of consciousness may occur. Neurological impairment may be the first symptom of hypoglycemia in patients with long-standing frequent episodes (so-called “hypoglycemia unawareness”). Rapid reversal of hypoglycemia through oral or intravenous glucose intake is critical for preventing permanent neurologic damage.

Chronic Complications of Diabetes

The morbidity associated with long-standing diabetes of either type is due to damage induced in large- and medium-sized muscular arteries (diabetic macrovascular disease) and in small vessels (diabetic microvascular disease) by chronic hyperglycemia. Macrovascular disease causes accelerated atherosclerosis among patients with diabetes, resulting in increased risk of myocardial infarction, stroke, and lower extremity ischemia. The effects of microvascular disease are most profound in the retina, kidneys, and peripheral nerves, resulting in *diabetic retinopathy, nephropathy, and neuropathy*, respectively (see later).

Pathogenesis

Persistent hyperglycemia (glucotoxicity) seems to be responsible for many of the long-term complications of diabetes. Much of the evidence supporting a role for glycemic control in ameliorating the long-term complications of diabetes has come from large randomized trials. The assessment of glycemic control in these trials has been based on the percentage of HbA1c (*glycated hemoglobin*) which is formed by nonenzymatic covalent addition of glucose moieties to hemoglobin in red cells. Unlike blood glucose levels, HbA1c provides a measure of glycemic control over the lifespan of a red cell (120 days) and is affected little by day-to-day variation in glucose levels. It is recommended that HbA1c be maintained below 7% in most patients with diabetes. The emergence of new technology, including continuous glucose-monitoring systems, has introduced a new goal, increasing “time-in-range,” which may be a better predictor of the risk of chronic complications than HbA1c level. It is important to stress, however, that hyperglycemia is not the only factor responsible for the long-term complications of diabetes, and that other underlying abnormalities, such as insulin resistance, and comorbidities like obesity, also play an important role.

At least four distinct mechanisms have been implicated in the deleterious effects of persistent hyperglycemia on peripheral tissues. **In each proposed mechanism, increased flux through metabolic pathways due to hyperglycemia is thought to generate harmful precursors that contribute to end-organ damage.**

- **Formation of advanced glycation end products.** Advanced glycation end products (AGEs) are formed because of nonenzymatic reactions between glucose-derived metabolites (glyoxal, methylglyoxal, and 3-deoxyglucosone) and the amino groups of intracellular and extracellular proteins. The rate of AGE formation is accelerated by hyperglycemia.

AGEs bind to a specific receptor (RAGE) that is expressed on inflammatory cells (macrophages and T cells), endothelium, and vascular smooth muscle. Endothelium-specific overexpression of RAGE in diabetic mice accelerates large vessel injury and microangiopathy, while RAGE-null mice show attenuation of these features. Antagonists of RAGE have emerged as potential therapeutic agents in diabetes and are being tested in clinical trials. The detrimental effects of the AGE-RAGE signaling axis within the vascular compartment include:

- Release of *cytokines and growth factors*, including transforming growth factor- β (TGF- β), which leads to deposition of excess basement membrane material, and vascular endothelial growth factor (VEGF), implicated in diabetic retinopathy (see later)
- Generation of *reactive oxygen species* (ROS) in endothelial cells
- Increased *procoagulant activity* on endothelial cells and macrophages
- Enhanced *proliferation of vascular smooth muscle cells* and *synthesis of extracellular matrix*
- **Activation of protein kinase C.** Calcium-dependent activation of intracellular protein kinase C (PKC) and the second messenger diacylglycerol (DAG) is an important signal transduction pathway. Intracellular hyperglycemia stimulates the *de novo* synthesis of DAG from glycolytic intermediates, and hence causes excessive PKC activation. The downstream effects of PKC activation are numerous, including production of VEGF, TGF- β , and the procoagulant protein plasminogen activator inhibitor-1 (PAI-1) (Chapter 4) by the vascular endothelium. It should be evident that some effects

of AGEs and activated PKC are overlapping, and both likely contribute to diabetic microangiopathy.

- **Oxidative stress and disturbances in polyol pathways.** Even in tissues that do not require insulin for glucose transport (e.g., nerves, lenses, kidneys, blood vessels), persistent hyperglycemia leads to an increase in intracellular glucose. This excess glucose is metabolized by the enzyme aldose reductase to sorbitol, a polyol, and eventually to fructose, in a reaction that uses the reduced form of nicotinamide-adenine dinucleotide phosphate (NADPH) as a cofactor. NADPH is also required by the enzyme glutathione reductase in a reaction that regenerates reduced glutathione (GSH). GSH is one of the important antioxidant mechanisms in the cell (Chapter 2), and any reduction in GSH increases cellular susceptibility to reactive oxygen species ("oxidative stress"). In the face of sustained hyperglycemia, progressive depletion of intracellular NADPH by aldose reductase compromises GSH regeneration, increasing cellular susceptibility to oxidative stress. In addition, sorbitol accumulation in the lens contributes to cataract formation.
- **Hexosamine pathways and generation of fructose-6-phosphate.** It has also been postulated that hyperglycemia induces flux of glycolytic intermediates through the hexosamine pathway, which results in cell damage and enhanced oxidative stress, which ultimately leads to insulin resistance and vascular complications.

Morphology of Chronic Complications of Diabetes

The important morphologic changes are related to the many late systemic complications of diabetes. As discussed earlier, these changes are seen in both T1D and T2D (Fig. 24.34).

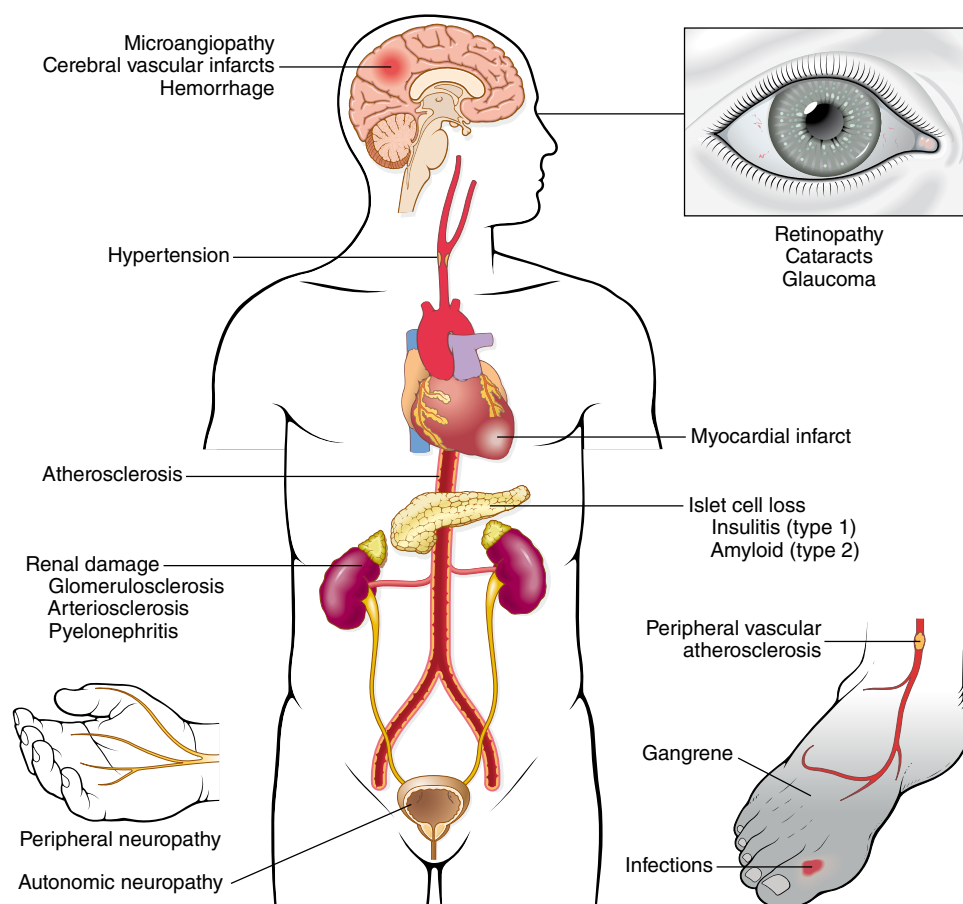


Fig. 24.34 Long-term complications of diabetes.

MORPHOLOGY

Alterations in the pancreas are varied and often subtle. Distinctive changes are more commonly associated with T1D than with T2D. One or more of the following alterations may be present:

- **Reduction in the number and size of islets.** This is most often seen in T1D, particularly with rapidly advancing disease. Most of the islets are small and inconspicuous.
- **Leukocytic infiltrates in the islets** (insulitis) are principally composed of T lymphocytes and are also seen in animal models of autoimmune diabetes (Fig. 24.35A). Lymphocytic infiltrates may be present in T1D at the time of clinical presentation.
- **In T2D there may be a subtle reduction in islet cell mass,** demonstrated only by special morphometric studies.
- **Amyloid deposition within islets in T2D** begins in and around capillaries and between cells. At advanced stages, the islets may be virtually obliterated (Fig. 24.35B); fibrosis may also be observed. Similar lesions may be found in older individuals without diabetes, apparently as part of normal aging.
- **An increase in the number and size of islets** is especially characteristic of nondiabetic newborns of mothers with diabetes. Presumably, fetal islets undergo hyperplasia in response to the maternal hyperglycemia.

Diabetic Macrovascular Disease. Diabetes exacts a heavy toll on the vascular system. *Endothelial dysfunction* (Chapter 11), which predisposes to atherosclerosis and other cardiovascular morbidities, is widespread in diabetes, because of the deleterious effects of persistent hyperglycemia and insulin resistance on the vascular compartment. The hallmark of diabetic macrovascular disease is *accelerated atherosclerosis* involving the aorta and large- and medium-sized arteries. The morphology of atherosclerosis in patients with diabetes is indistinguishable from that in individuals without diabetes (Chapter 11). **Myocardial infarction, caused by atherosclerosis of the coronary arteries, is the most common cause of death in diabetes.** *Gangrene of the lower extremities*, because of advanced vascular disease, is about 100 times more common in diabetes patients than in the general population. The larger renal arteries are also subject to severe atherosclerosis, but the most damaging effect of diabetes on the kidneys is exerted at the level of the glomeruli and the microcirculation (discussed later).

Diabetic Microangiopathy. One of the most consistent morphologic features of diabetes is *diffuse thickening of basement membranes*. The thickening is most evident in the capillaries of the skin, skeletal muscle, retina, renal glomeruli, and renal medulla. However, it may also be seen in such nonvascular structures as renal tubules, Bowman capsule, peripheral nerves, and placenta. It should be noted that despite the increase in the thickness of basement membranes, capillaries in patients with diabetes are leakier than normal to plasma proteins. **The microangiopathy underlies the development of diabetic nephropathy, retinopathy, and some forms of neuropathy.**

Diabetic Nephropathy. The kidneys are prime targets of diabetes. Renal failure is second only to myocardial infarction as a cause of death from this disease. **Three lesions are encountered: (1) glomerular lesions; (2) renal vascular lesions, principally arteriosclerosis; and (3) pyelonephritis, including necrotizing papillitis.**

The most important glomerular lesions are capillary basement membrane thickening, diffuse mesangial sclerosis, and nodular glomerulosclerosis.

- **Capillary basement membrane thickening.** Widespread thickening of the capillary glomerular basement membrane (GBM) occurs in

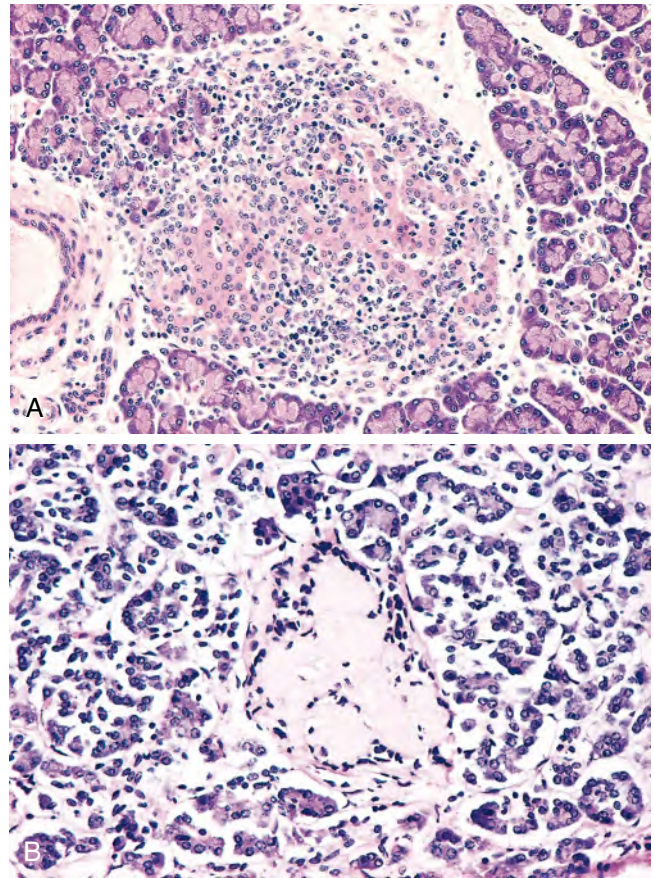


Fig. 24.35 (A) Insulitis, shown here from a rat (BB) model of autoimmune diabetes, also seen in type 1 human diabetes. (B) Amyloidosis of a pancreatic islet in type 2 diabetes. (A, Courtesy Dr. Arthur Like, University of Massachusetts, Worcester.)

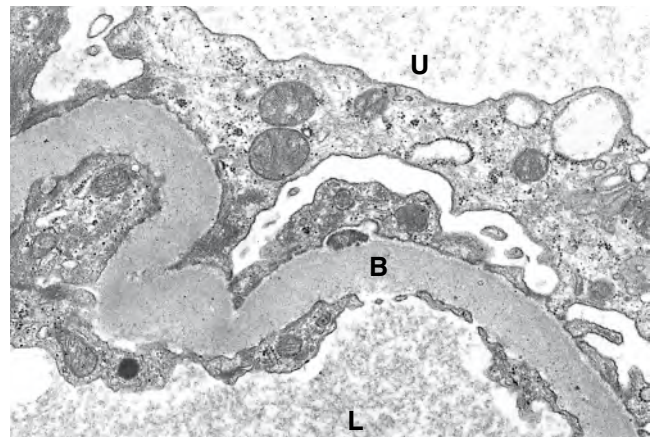


Fig. 24.36 Electron micrograph of a renal glomerulus showing markedly thickened glomerular basement membrane (B) in diabetes. L, Glomerular capillary lumen; U, urinary space. (Courtesy Dr. Michael Kashgarian, Department of Pathology, Yale University School of Medicine, New Haven, CT.)

virtually all cases of diabetic nephropathy and is part and parcel of diabetic microangiopathy. Capillary basement membrane thickening is best appreciated by electron microscopy (Fig. 24.36). Morphometric studies demonstrate that thickening begins as early as 2 years after the onset of T1D and by 5 years amounts to about a 30%

increase. These progressive changes in the GBM are usually accompanied by mesangial widening and thickening of the tubular basement membranes (Fig. 24.37).

- **Diffuse mesangial sclerosis.** This lesion consists of diffuse increase in mesangial matrix. The matrix depositions are PAS-positive (Fig. 24.38). As the disease progresses, the mesangial matrix deposits may take on a nodular appearance. The progressive expansion of the mesangium has been shown to correlate well with measures of deteriorating renal function such as increasing proteinuria.
- **Nodular glomerulosclerosis.** This is also known as intercapillary glomerulosclerosis or *Kimmelstiel-Wilson disease*. The glomerular lesions take the form of ovoid or spherical, often laminated, PAS-positive nodules of matrix situated in the periphery of the glomerulus. They lie within the mesangial core of the glomerular lobules and may be surrounded by patent peripheral capillary loops (see Fig. 24.38) or loops that are markedly dilated. The nodules often show features of mesangiolysis, defined by disruption of anchoring interactions between the capillaries and the mesangial stalks. The loss of these support structures may lead to the formation of capillary microaneurysms. While not all the lobules in individual glomeruli are involved by nodular lesions, even uninvolved lobules and glomeruli show striking diffuse mesangial sclerosis. The nodular

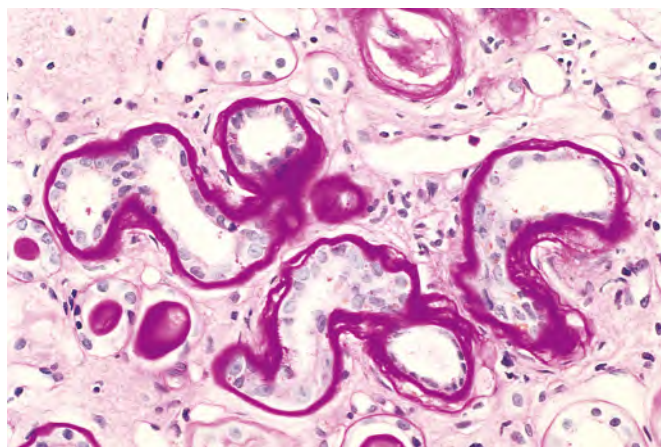


Fig. 24.37 Renal cortex showing thickening of tubular basement membranes in a diabetic patient (periodic acid–Schiff stain).

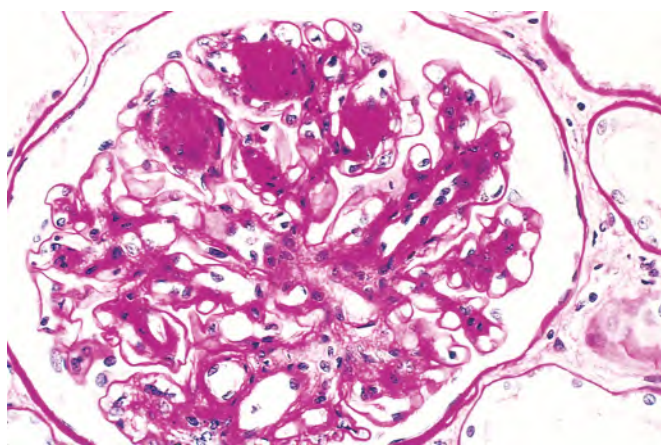


Fig. 24.38 Diffuse and nodular diabetic glomerulosclerosis (periodic acid–Schiff [PAS] stain). Note the diffuse increase in mesangial matrix and characteristic acellular PAS-positive nodules.

lesions are frequently accompanied by prominent accumulations of hyaline material in capillary loops (“fibrin caps”) or adherent to Bowman capsules (“capsular drops”). Both afferent and efferent glomerular hilar arterioles show hyalinosis. Because of the glomerular and arteriolar lesions, the kidney becomes ischemic, develops tubular atrophy and interstitial fibrosis, and usually contracts in size (Fig. 24.39). Approximately 15% to 30% of individuals with long-term diabetes develop nodular glomerulosclerosis, and in most instances, it is associated with renal failure.

Renal atherosclerosis and arteriolosclerosis also contribute to renal dysfunction in patients with diabetes. Hyaline arteriolosclerosis affects not only the afferent but also the efferent arteriole. Such efferent arteriolosclerosis is rarely, if ever, encountered in individuals who do not have diabetes.

Pyelonephritis is an acute or chronic inflammation of the kidneys that usually begins in the interstitial tissue and then spreads to affect the tubules. Both the acute and chronic forms of this disease are more common and more severe in patients with diabetes than in the general population. One special pattern of acute pyelonephritis, papillary necrosis, is much more prevalent in diabetes patients than in those without diabetes.

Diabetic Ocular Complications. The eye is profoundly affected by diabetes mellitus. The architecture and microanatomy of the eye are discussed in Chapter 29.

Diabetes-induced hyperglycemia leads to acquired opacification of the lens, a condition known as *cataract*. Long-standing diabetes is also associated with increased intraocular pressure (**glaucoma**; see later) and resulting damage to the optic nerve.

The most profound ocular changes of diabetes are seen in the retina. The retinal vasculopathy of diabetes mellitus can be classified into background (preproliferative) diabetic retinopathy and proliferative diabetic retinopathy (Chapter 29).

Diabetic Neuropathy. The prevalence of peripheral neuropathy in individuals with diabetes depends on the duration of the disease; up to



Fig. 24.39 Nephrosclerosis in a patient with long-standing diabetes. The kidney has been bisected to demonstrate both diffuse granular transformation of the surface (*left*) and marked thinning of the cortical tissue (*right*). Additional features include some irregular depressions, the result of pyelonephritis, and an incidental cortical cyst (*far right*).

50% of diabetes patients overall, and up to 80% of those who have had the disease for more than 15 years have clinical peripheral neuropathy. This is discussed further in Chapter 27.

Clinical Manifestations of Chronic Diabetes

The long-term effects of diabetes, rather than acute metabolic complications, are responsible for most of the morbidity and mortality of this disease. These complications most often first appear 15 to 20 years after the onset of hyperglycemia. The severity of chronic complications is related to both the degree and the duration of hyperglycemia, as evidenced by the attenuation of end-organ damage by effective glycemic control in prospective studies.

- **Vascular disease.** Macrovascular complications such as myocardial infarction, renal vascular insufficiency, and cerebrovascular accidents are the most common causes of mortality in long-standing diabetes. Patients with diabetes have a two to four times greater incidence of coronary artery disease and a four-fold higher risk of dying from cardiovascular complications than age-matched individuals without diabetes. An elevated risk for cardiovascular disease is even observed in patients with prediabetes. Significantly, myocardial infarction is almost as common in females with diabetes as in males. In contrast, myocardial infarction is less common in females of reproductive age without diabetes. Diabetes is often accompanied by underlying conditions that favor the development of adverse cardiovascular events. For example, *hypertension* is found in approximately 75% of individuals with T2D and potentiates the effects of hyperglycemia and insulin resistance on endothelial dysfunction and atherosclerosis. Another cardiovascular risk frequently seen in diabetes patients is *dyslipidemia*, which includes both increased triglycerides and LDL levels and decreased levels of the “protective” lipoprotein, high-density lipoprotein (HDL, Chapter 11). Insulin resistance is believed to contribute to “diabetic dyslipidemia” by favoring the hepatic production of atherogenic lipoproteins and by suppressing the uptake of circulating lipids in peripheral tissues.
- **Diabetic nephropathy.** Diabetic nephropathy is a leading cause of end-stage renal disease in the United States. Diabetic nephropathy ranges from asymptomatic albuminuria to end-stage renal disease requiring dialysis or transplantation. Approximately 30% to 40% of all patients with diabetes develop clinical evidence of nephropathy. The likelihood of diabetic nephropathy is influenced by ethnicity; for example, Native Americans, Hispanic people, and African Americans have a greater risk of developing end-stage renal disease than do non-Hispanic individuals of European descent with T2D. These differences are suspected to be genetic in origin, but the responsible genetic variants have yet to be identified. The earliest manifestation of diabetic nephropathy is the appearance of low amounts of albumin (*microalbuminuria*, now termed *moderately increased albuminuria*) in the urine (>30 mg and <300 mg per day). Without specific interventions, approximately 80% of patients with T1D and 20% to 40% of patients with T2D will develop overt nephropathy with *macroalbuminuria* (now called *severely increased albuminuria*) (>300 mg per day of urinary albumin) over 10 to 15 years, usually accompanied by the appearance of hypertension.
- **Diabetic ocular diseases.** Visual impairment, including total blindness, is one of the more feared consequences of long-standing diabetes. Approximately 60% to 80% of patients develop some form of *diabetic retinopathy* and diabetic retinopathy is the leading cause of adult blindness in the United States. The fundamental lesion of retinopathy—neovascularization—is attributable to hypoxia-induced expression of VEGF in the retina. Current treatment for this condition includes administration of antiangiogenic agents

that block the action of VEGF. As stated earlier, patients with diabetes also have an increased propensity for *glaucoma* and *cataract formation*, which also contribute to visual impairment.

- **Diabetic neuropathy.** Diabetic neuropathy can result in damage to the central nervous system, peripheral sensorimotor nerves, and the autonomic nervous system. It most frequently takes the form of a *distal symmetric polyneuropathy* of the lower extremities that affects both motor and sensory function. Over time, the upper extremities may be involved as well, thus approximating a “glove-and-stocking” pattern of polyneuropathy. Other forms include *autonomic neuropathy*, which produces disturbances in bowel and bladder function and sometimes erectile dysfunction, and *diabetic mononeuropathy*, which may manifest as sudden footdrop, wrist-drop, or isolated cranial nerve palsies.
- **Infections.** Patients with diabetes exhibit enhanced susceptibility to infections of the skin and to tuberculosis, pneumonia, and pyelonephritis. Such infections cause the deaths of about 5% of these patients. In an individual with diabetic neuropathy, a trivial infection in a toe may be the first event in a long succession of complications (gangrene, bacteremia, pneumonia) that ultimately leads to death. The basis of enhanced susceptibility is multifactorial and includes a decrease in neutrophil function (due to altered chemotaxis, adherence to the endothelium, phagocytosis, and microbicidal activity) and impaired cytokine production by macrophages. The vascular compromise also impairs the delivery of immune cells and molecules to sites of infection.

The staggering societal and economic impact of diabetes has already been discussed. For individuals with T2D, dietary modifications, exercise, and weight-loss regimens can reduce insulin resistance and hyperglycemia, at least early in the disease. However, all patients will ultimately require some form of therapeutic intervention to maintain glycemic control.

KEY CONCEPTS

Diabetes Mellitus: Pathogenesis and Long-Term Complications

- T1D is an autoimmune disease characterized by progressive destruction of islet β cells, leading to absolute insulin deficiency. T1D stems from a failure of self-tolerance in T cells. Circulating autoantibodies to islet cell antigens (including insulin) are often detected in affected patients.
- T2D has no autoimmune basis; instead, its central pathogenic features are insulin resistance and β -cell dysfunction, resulting in relative insulin deficiency.
- Obesity contributes to insulin resistance (and hence T2D) through multiple factors, including excess free fatty acids (FFAs), cytokines released from adipose tissues (adipocytokines), and inflammation.
- Monogenic forms of diabetes are less common and are caused by single-gene defects that result in primary β -cell dysfunction or lead to abnormalities of insulin receptor signaling (e.g., insulin receptor gene mutations).
- The long-term complications of diabetes are similar in both types and involve four potential mechanisms resulting from sustained hyperglycemia: formation of AGEs, activation of protein kinase C (PKC), disturbances in the polyol pathways, and overload of the hexosamine pathway.
- Long-term complications of diabetes include both large-vessel disease (macroangiopathy), such as atherosclerosis, ischemic heart disease, and lower extremity ischemia, and small vessel disease (microangiopathy), the latter manifesting mainly as retinopathy, nephropathy, and neuropathy.

PANCREATIC NEUROENDOCRINE NEOPLASMS

The preferred term for tumors of the pancreatic islet cells (“islet cell tumors”) is *pancreatic neuroendocrine neoplasms* or *PanNENs*. They are rare in comparison to tumors of the exocrine pancreas, accounting for 2% of pancreatic neoplasms. PanNENs can occur anywhere in the pancreas or the adjacent peripancreatic tissues. They resemble their counterparts found elsewhere in the alimentary tract (Chapter 17). These tumors may be single or multiple and of varying histologic grades. Pancreatic endocrine neoplasms often elaborate pancreatic hormones but are sometimes nonfunctional.

Like other endocrine neoplasms, it is difficult to predict the behavior of a PanNEN based on its light microscopic appearance alone, although tumors with a higher proliferation index are more likely to be aggressive. PanNENs are well-differentiated neoplasms that are graded based on mitotic activity and expression of the cell cycle marker Ki-67, which is used to determine a proliferative index. Pancreatic neuroendocrine carcinomas can be of small cell or large cell type. They are generally poorly differentiated neoplasms with a Ki-67 proliferation index >20%. The functional status of the tumor has some impact on prognosis, in that approximately 90% of insulin-producing tumors are benign, and 60% to 90% of other functioning and nonfunctioning pancreatic endocrine neoplasms are malignant. Fortunately, insulinomas are the most common subtype of pancreatic endocrine neoplasms.

Pathogenesis

The genome of sporadic PanNENs has been sequenced, with identification of recurrent somatic alterations in three major genes or pathways:

- *MEN1*, which causes familial MEN syndrome, type 1, also is mutated in some sporadic neuroendocrine tumors.
- Loss-of-function mutations in tumor suppressor genes such *PTEN* and *TSC2* (Chapter 7), which result in oncogenic activation of the molecular target of rapamycin (mTOR) signaling pathway.
- Inactivating mutations in two genes, alpha-thalassemia/mental retardation syndrome, X-linked (*ATRX*) and death-domain-associated protein (*DAXX*), which have multiple cellular functions, including telomere maintenance. PanNENs with *DAXX* or *ATRX* mutations demonstrate a phenomenon known as “alternative lengthening of telomeres” (ALT), which allows telomeres to be maintained in neoplastic cells that do not express telomerase (Chapter 7). Of note, nearly one-half of PanNENs have a somatic mutation in either *ATRX* or *DAXX*, but not both, consistent with their function in the same oncogenic pathway.

The three most common and distinctive clinical syndromes associated with functional pancreatic endocrine neoplasms are (1) *hyperinsulinism*, (2) *hypergastrinemia* and the *Zollinger-Ellison syndrome*, and (3) *MEN* (described later).

Hyperinsulinism (Insulinoma)

β -cell tumors (insulinomas) are the most common pancreatic endocrine neoplasms and often produce sufficient insulin to induce clinically significant hypoglycemia. The characteristic clinical picture is dominated by hypoglycemic episodes, which occur when the blood glucose level falls below 50 mg/dL. Clinical manifestations include confusion, stupor, and loss of consciousness. These episodes are precipitated by fasting or exercise and are promptly relieved by feeding or parenteral administration of glucose.

MORPHOLOGY

Insulinomas are most often found within the pancreas and are usually benign. Most are solitary, although multiple tumors may be encountered. *Bona fide* carcinomas make up only about 10% of cases. On rare occasions an insulinoma may arise in ectopic pancreatic tissue. In such cases, electron microscopy reveals the distinctive granules of β cells (see Fig. 24.27).

Solitary tumors are usually small (often <2 cm in diameter), encapsulated, pale to red-brown nodules located anywhere in the pancreas. Histologically, these benign tumors look remarkably like giant islets, with preservation of the regular cords of monotonous cells and their orientation to the vasculature. Malignant tumors may be deceptively encapsulated. **Deposition of amyloid** is a characteristic feature of many insulinomas (Fig. 24.40).

Hyperinsulinism may also be caused by **focal or diffuse hyperplasia of the islets**. This change is found occasionally in adults but is far more commonly encountered as congenital hyperinsulinism with hypoglycemia in neonates and infants. Several clinical scenarios may result in islet hyperplasia (previously known as nesidioblastosis), including maternal diabetes, Beckwith-Wiedemann syndrome (Chapter 10), and rare mutations in the β -cell K^{+} -channel protein or sulfonylurea receptor. In maternal diabetes, the fetal islets respond to hyperglycemia by increasing their size and number. In the postnatal period, these hyperactive islets may be responsible for serious episodes of hypoglycemia. This phenomenon is usually transient.

Clinical Features

While up to 80% of islet cell tumors demonstrate excessive insulin secretion, the hypoglycemia is mild in all but about 20%, and many cases never become clinically symptomatic. The critical laboratory findings in insulinomas are inappropriate high circulating levels of insulin in the setting of hypoglycemia. The presence of elevated C-peptide confirms endogenous hyperinsulinism. Surgical removal of the tumor is usually followed by prompt reversal of the hypoglycemia.

It is important to note that there are many other causes of hypoglycemia besides insulinoma. The differential diagnosis includes such conditions as adrenal insufficiency, diffuse liver disease, inherited glycogenoses, ectopic production of insulin by certain tumors, and use of sulfonylurea medications. Depending on the clinical circumstances, hypoglycemia induced by self-injection of insulin should also be considered.

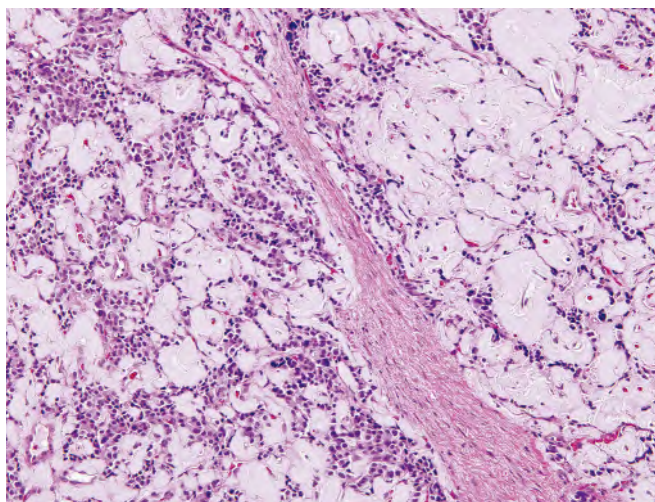


Fig. 24.40 Pancreatic neuro endocrine neoplasm (“islet cell tumor”). The neoplastic cells are monotonous and demonstrate minimal pleomorphism or mitotic activity. There is abundant amyloid deposition (pink material), characteristic of an insulinoma. Clinically, the patient had episodic hypoglycemia.

Zollinger-Ellison Syndrome (Gastrinoma)

Marked hypersecretion of gastrin usually has its origin in gastrin-producing tumors (*gastrinomas*), which are just as likely to arise in the duodenum and peripancreatic soft tissues as in the pancreas (so-called gastrinoma triangle). There has been lack of agreement regarding the cell of origin of these tumors, although it seems likely that endocrine cells of either the gut or the pancreas could be the source. Zollinger and Ellison first called attention to the **association of pancreatic islet cell lesions, hypersecretion of gastric acid, and severe peptic ulceration**, which are present in 90% to 95% of patients.

MORPHOLOGY

More than half of gastrin-producing tumors are locally invasive or have already metastasized at the time of diagnosis. In approximately 25% of patients, gastrinomas arise in conjunction with other endocrine tumors, as part of the MEN-1 syndrome (see later). MEN-1-associated gastrinomas are frequently multifocal, while sporadic gastrinomas are usually single. As with insulin-secreting tumors of the pancreas, gastrin-producing tumors are histologically bland and rarely show marked anaplasia.

In the Zollinger-Ellison syndrome, hypergastrinemia gives rise to excessive gastric acid secretion, which in turn causes **peptic ulceration** (Chapter 17). The duodenal and gastric ulcers are often multiple; although they are identical to those found in the general population, they are often unresponsive to therapy. In addition, ulcers may occur in unusual locations such as the jejunum.

Clinical Features

Zollinger-Ellison syndrome should be suspected in patients with intractable jejunal ulcers. Treatment involves control of gastric acid secretion by use of H^+K^+ -ATPase inhibitors and excision of the neoplasm. Total resection of the neoplasm, when possible, eliminates the syndrome. Patients with hepatic metastases have a shortened life expectancy, with progressive tumor growth leading to liver failure, usually within 10 years.

Other Rare Pancreatic Endocrine Neoplasms

- α -cell tumors (*glucagonomas*) are associated with increased serum levels of glucagon and a syndrome consisting of mild diabetes mellitus, a characteristic skin rash (*necrolytic migratory erythema*), and anemia. They occur most frequently in perimenopausal and postmenopausal females and are characterized by extremely high plasma glucagon levels.
- δ -cell tumors (*somatostatinomas*) are associated with diabetes mellitus, cholelithiasis, steatorrhea, and hypochlorhydria. They may be very difficult to localize preoperatively. High plasma somatostatin levels are required for diagnosis.
- *VIPoma* induces a characteristic syndrome (*watery diarrhea, hypokalemia, achlorhydria, or WDHA syndrome*) that is caused by release of vasoactive intestinal peptide (VIP) from the tumor. Some of these tumors are locally invasive and can metastasize. A VIP assay should be performed on all patients with severe secretory diarrhea. Neural crest tumors, such as neuroblastomas, ganglioneuroblastomas, and ganglioneuromas (Chapter 10), and pheochromocytomas (see later) can also be associated with the *VIPoma* syndrome.
- *Pancreatic carcinoid tumors* producing serotonin and an atypical carcinoid syndrome are exceedingly rare.
- *Pancreatic polypeptide-secreting endocrine tumors* present as mass lesions, as even high plasma levels of this hormone fail to cause symptoms.

- Some pancreatic and extrapancreatic endocrine tumors produce two or more hormones. In addition to insulin, glucagon, and gastrin, PanNENs may produce corticotropin, MSH, ADH, serotonin, and norepinephrine. These *multihormonal tumors* must be distinguished from the MEN syndromes (discussed later), in which multiple hormones are produced by tumors in several different glands.

ADRENAL GLANDS

The adrenal glands are paired endocrine organs consisting of a cortex and a medulla, which differ in their development, structure, and function. In essence, the cortex and medulla are two glands packaged into one structure. The adrenal cortex has three zones. Beneath the capsule is the narrow layer of zona glomerulosa. An equally narrow zona reticularis abuts the medulla. Intervening is the broad zona fasciculata, which makes up about 75% of the total cortex. The *adrenal cortex* synthesizes three different types of steroids: (1) *glucocorticoids* (principally cortisol), which are synthesized primarily in the zona fasciculata and to a lesser degree in the zona reticularis; (2) *mineralocorticoids*, the most important being aldosterone, which is generated in the zona glomerulosa; and (3) *sex steroids* (estrogens and androgens), which are produced largely in the zona reticularis. The *adrenal medulla* is composed of chromaffin cells, which synthesize and secrete *catecholamines*, mainly epinephrine. Catecholamines have many effects that allow rapid adaptations to changes in the environment.

ADRENAL CORTEX

Diseases of the adrenal cortex can be divided into those associated with hyperfunction and those associated with hypofunction.

Adrenocortical Hyperfunction (Hyperadrenalism)

The syndromes of adrenal hyperfunction are caused by overproduction of the three major hormones of the adrenal cortex (1) *Cushing syndrome*, characterized by an excess of cortisol; (2) *hyperaldosteronism* because of excessive aldosterone; and (3) *adrenogenital or virilizing syndromes* caused by an excess of androgens. The clinical features of these syndromes overlap somewhat because of the overlapping functions of some of the adrenal steroids.

Hypercortisolism (Cushing Syndrome)

Pathogenesis

This disorder is caused by conditions that produce elevated glucocorticoid levels. Cushing syndrome can be broadly divided into *exogenous* and *endogenous* causes. Most cases of Cushing syndrome are the result of the administration of exogenous glucocorticoids (*iatrogenic* Cushing syndrome). The endogenous causes can, in turn, be divided into those that are *corticotropin dependent* and those that are *corticotropin independent* (Fig 24.41).

Corticotropin-secreting pituitary neuroendocrine tumors account for 60% to 70% of cases of endogenous hypercortisolism. The pituitary form of this syndrome is referred to as *Cushing disease*. The disorder affects females about four times more frequently than males and occurs most frequently in young adults. In most cases, it is caused by an *corticotropin-producing pituitary neuroendocrine tumor*. In some cases, there is an underlying macroadenoma, and rarely there is *corticotroph cell hyperplasia* without a discrete adenoma. Corticotroph cell hyperplasia may be primary or arise secondarily from excessive stimulation of corticotropin release by a hypothalamic or other corticotrophin-releasing hormone (CRH)-producing tumor. The adrenal glands in individuals with Cushing disease are characterized by

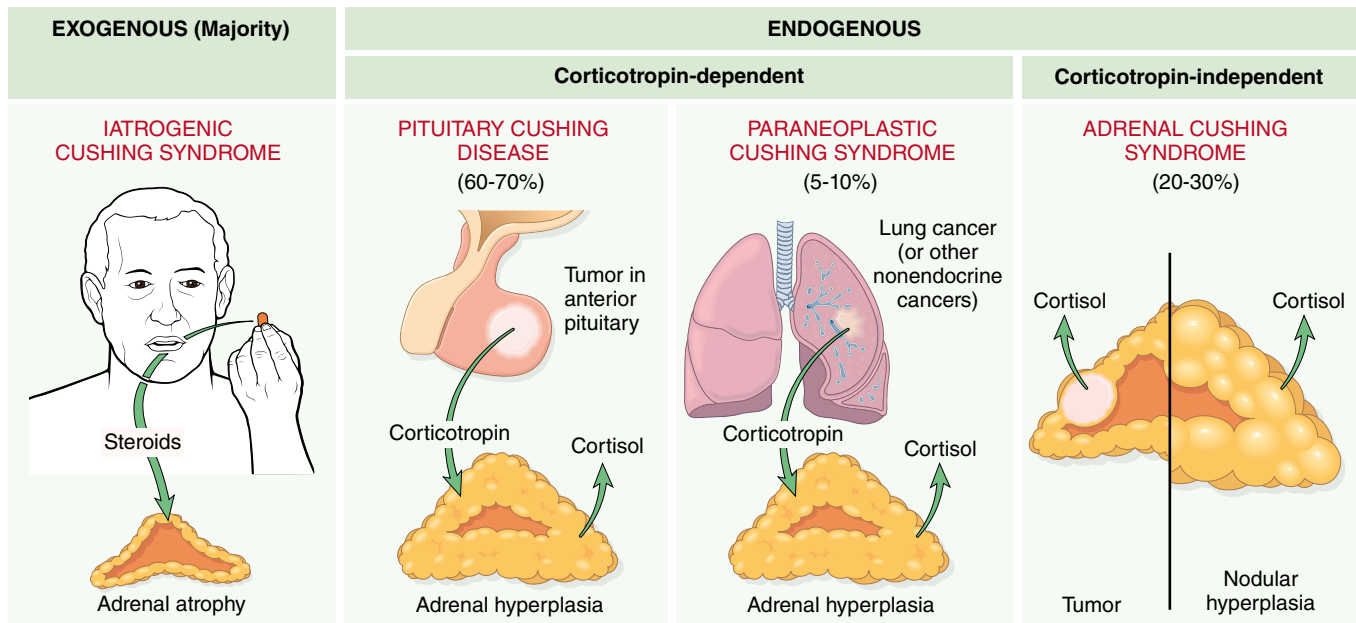


Fig. 24.41 Causes of Cushing syndrome. The three endogenous forms, as well as the more common exogenous (iatrogenic) form.

variable degrees of *nodular cortical hyperplasia* (discussed later) caused by the elevated levels of corticotropin. The cortical hyperplasia, in turn, is responsible for hypercortisolism.

Secretion of ectopic corticotropin by nonpituitary tumors accounts for about 5% to 10% of corticotropin-dependent Cushing syndrome cases. In many instances, the responsible tumor is a *small-cell carcinoma of the lung*, although other neoplasms, including carcinoid, medullary carcinoma of the thyroid, and PanNENs have been associated with the syndrome. In addition to tumors that elaborate ectopic corticotropin, occasionally a neuroendocrine neoplasm may produce ectopic CRH, which, in turn, causes corticotropin secretion and hypercortisolism. As in the pituitary variant, the adrenal glands undergo bilateral cortical hyperplasia, but the rapid downhill course of patients with these cancers often limits the extent of the adrenal enlargement. This variant of Cushing syndrome is more common in males and usually occurs in the 40s and 50s.

Primary adrenal neoplasms, such as adrenal adenoma (~10% to 20%) and carcinoma (~5% to 7%) are the most common underlying causes of corticotropin-independent Cushing syndrome. The biochemical *sine qua non* of corticotropin-independent Cushing syndrome is elevated serum levels of cortisol with low levels of corticotropin. Cortical carcinomas tend to produce more marked hypercortisolism than adenomas or hyperplasias. Gene-sequencing studies have helped define the molecular features of adrenocortical carcinomas, including recurrent activating mutations of the beta-catenin gene (*CTNNB1*) and inactivating mutations of *TP53*, *MEN1*, and *PRKARIA*. *PRKARIA* mutations, which act by raising cAMP levels (discussed later), are observed in both adrenocortical adenomas and carcinomas, as well as in micronodular hyperplasia (see earlier), indicating a common molecular mechanism is at play in these cortisol-secreting lesions.

Primary cortical hyperplasia (i.e., corticotropin-independent hyperplasia) is less common than corticotropin-dependent adrenal cortical hyperplasia. In *bilateral macronodular adrenal hyperplasia* (BMAH), the nodules are usually greater than 10 mm in diameter. Familial forms

associated with germline mutations in the armadillo repeat containing 5 (*ARMC5*) gene, a putative tumor suppressor gene, have been reported; of note, up to 50% of apparently sporadic BMAH cases harbor germline *ARMC5* mutations with somatic loss of the second allele in the hyperplastic nodules. In true sporadic cases of BMAH, nodular hyperplasia is not entirely “autonomous.” Specifically, cortisol production is regulated by hormones other than corticotropin because of ectopic overexpression of the corresponding G-protein-coupled hormone receptors in the hyperplastic adrenocortical cells. Implicated hormones other than corticotropin include gastric inhibitory peptide (GIP), LH, and ADH (vasopressin).

Corticotropin-independent bilateral hyperplasia can also be *micronodular* (<10 mm in size), and these arise mainly in two settings: primary pigmented nodular adrenocortical disease or as part of the so-called *Carney complex*, a multisystem syndrome resulting in both endocrine and nonendocrine neoplasms. Both primary pigmented nodular adrenocortical disease and Carney complex are most commonly associated with mutations of the regulatory subunit of cAMP-dependent protein kinase (encoded by the *PRKARIA* gene), which, like activating *GNAS* mutations, lead to increased intracellular cAMP levels.

MORPHOLOGY

The main lesions of Cushing syndrome are found in the pituitary and adrenal glands. The pituitary shows changes regardless of the cause. The most common alteration, resulting from high levels of endogenous or exogenous glucocorticoids, is termed **Crooke hyaline change**. In this condition, the normal granular, basophilic cytoplasm of the corticotropin-producing cells in the anterior pituitary becomes homogeneous and paler. This alteration is the result of the accumulation of intermediate keratin filaments in the cytoplasm, a finding that is also seen in Crooke cell corticotroph adenoma (discussed earlier).

Depending on the cause of the hypercortisolism, the **adrenals** show one of the following abnormalities: (1) cortical atrophy, (2) diffuse hyperplasia, (3) macronodular or micronodular hyperplasia, or (4) an adenoma or carcinoma. In patients in whom the syndrome results from exogenous glucocorticoids, suppression of endogenous corticotropin results in bilateral **cortical atrophy** due to a lack of stimulation of the *zonae fasciculata* and *reticularis* by corticotropin. The *zona glomerulosa* is of normal thickness in such cases, because this portion of the cortex functions independently of corticotropin. In contrast, in cases of endogenous hypercortisolism, the adrenals either are hyperplastic or contain a cortical neoplasm. **Diffuse hyperplasia** is found in individuals with corticotropin-dependent Cushing syndrome (Fig. 24.42). Both glands are enlarged, either subtly or markedly, weighing up to 30 g. The adrenal cortex is diffusely thickened and variably nodular, although the latter is not as pronounced as that seen in cases of corticotropin-independent nodular hyperplasia. Microscopically, the hyperplastic cortex demonstrates an expanded “lipid-poor” *zona reticularis* comprised of compact, eosinophilic cells, surrounded by an outer zone of vacuolated “lipid-rich” cells, resembling those seen in the *zona fasciculata*. Any nodules present are usually composed of vacuolated “lipid-rich” cells, which account for the yellow color of diffusely hyperplastic glands. In contrast, in **macronodular hyperplasia** the adrenals are almost entirely replaced by prominent nodules of varying sizes (10 to 30 mm), which contain an admixture of lipid-poor and lipid-rich cells. Unlike diffuse hyperplasia, the areas between the macroscopic nodules also demonstrate evidence of microscopic nodularity. **Micronodular hyperplasia** is composed of 1- to 3-mm (typically <10 mm) darkly pigmented (brown to black) micronodules with atrophic intervening areas (Fig. 24.43). The pigment is believed to be lipofuscin, a wear-and-tear pigment (Chapter 2).

Primary adrenocortical neoplasms causing Cushing syndrome may be malignant or benign. Functional adenomas or carcinomas are not morphologically distinct from nonfunctioning adrenal neoplasms (described later). Both the benign and malignant lesions are more common in females in their 30s to 50s. Adrenocortical **adenomas** are yellow tumors surrounded by thin or well-developed capsules, and most weigh less than 30 g. Microscopically, they are composed of cells that are similar to those encountered in the normal *zona fasciculata*. The **carcinomas** associated with Cushing syndrome, by contrast, tend to be larger than the adenomas. These tumors (discussed later) are unencapsulated masses frequently exceeding 200 to 300 g in weight that have all the anaplastic characteristics of cancer. With functioning tumors, both benign and malignant, the adjacent adrenal cortex and that of the contralateral adrenal gland are atrophic, because of suppression of endogenous corticotropin by high cortisol levels.

Clinical Features

Cushing syndrome develops slowly and its onset may be subtle. Early stages may present with hypertension and weight gain (Table 24.8). With time, the more characteristic central pattern of adipose tissue deposition becomes apparent in the form of truncal obesity, moon facies, and accumulation of fat in the posterior neck and back (*buffalo hump*). Hypercortisolism causes selective atrophy of fast-twitch (type 2) myofibers, resulting in decreased muscle mass and proximal limb weakness. Glucocorticoids induce gluconeogenesis and inhibit the uptake of glucose by cells, with resultant hyperglycemia, glucosuria, and polydipsia (*secondary diabetes*). The catabolic effects cause loss of collagen and resorption of bones. Consequently, the skin is thin, fragile, and easily bruised; wound healing is poor; and cutaneous striae are particularly



Fig. 24.42 Diffuse hyperplasia of the adrenal gland contrasted with a normal adrenal gland (*top*). In cross-section, the hyperplastic adrenal cortex is yellow and thickened, and a subtle nodularity is seen (contrast with Fig. 24.43) in this gland from a patient with corticotropin-dependent Cushing syndrome.

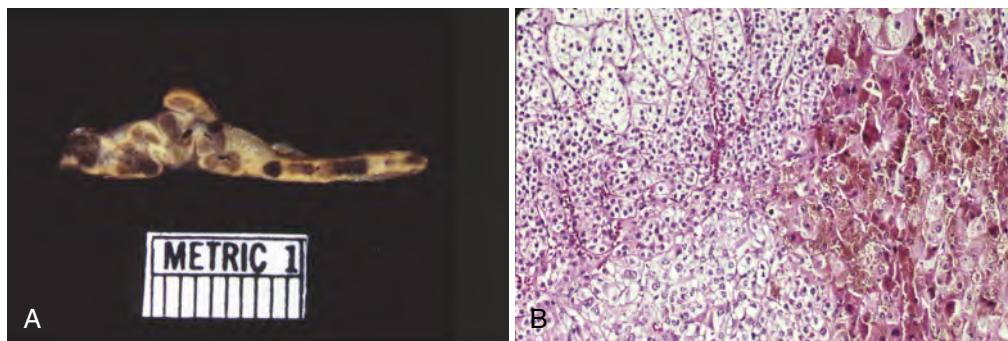


Fig. 24.43 (A) Micronodular adrenocortical hyperplasia with prominent pigmented nodules in the adrenal gland. (B) On histologic examination, the nodules are composed of cells containing lipofuscin pigment, seen in the right part of the field. (Courtesy Dr. Aidan Carney, Department of Medicine, Mayo Clinic, Rochester, MN.)

TABLE 24.8 Clinical Features of Cushing Syndrome

Feature	Percent
Obesity or weight gain	95% ^a
Facial plethora	90%
Rounded face	90%
Decreased libido	90%
Thin skin	85%
Decrease in linear growth in children	70%–80%
Menstrual irregularity	80%
Hypertension	75%
Hirsutism	75%
Depression/emotional lability	70%
Easy bruising	65%
Glucose intolerance	60%
Weakness	60%
Osteopenia or fracture	50%
Nephrolithiasis	50%

^a100% in children.

Modified from Newell-Price J, Bertagna X, Grossman AB, Nieman LK. Cushing syndrome. *Lancet*. 2006;367(9522):1605-1617.



Fig. 24.44 A patient with Cushing syndrome demonstrating central obesity, “moon facies,” and abdominal striae. (From Lloyd RV, Douglas BR, Young WF Jr, eds. *Atlas of Nontumor Pathology: Endocrine Diseases*. American Registry of Pathology: 2002.)

common in the abdominal area (Fig. 24.44). Bone resorption results in the development of *osteoporosis*, with consequent backache and increased susceptibility to fractures. Persons with Cushing syndrome are at increased risk for a variety of infections, because glucocorticoids suppress the immune response. Additional manifestations include several *mental disturbances*, including mood swings,

depression, and frank psychosis, as well as *hirsutism* and *menstrual abnormalities*.

The laboratory diagnosis of Cushing syndrome is based on the following: (1) the 24-hour urine free-cortisol concentration, which is increased; and (2) loss of normal diurnal pattern of cortisol secretion (thus midnight salivary cortisol levels are elevated). In addition, a 1 mg (low-dose) dexamethasone suppression test can also identify hypercortisolism. Determining the cause of Cushing syndrome depends on measurement of the serum corticotropin. Corticotropin levels are low in corticotropin-independent adrenal adenomas and carcinomas, and in exogenous causes. In contrast, corticotropin levels are high or normal in the setting of Cushing disease or ectopic corticotropin (or CRH) production. High-dose dexamethasone suppression tests can help distinguish between the two main types of corticotropin-dependent Cushing syndrome, Cushing disease and the ectopic production of corticotropin. Although patients with both disorders generally have abnormally high levels of cortisol in response to the low-dose (1 mg dexamethasone) test, patients with Cushing disease generally suppress appropriately with higher doses of dexamethasone, whereas patients with ectopic Cushing do not. However, these tests are far from perfect, and some patients require inferior petrosal sinus sampling to distinguish between them.

KEY CONCEPTS

Hypercortisolism (Cushing Syndrome)

- The most common cause of hypercortisolism is exogenous administration of steroids.
- Endogenous hypercortisolism most often is secondary to an corticotropin-producing pituitary neuroendocrine tumor (Cushing disease), followed by primary adrenal neoplasms (corticotropin-independent hypercortisolism) and paraneoplastic corticotropin production by tumors (e.g., small-cell lung cancer).
- The morphologic features in the adrenal vary from bilateral cortical atrophy (in exogenous steroid-induced disease), to bilateral diffuse or nodular hyperplasia (most common finding in endogenous Cushing syndrome), to an adrenocortical neoplasm.

Primary Hyperaldosteronism

Hyperaldosteronism is the generic term for a group of closely related conditions characterized by chronic excess aldosterone secretion. Hyperaldosteronism may be primary or it may be secondary to an extra-adrenal cause. *Primary hyperaldosteronism* stems from an autonomous overproduction of aldosterone, with resultant hypertension, suppression of the renin-angiotensin system, and decreased plasma renin activity. Primary hyperaldosteronism is caused by one of three conditions (Fig. 24.45):

- *Bilateral idiopathic hyperaldosteronism*, characterized by bilateral nodular hyperplasia of aldosterone-secreting zona glomerulosa cells of the adrenal glands, is the most common underlying cause, accounting for about 60% of cases, most of which are sporadic. Individuals with bilateral idiopathic hyperaldosteronism tend to be older and to have less severe hypertension than those presenting with adrenal neoplasms.
- *Adrenocortical neoplasm*, either an aldosterone-producing adenoma (the most common cause) or, rarely, an adrenocortical carcinoma. In approximately 35% of cases, primary hyperaldosteronism is caused by a solitary aldosterone-secreting adenoma, a condition referred to as *Conn syndrome*. This condition occurs most frequently in mid-adult life and is more common in females than in

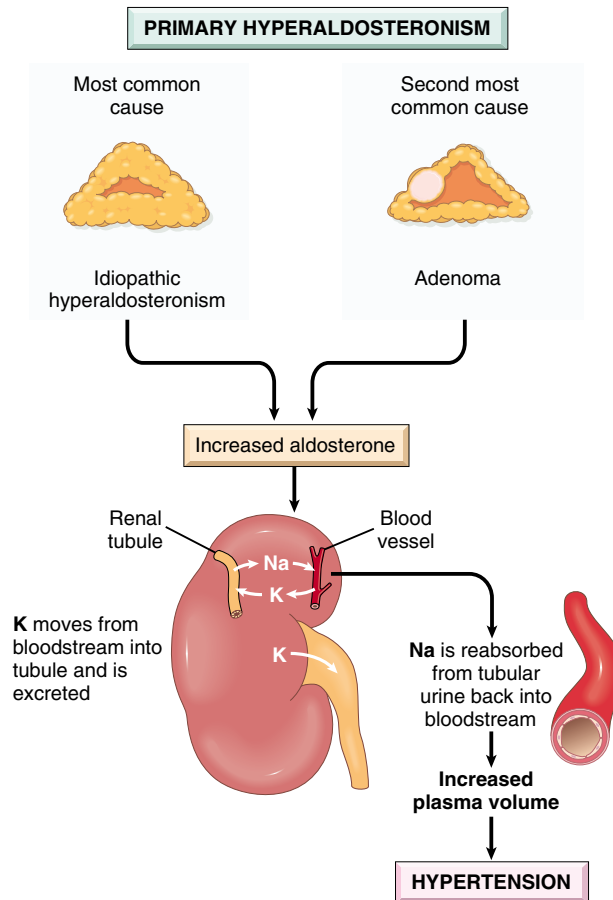


Fig. 24.45 The major causes of primary hyperaldosteronism and its principal effects on the kidney. K, Potassium; Na, sodium.

males (2:1). Gene-sequencing studies have elucidated the genomic landscape of aldosterone-producing adenomas. Up to 50% harbor somatic mutations of *KCNJ5*, which encodes a potassium ion channel protein known as GIRK4 expressed in zona glomerulosa cells. *KCNJ5* mutations result in loss of selectivity of the channel to potassium ions, leading to nonspecific influx of sodium, and subsequent calcium-dependent activation of the aldosterone synthase enzyme. The functional consequence of these mutations is to maintain cells in a chronic state of depolarization, resulting in autonomous aldosterone synthesis.

- *Familial hyperaldosteronism* accounts for ~5% of cases. Four distinct subtypes (FH-I to FH-IV) have been described, of which FH-I, or *glucocorticoid-remediable aldosteronism*, is the most common. FH-I stems from a rearrangement in chromosome 8 that places *CYP11B2* (the gene encoding aldosterone synthase, the enzyme that carries out the last step in aldosterone synthesis) under the control of the corticotropin-responsive *CYP11B1* gene promoter. corticotropin thus stimulates the synthesis of aldosterone synthase from the chimeric gene.

In *secondary hyperaldosteronism*, in contrast, aldosterone release occurs in response to physiologic activation of the renin-angiotensin system (Chapter 11). It is characterized by increased levels of plasma renin and is encountered in conditions such as:

- Decreased renal perfusion (arteriolar nephrosclerosis, renal artery stenosis)
- Arterial hypovolemia and edema (congestive heart failure, cirrhosis, nephrotic syndrome)

- Pregnancy (due to estrogen-induced increases in plasma renin substrate)

MORPHOLOGY

Aldosterone-producing adenomas are almost always solitary, small (<2 cm in diameter), well-circumscribed lesions, more often found on the left than on the right. They tend to occur in the 30s and 40s, and in females more often than in males. They are often buried within the gland and do not produce visible enlargement, making them difficult to locate by imaging. They are bright yellow on cut section and are composed of lipid-laden cortical cells that more closely resemble fasciculata cells than glomerulosa cells (the normal source of aldosterone). In general, the cells tend to be uniform in size and shape; occasionally, there is modest nuclear and cellular pleomorphism (see Fig. 24.51). A characteristic feature of aldosterone-producing adenomas is the presence of eosinophilic, laminated cytoplasmic inclusions, known as **spironolactone bodies**, found after treatment with the antihypertensive drug spironolactone. In contrast to cortical adenomas associated with Cushing syndrome, those associated with hyperaldosteronism do not usually suppress corticotropin secretion. Therefore, the adjacent adrenal cortex and that of the contralateral gland are not atrophic.

Bilateral idiopathic hyperaldosteronism is marked by diffuse and focal hyperplasia of cells resembling those of the normal zona glomerulosa. The hyperplasia is often wedge-shaped, extending from the periphery toward the center of the gland. The enlargement may be subtle, and as a rule an adrenocortical adenoma must be carefully excluded as the cause for hyperaldosteronism.

Clinical Features

Primary hyperaldosteronism is the most common cause of secondary hypertension (i.e., hypertension secondary to an identifiable cause). The prevalence of hyperaldosteronism increases with the severity of hypertension, reaching 20% or more in some studies of patients who are classified as having treatment-resistant hypertension. Through its effects on the renal mineralocorticoid receptor, aldosterone promotes sodium reabsorption, which secondarily increases the reabsorption of water, expanding the extracellular fluid volume and elevating cardiac output.

The long-term effects of hyperaldosteronism-induced hypertension are cardiovascular compromise (e.g., left ventricular hypertrophy and reduced diastolic volumes) and an increase in the prevalence of adverse events such as stroke and myocardial infarction. Hypokalemia is not a mandatory feature but, when present, results from renal potassium wasting and can cause a variety of neuromuscular manifestations, including weakness, paresthesias, visual disturbances, and occasionally tetany.

The diagnosis of primary hyperaldosteronism is confirmed by elevated ratios of plasma aldosterone concentration to plasma renin activity; if this screening test is positive, a confirmatory aldosterone suppression test is performed, as many other disorders can alter the plasma aldosterone and renin ratios.

In primary hyperaldosteronism, the therapy varies according to cause. Adenomas are amenable to surgical excision. In contrast, patients with bilateral hyperplasia are best managed medically with an aldosterone antagonist such as spironolactone or eplerenone, though

potassium-sparing diuretics have also been used. In contrast, the treatment of secondary hyperaldosteronism rests on correcting the underlying cause of the hyperactivity of the renin-angiotensin system.

Adrenogenital Syndromes

Disorders of sexual differentiation, such as *virilization* or *feminization*, can be caused by primary gonadal disorders (Chapter 22) and several primary adrenal disorders. The adrenal cortex secretes dehydroepiandrosterone and androstenedione, two compounds that can be converted to testosterone in peripheral tissues. Unlike gonadal androgens, corticotropin regulates adrenal androgen formation (Fig. 24.46); thus, excess secretion can occur either as a “pure” syndrome or as a component of Cushing disease. The adrenal causes of androgen excess include *adrenocortical neoplasms* and a group of disorders that have been designated *congenital adrenal hyperplasia* (CAH).

Adrenocortical neoplasms associated with virilization are more likely to be androgen-secreting adrenal carcinomas than adenomas. Such tumors are often also associated with hypercortisolism (“mixed syndrome”). They are morphologically identical to other cortical neoplasms and are discussed later.

Congenital adrenal hyperplasia stems from several autosomal recessive, inherited metabolic errors, each characterized by a deficiency of a particular enzyme involved in the biosynthesis of cortisol (Fig. 24.46). Steroid precursors that build behind the defective step in the pathway are channeled into other pathways, resulting in increased production of androgens, which accounts for *virilization*. Simultaneously,

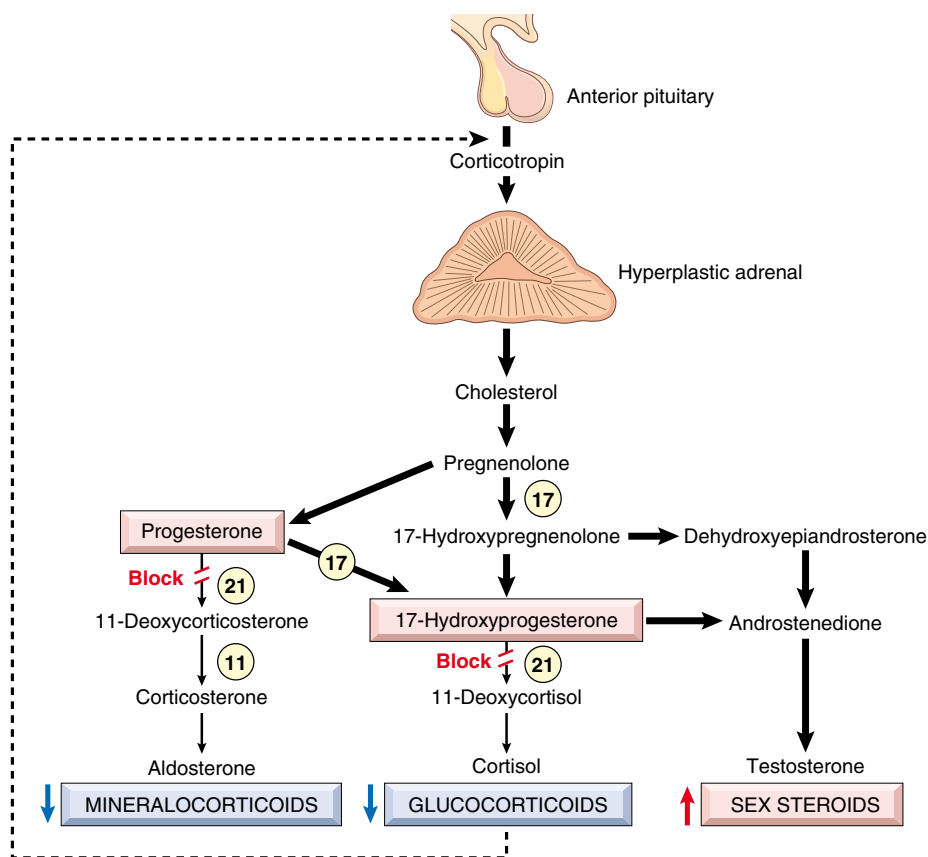


Fig. 24.46 Consequences of 21-hydroxylase deficiency. 21-hydroxylase deficiency impairs the synthesis of both cortisol and aldosterone at different steps (shown as *Block* in the biosynthesis pathway). The resultant decrease in feedback inhibition (*dashed line*) causes increased secretion of adrenocorticotrophic hormone, resulting ultimately in adrenal hyperplasia and increased synthesis of testosterone. The sites of action of 11-, 17-, and 21-hydroxylase are shown as numbers in circles.

the deficiency of cortisol leads to increased secretion of corticotropin, culminating in adrenal hyperplasia. Certain enzyme defects may also impair aldosterone secretion, adding *salt wasting* to the virilizing syndrome. Other enzyme deficiencies may be incompatible with life or, in rare instances, may involve only the aldosterone pathway.

21-hydroxylase deficiency (caused by mutations of *CYP21A2*) is by far the most common genetic cause of CAH, accounting for over 90% of cases. Fig. 24.46 illustrates normal adrenal steroidogenesis and the consequences of 21-hydroxylase deficiency, which varies in severity depending on the nature of the *CYP21A2* mutation. Three distinctive syndromes have been described: (1) salt-wasting (“classic”) CAH, (2) simple virilizing adrenogenitalism, and (3) “nonclassic” hyperandrogenism.

- The *salt-wasting syndrome* results from an inability to convert progesterone into deoxycorticosterone because of a total lack of 21-hydroxylase. Thus, there is virtually no synthesis of mineralocorticoids, and, concomitantly, there is a block in the conversion of hydroxyprogesterone into deoxycortisol resulting in deficient cortisol synthesis. This pattern usually comes to light soon after birth, as in utero, electrolyte and fluid balance are maintained by the maternal kidneys. There is salt wasting, hyponatremia, and hyperkalemia, which induce acidosis, hypotension, cardiovascular collapse, and possibly death. The concomitant block in cortisol synthesis and excess production of androgens, however, lead to virilization, which can be recognized in the female at birth or in utero. Males with this disorder are generally unrecognized at birth but come to clinical attention 5 to 15 days later because of salt loss, hypotension, and other abnormalities.
- *Simple virilizing adrenogenital syndrome without salt wasting* (presenting as genital ambiguity) occurs in approximately one-third of patients with 21-hydroxylase deficiency. These patients generate sufficient mineralocorticoid to prevent a salt-wasting “crisis.” However, the lowered glucocorticoid level fails to cause feedback inhibition of corticotropin secretion. Thus, the level of testosterone is increased, with resultant progressive virilization.
- *Nonclassic or late-onset hyperandrogenism* is significantly more common than the classic patterns already described. There is only a partial deficiency of 21-hydroxylase, which accounts for the later onset. Individuals with this syndrome may present with mild manifestations, such as hirsutism, acne, and menstrual irregularities. Nonclassic CAH cannot be diagnosed on routine newborn screening, and the diagnosis is usually made by demonstration of biosynthetic defects in steroidogenesis.

MORPHOLOGY

In all cases of CAH, the adrenals are bilaterally hyperplastic, sometimes increasing to 10 to 15 times their normal weight because of the sustained elevation in corticotropin. The adrenal cortex is thickened and nodular, and on cut section the widened cortex appears brown because of lipid depletion. The proliferating cells are mostly compact, eosinophilic, lipid-depleted cells intermixed with lipid-laden clear cells. Hyperplasia of corticotroph (corticotropin-producing) cells is present in the anterior pituitary in most persons with CAH.

Clinical Features

The clinical features of these disorders are determined by the specific enzyme deficiency and include abnormalities related to androgen excess, with or without aldosterone and glucocorticoid deficiency. In addition, high levels of intraadrenal glucocorticoids are required to facilitate medullary catecholamine (epinephrine and norepinephrine) synthesis. In patients with severe salt-wasting 21-hydroxylase

deficiency, a combination of low cortisol levels and developmental defects of the medulla (*adrenomedullary dysplasia*) affects catecholamine secretion as well, further predisposing these individuals to hypotension and circulatory collapse.

Depending on the nature and severity of the enzymatic defect, the onset of clinical symptoms may occur in the perinatal period, later in childhood, or, less commonly, in adulthood. For example, in 21-hydroxylase deficiency excessive androgenic activity causes signs of masculinization in females ranging from clitoral hypertrophy and other differences in sexual development (DSD) in infants to oligomenorrhea, hirsutism, and acne in postpubertal females. In males, androgen excess is associated with enlargement of the external genitalia and other evidence of precocious puberty in prepubertal patients and oligospermia in older males.

CAH should be suspected in any neonate with ambiguous genitalia. Severe enzyme deficiency in infancy can be a life-threatening condition with vomiting, dehydration, and salt wasting. Individuals with CAH are treated with exogenous glucocorticoids, which, in addition to providing glucocorticoids, also suppress corticotropin levels and thus decrease the excessive synthesis of the steroid hormones responsible for many of the clinical abnormalities. Mineralocorticoid supplementation is required in the salt-wasting variants of CAH. With the availability of routine neonatal metabolic screens for CAH and the feasibility of molecular testing for antenatal detection of 21-hydroxylase mutations, the outcome for even the most severe variants has improved significantly.

KEY CONCEPTS

Adrenogenital Syndromes

- The adrenal cortex can secrete excess androgens in two settings: adrenocortical neoplasms (usually virilizing carcinomas) or congenital adrenal hyperplasia (CAH).
- CAH consists of a group of autosomal recessive disorders characterized by defects in steroid biosynthesis, usually cortisol; the most common subtype is caused by deficiency of the enzyme 21-hydroxylase.
- Reduction in cortisol production causes a compensatory increase in corticotropin secretion, which, in turn, stimulates androgen production. Androgens have virilizing effects, including masculinization in females (ambiguous genitalia, oligomenorrhea, hirsutism) and precocious puberty in males. Reductions in cortisol and aldosterone lead to salt (sodium) wasting and hypotension.
- Bilateral hyperplasia of the adrenal cortex is characteristic, and a subset of 21-hydroxylase-deficient patients also demonstrates adrenomedullary dysplasia.

Adrenocortical Insufficiency

Adrenocortical insufficiency, or hypofunction, may be caused by primary adrenal disease (primary adrenal insufficiency) or decreased stimulation of the adrenals due to a deficiency of corticotropin (secondary adrenal insufficiency) (Table 24.9). Three major patterns of adrenocortical insufficiency are recognized: (1) primary acute adrenocortical insufficiency (adrenal crisis), (2) primary chronic adrenocortical insufficiency (*Addison disease*), and (3) secondary adrenocortical insufficiency.

Primary Acute Adrenocortical Insufficiency

Acute adrenal cortical insufficiency occurs in a variety of clinical settings.

- In individuals with chronic adrenocortical insufficiency, a *crisis* may be precipitated by any form of stress that requires an immediate increase in steroid output to maintain homeostasis.

TABLE 24.9 Adrenocortical Insufficiency

Primary Insufficiency
Loss of Cortical Cells
Congenital adrenal hypoplasia X-linked adrenal hypoplasia (<i>NR0B1</i> mutation)
Adrenoleukodystrophy (<i>ABCD1</i> mutation)
Autoimmune adrenal insufficiency
Autoimmune polyendocrinopathy syndrome type 1 (<i>AIRE1</i> gene)
Autoimmune polyendocrinopathy syndrome types 2 and 4 (polygenic)
Infection
Acquired immune deficiency syndrome
Tuberculosis
Fungi
Acute hemorrhagic necrosis (Waterhouse-Friderichsen syndrome)
Amyloidosis, sarcoidosis, hemochromatosis
Metastatic carcinoma
Metabolic Failure in Hormone Production
Congenital adrenal hyperplasia (cortisol and aldosterone deficiency with virilization)
Drug- and steroid-induced inhibition of corticotropin or cortical cell function
Secondary Insufficiency
Hypothalamic Pituitary Disease
Neoplasm, inflammation (sarcoidosis, tuberculosis, pyogenic bacteria, fungi)
Hypothalamic Pituitary Suppression
Long-term steroid administration
Steroid-producing neoplasms

- In patients maintained on exogenous corticosteroids, in whom *rapid withdrawal of steroids* or failure to increase steroid doses in response to an acute stress may precipitate an adrenal crisis because of the inability of the atrophic adrenals to produce glucocorticoid hormones.
- As a result of *massive adrenal hemorrhage*, which damages the adrenal cortex sufficiently to cause acute adrenocortical insufficiency—as occurs in newborns following prolonged and difficult delivery with considerable trauma and hypoxia. It also occurs in some patients on anticoagulant therapy, in postsurgical patients who develop disseminated intravascular coagulation and consequent hemorrhagic infarction of the adrenals, and as a complication of disseminated bacterial infection; in this last setting, it is called *Waterhouse-Friderichsen syndrome*.
- Acute adrenal insufficiency can occur as a rare but potentially fatal condition during the use of immune checkpoint blockade as part of cancer immunotherapy. These medications can lead to either primary adrenal insufficiency (via development of adrenalitis) or secondary adrenal insufficiency (via development of hypophysitis).

Waterhouse-Friderichsen Syndrome

This uncommon but catastrophic syndrome is characterized by:

- Overwhelming bacterial infection, classically *Neisseria meningitidis* septicemia but occasionally caused by other highly virulent organisms, such as *Pseudomonas* species, pneumococci, *Haemophilus influenzae*, or even staphylococci
- Rapidly progressive hypotension leading to shock
- Disseminated intravascular coagulation associated with widespread purpura, particularly of the skin (Fig. 24.47)



Fig. 24.47 Diffuse purpuric rash in a patient with Waterhouse-Friderichsen syndrome. (From Vincentelli C, Molina EG, Robinson MJ. Fatal pneumococcal Waterhouse-Friderichsen syndrome in a vaccinated adult with congenital asplenia. *Am J Emerg Med*. 2009;27(6):751.e3-751.e5.)

- Rapidly developing adrenocortical insufficiency associated with massive bilateral adrenal hemorrhage

Waterhouse-Friderichsen syndrome can occur at any age but is more common in children. The basis for the adrenal hemorrhage is uncertain but may be due to direct bacterial seeding of small vessels in the adrenal, the development of disseminated intravascular coagulation, or endothelial dysfunction caused by microbial products and inflammatory mediators. Whatever the basis, the adrenals are converted to sacs of clotted blood, which obscures virtually all of the underlying detail. Histologic examination reveals that the hemorrhage starts within the medulla near thin-walled venous sinusoids and then spreads into the cortex, often leaving islands of recognizable cortical cells (Fig. 24.48). Prompt recognition and appropriate therapy must be instituted immediately, or death follows within hours to a few days.

Primary Chronic Adrenocortical Insufficiency (Addison Disease)

In an article published in 1855, Thomas Addison described a group of patients with a constellation of symptoms, including “general languor and debility, remarkable feebleness of the heart’s action, and a peculiar change in the color of the skin” associated with disease of the “suprarenal capsules” or, in modern parlance, the adrenal glands. Addison disease, or chronic adrenocortical insufficiency, is an uncommon disorder resulting from progressive destruction of the adrenal cortex. Typically, clinical manifestations of adrenocortical insufficiency do not appear until at least 90% of the adrenal cortex has been compromised. The causes of chronic adrenocortical insufficiency are listed in Table 24.9. Although all races and both sexes may be affected, certain causes of Addison disease (e.g., autoimmune adrenalitis) are much more common in individuals of European descent and in females.

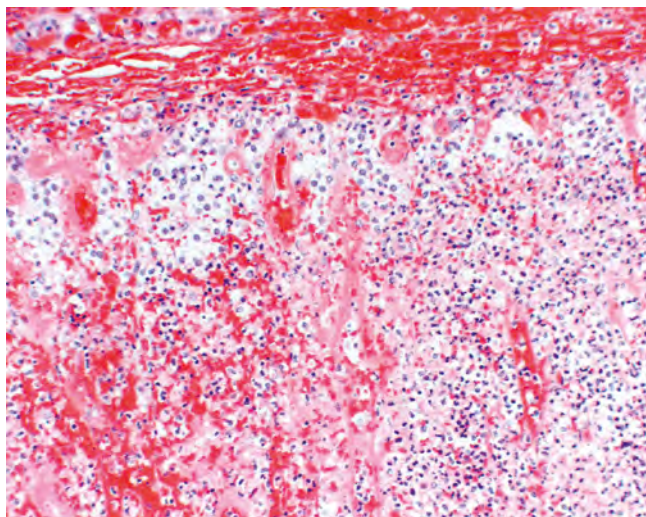


Fig. 24.48 Waterhouse-Friderichsen syndrome. At autopsy, the adrenals were grossly hemorrhagic and shrunken; microscopically, little residual cortical architecture is discernible.

Pathogenesis

A large number of diseases may affect the adrenal cortex, including lymphoma, amyloidosis, sarcoidosis, hemochromatosis, fungal infections, and adrenal hemorrhage, but more than 90% of all cases are attributable to one of four disorders: autoimmune adrenalitis, tuberculosis, AIDS, or metastatic cancers.

- **Autoimmune adrenalitis** accounts for 80% to 90% of cases of primary adrenal insufficiency in high-income countries. As the name implies, there is autoimmune destruction of steroidogenic cells, and autoantibodies to several key steroidogenic enzymes (21-hydroxylase, 17-hydroxylase) are detected in most patients. Autoimmune adrenalitis occurs as an isolated condition in 40% of cases, or as part of an autoimmune polyendocrinopathy syndrome in the remaining 60%. As noted earlier, autoimmune adrenalitis can occur due to immune checkpoint blockade as part of cancer chemotherapy. There are several distinct syndromes of autoimmune polyendocrinopathies:
 - **APS-1** (autoimmune polyglandular syndrome type 1) is characterized by a combination of organ-specific autoimmune disorders (autoimmune adrenalitis, autoimmune hypoparathyroidism, idiopathic hypogonadism, pernicious anemia) that result in immune destruction of target organs. Other features include mucocutaneous candidiasis and abnormalities of skin, dental enamel, and nails (ectodermal dystrophy). APS-1 is caused by mutations in the autoimmune regulator (*AIRE*) gene on chromosome 21q22. *AIRE* is expressed primarily in the thymus, where it functions as a transcription factor that promotes the expression of many peripheral tissue antigens. Self-reactive T cells that recognize these antigens are eliminated (Chapter 6). In the absence of *AIRE* function, central T-cell tolerance to peripheral tissue antigens is compromised, promoting autoimmunity. Individuals with APS-1 develop autoantibodies against IL-17 and IL-22, which are the principal effector cytokines secreted by Th17 T cells (Chapter 6). Because these two Th17-derived cytokines are crucial for defense against fungal infections, it is not surprising that patients develop candidiasis.
 - **APS-2** (autoimmune polyglandular syndrome type 2) usually starts in the fourth decade of life and presents as a combination of adrenal insufficiency and autoimmune thyroiditis, with or

without T1D. Unlike in APS-1, mucocutaneous candidiasis, ectodermal dysplasia, and autoimmune hypoparathyroidism do not develop. Overall, APS-2 is more prevalent than APS-1. APS-4 is a rare condition characterized by adrenalitis plus other autoimmune phenomena (e.g., gastritis, vitiligo, alopecia, pernicious anemia) but neither thyroiditis nor T1D. The pathogenesis of these other autoimmune polyendocrinopathies is not understood.

- **Infections**, particularly tuberculosis and those produced by fungi, may also cause primary chronic adrenocortical insufficiency. **Tuberculous adrenalitis**, which once accounted for as much as 90% of cases of Addison disease, has become less common with the development of antimycobacterial drugs. With the resurgence of tuberculosis in urban centers and the persistence of the disease in low-resource countries, however, this cause of adrenal insufficiency must be kept in mind. When present, tuberculous adrenalitis is usually associated with active infection in other sites, particularly in the lungs and genitourinary tract. Among fungi, disseminated infections caused by *Histoplasma capsulatum* and *Coccidioides immitis* may result in chronic adrenocortical insufficiency. AIDS patients are at risk for developing adrenal insufficiency from several infectious (cytomegalovirus, *Mycobacterium avium-intracellulare*) and noninfectious (Kaposi sarcoma) complications.
- **Metastatic neoplasms** involving the adrenals are another cause of adrenal insufficiency. The adrenals are a fairly common site for metastases in patients with disseminated tumors. Although adrenal function is preserved in most such patients, metastases occasionally destroy enough adrenal cortex to produce adrenal insufficiency. Carcinomas of the lung and breast are the source of most metastases, although many other neoplasms, including gastrointestinal carcinomas, malignant melanoma, and hematopoietic neoplasms, may also metastasize to the adrenals.
- **Genetic causes of adrenal insufficiency** include congenital adrenal hypoplasia (**adrenal hypoplasia congenita**) and **adrenoleukodystrophy**. Adrenoleukodystrophy is caused by mutations of the ATP-binding cassette, subfamily D, member 1 (*ABCD1*) gene and is further described in Chapter 28. Congenital adrenal hypoplasia is a rare X-linked disease caused by mutations in the *NR0B1* gene that encodes DAX1, a transcription factor implicated in adrenal development.

MORPHOLOGY

The anatomic changes in the adrenal glands depend on the underlying disease. **Primary autoimmune adrenalitis** is characterized by irregularly shrunken glands, which may be difficult to identify within the suprarenal adipose tissue. Histologically, the cortex contains only scattered residual cortical cells in a collapsed network of connective tissue. A variable lymphoid infiltrate is present in the cortex and may extend into the adjacent medulla, although the medulla is otherwise preserved (Fig. 24.49). In cases of **tuberculous and fungal disease**, the adrenal architecture is effaced by a granulomatous inflammatory reaction identical to that encountered in other sites of infection. When hypoadrenalism is caused by **metastatic carcinoma**, the adrenals are enlarged, and the normal architecture is obscured by the infiltrating neoplasm.

Clinical Features

Addison disease begins insidiously and does not come to attention until the levels of circulating glucocorticoids and mineralocorticoids are significantly decreased. The initial manifestations include progressive weakness and easy fatigability, which may be dismissed as

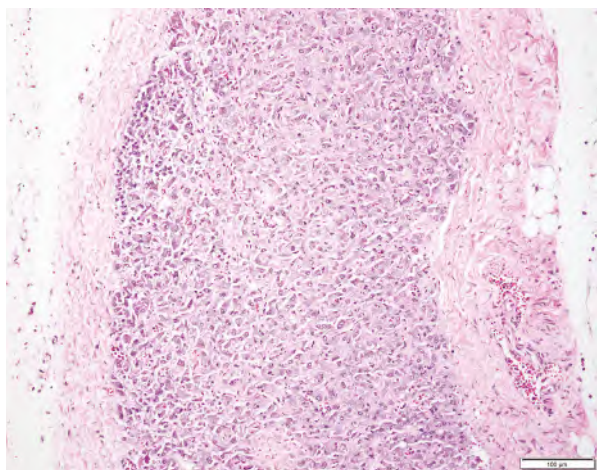


Fig. 24.49 Autoimmune adrenalitis. Although an immune infiltrate may be present, it is not evident here. Rather, due to autoimmune destruction, the adrenal cortex has been completely replaced by connective tissue, while the medulla remains largely preserved.

nonspecific complaints. Gastrointestinal disturbances are common and include anorexia, nausea, vomiting, weight loss, and diarrhea. In individuals with primary adrenal disease, hyperpigmentation of the skin, particularly of sun-exposed areas and at pressure points, such as the neck, elbows, knees, and knuckles, is quite characteristic. This is caused by elevated levels of pro-opiomelanocortin (POMC), which is derived from the anterior pituitary and is a precursor of both corticotropin and melanocyte stimulating hormone (MSH). By contrast, hyperpigmentation is not seen in persons with adrenocortical insufficiency caused by pituitary or hypothalamic disease. Decreased mineralocorticoid activity in persons with primary adrenal insufficiency results in potassium retention and sodium loss, with consequent hyperkalemia, hyponatremia, volume depletion, and hypotension. Patients with secondary adrenal insufficiency (from a pituitary cause) or tertiary adrenal insufficiency (from a hypothalamic cause) do not generally exhibit hyperkalemia, because the renin-angiotensin-aldosterone system is intact. Hypoglycemia may occasionally occur because of glucocorticoid deficiency and impaired gluconeogenesis. Stresses such as infections, trauma, or surgical procedures in such patients can precipitate an acute adrenal crisis, manifested by intractable vomiting, abdominal pain, hypotension, coma, and vascular collapse. Death occurs rapidly unless corticosteroid therapy begins immediately.

KEY CONCEPTS

Adrenocortical Insufficiency (Hypoadrenalism)

- Primary adrenocortical insufficiency can be acute or chronic (Addison disease).
- Chronic adrenal insufficiency in well-resourced countries most often is secondary to autoimmune adrenalitis, which can occur as an isolated condition or in the context of one of two autoimmune polyendocrine syndromes: APS-1 (caused by mutations in the *AIRE* gene) or APS-2.
- Tuberculosis and infections due to opportunistic pathogens associated with AIDS and tumors metastatic to the adrenals are the other important causes of chronic hypoadrenalism.
- Patients typically present with fatigue, weakness, and gastrointestinal disturbances. Primary adrenocortical insufficiency is also characterized by high corticotropin levels with associated skin pigmentation.

Secondary Adrenocortical Insufficiency

Any disorder of the hypothalamus and pituitary, such as metastatic cancer, infection, infarction, or irradiation, that reduces the output of corticotropin leads to a syndrome of hypoadrenalism that has many similarities to Addison disease. Analogously, prolonged administration of exogenous glucocorticoids suppresses the output of corticotropin and adrenal function. As noted earlier, secondary adrenal insufficiency can also develop as part of hypophysitis due to immune checkpoint blockade. With secondary disease, the hyperpigmentation of primary Addison disease is lacking. The manifestations also differ in that secondary hypoadrenalism is characterized by deficient cortisol and androgen output but normal or near-normal aldosterone synthesis. Thus, in adrenal insufficiency secondary to pituitary malfunction, hyperkalemia is not seen.

Corticotropin deficiency can occur alone, but in some instances, it is only one component of panhypopituitarism, associated with multiple trophic hormone deficiencies. Secondary disease can be differentiated from Addison disease by demonstration of low (or inappropriately normal) levels of plasma corticotropin in the former.

Synthetic corticotropin (1-24), or cosyntropin, can be used to diagnose adrenal insufficiency. In patients with primary disease, the destruction of the adrenal cortex reduces the response to exogenously administered corticotropin. In those with long-standing secondary hypofunction, responses to cosyntropin are also diminished, whereas cortisol responses can be normal in the setting of recent-onset secondary adrenal insufficiency.

MORPHOLOGY

In cases of hypoadrenalism secondary to hypothalamic or pituitary disease (**secondary hypoadrenalism**), depending on the severity of corticotropin deficiency, the adrenals may be moderately to markedly decreased in size. The small, flattened glands usually retain their yellow color as a result of a small amount of residual lipid. The cortex may be reduced to a thin ribbon composed largely of zona glomerulosa. The medulla is not affected.

Adrenocortical Neoplasms

It should be evident from the preceding sections that functional adrenal neoplasms may be responsible for any of the various forms of hyperadrenalism. While most cortical neoplasms are sporadic, two familial cancer syndromes are associated with a predisposition for developing adrenocortical carcinomas: Li-Fraumeni syndrome, caused by germline *TP53* mutations (Chapter 7), and Beckwith-Wiedemann syndrome, a disorder of epigenetic imprinting involving the gene for insulin-like growth factor 2 (IGF-2) (Chapter 10). Interestingly, somatic *TP53* mutations and overexpression of IGF-2 also occur in sporadic adrenocortical carcinomas, reiterating an oft-observed phenomenon of shared molecular mechanisms in familial and sporadic neoplasms.

Functional adenomas are most commonly associated with hyperaldosteronism and Cushing syndrome, whereas a virilizing neoplasm is more likely to be a carcinoma. However, not all adrenocortical neoplasms elaborate steroid hormones. Determination of functionality is based on clinical evaluation and measurement of hormones or hormone metabolites in the blood.

MORPHOLOGY

Most **adrenocortical adenomas** are clinically silent and are usually incidental findings at autopsy or during abdominal imaging for an unrelated cause (see the discussion of adrenal “incidentalomas” later). The typical cortical adenoma

is a well-circumscribed nodular lesion up to 2.5 cm in diameter that expands the adrenal (Fig. 24.50). In contrast to a functional adenoma, which is associated with atrophy of the adjacent cortex, the cortex adjacent to a nonfunctional adenoma is normal. On cut surface, adenomas are usually yellow to yellow-brown because of the presence of lipids.

Microscopically, adenomas are composed of cells similar to those populating the normal adrenal cortex. The nuclei tend to be small, although some degree of pleomorphism may be encountered, even in benign lesions ("random endocrine atypia"). The cytoplasm of the neoplastic cells ranges from eosinophilic to vacuolated, depending on their lipid content (Fig. 24.51). Mitotic activity is generally inconspicuous, with a Ki-67 labeling index less than 5% (in contrast to adrenocortical carcinomas). Adrenocortical lesions—both adenomas and carcinomas—express steroidogenic factor-1 (SF-1) and inhibin- α .

Adrenocortical carcinomas are rare neoplasms that have a bimodal distribution in the first and fifth decades of life. They are more likely to be functional than adenomas and are often associated with virilism or other clinical manifestations of hyperadrenalism. In most cases, adrenocortical carcinomas are large, invasive lesions, many exceeding 20 cm in diameter, which efface the native adrenal gland (Fig. 24.52). The less common, smaller, and better-circumscribed lesions may be difficult to distinguish from an adenoma. On cut surface, adrenocortical carcinomas are typically variegated, poorly demarcated lesions containing areas of necrosis, hemorrhage, and cystic change. Adrenal cancers have a strong tendency to invade the adrenal vein, vena cava, and lymphatics. Metastases to regional and periaortic nodes are common, as is distant hematogenous spread to the lungs and other viscera. The median patient survival is about 2 years.

Microscopically, adrenocortical carcinomas may be composed of well-differentiated cells, resembling those seen in cortical adenomas, or bizarre, monstrous giant cells (Fig. 24.53), which may be difficult to distinguish from those of an undifferentiated carcinoma metastatic to the adrenal. Between these extremes are cancers with moderate degrees of anaplasia, some composed predominantly of spindle cells. In addition to higher Ki-67 index, carcinomas also have significantly higher IGF-2 levels compared to adrenocortical adenomas.

Carcinomas, particularly those of bronchogenic origin, may metastasize to the adrenals and immunostains may be needed for definitive diagnosis. Of note, metastases to the adrenal cortex are significantly more common than primary adrenocortical carcinomas.

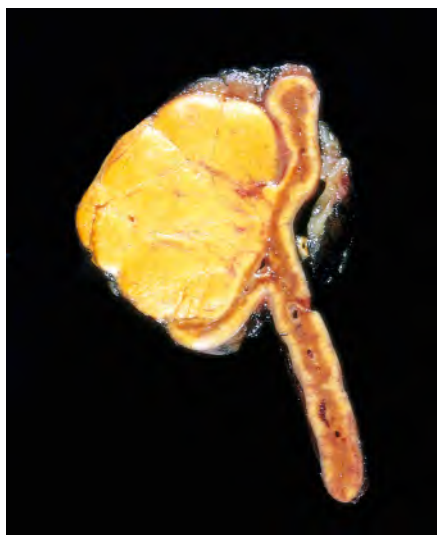


Fig. 24.50 Adrenal cortical adenoma. The adenoma is distinguished from nodular hyperplasia by its solitary, circumscribed nature. The functional status of an adrenal cortical adenoma cannot be predicted from its gross or microscopic appearance.

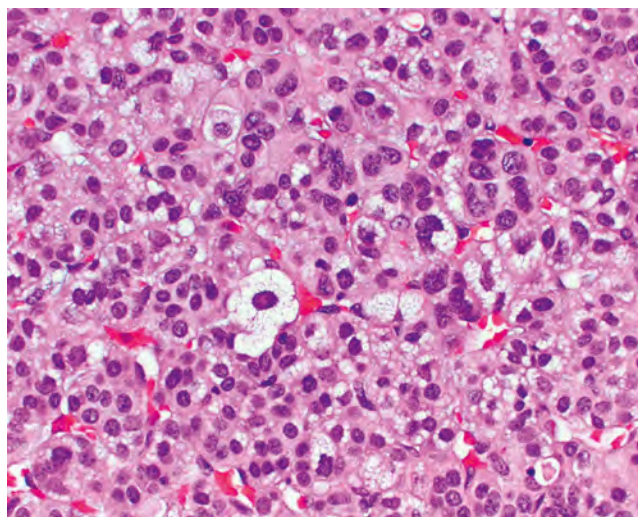


Fig. 24.51 Histologic features of an adrenal cortical adenoma. The neoplastic cells are vacuolated because of the presence of intracytoplasmic lipid. There is mild nuclear pleomorphism. Mitotic activity and necrosis are not seen.

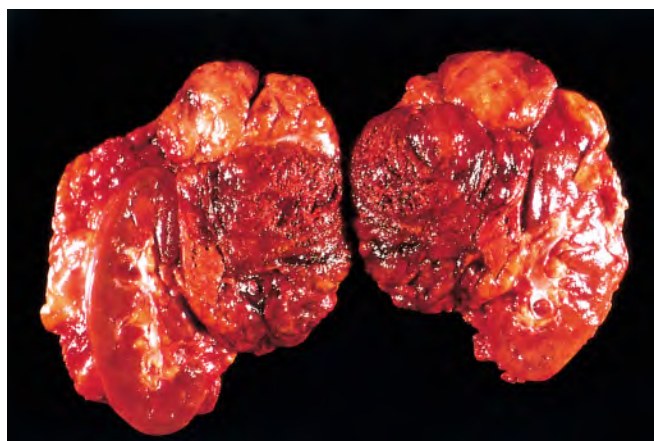


Fig. 24.52 Adrenal carcinoma. The hemorrhagic and necrotic tumor dwarfs the kidney and compresses the upper pole.

Other Adrenal Lesions

Adrenal cysts are relatively uncommon; however, with the use of sophisticated abdominal imaging techniques, the frequency of detection of these lesions is increasing. Larger cysts may produce an abdominal mass and flank pain. Both cortical and medullary neoplasms may undergo necrosis and cystic degeneration and may present as "nonfunctional cysts."

Adrenal myelolipomas are unusual benign lesions composed of mature fat and hematopoietic cells. Although most of these lesions represent incidental findings, occasional myelolipomas may reach massive proportions. Histologically, mature adipocytes are admixed with aggregates of differentiating hematopoietic cells belonging to all three lineages. Foci of myelolipomatous change may be seen in cortical tumors and in adrenals with cortical hyperplasia.

The term *adrenal incidentaloma* has crept into the medical lexicon as widespread medical imaging has led to the incidental discovery of adrenal masses in asymptomatic individuals or in individuals in whom the presenting complaint is not directly related to the adrenal gland. The estimated population prevalence of "incidentalomas"

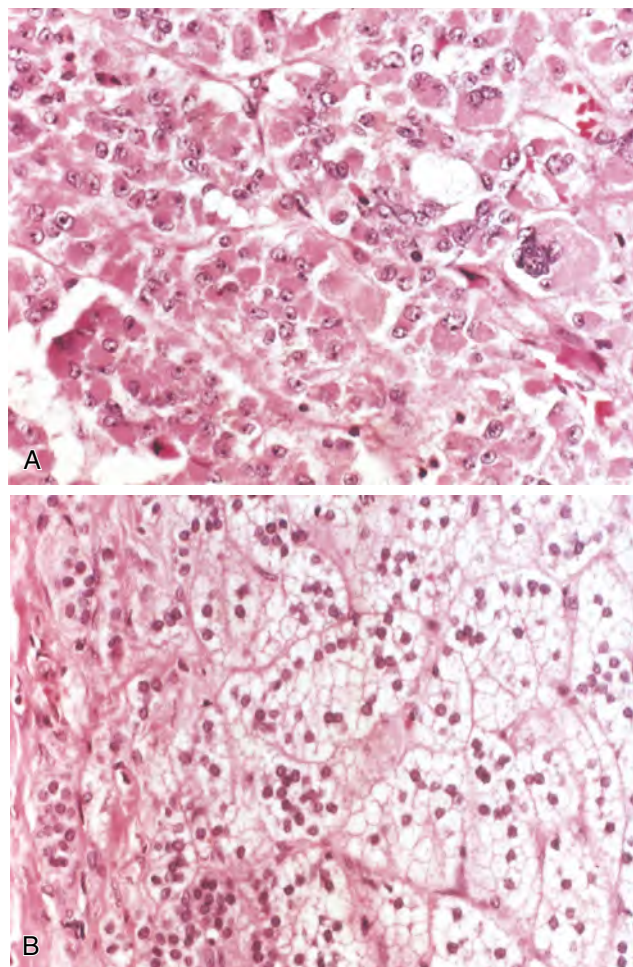


Fig. 24.53 Adrenal carcinoma (A) revealing marked anaplasia, contrasted with normal adrenal cortical cells (B, left side) and medullary cells (B, right side).

discovered by imaging is approximately 4%, with an age-dependent increase in prevalence. Fortunately, many of these lesions are small nonsecreting cortical adenomas of no clinical importance, though others are found to secrete elevated levels of adrenal hormones, particularly cortisol. Larger lesions (particularly those greater than 4 cm in size) or more are likely to be adrenal cancer, but fortunately this is rare. Imaging characteristics, for example Hounsfield units (a measure of radiodensity) identified on CT scans, can help identify suspicious lesions. Recently, it has been shown that patients with adrenal incidentalomas may secrete mildly elevated levels of cortisol, and this is the most common endocrine abnormality in such patients. This process has been termed *subclinical Cushing syndrome* or *mild autonomous cortisol secretion* (MACS). Which patients should undergo surgical resection of their adrenal lesions is an active area of investigation.

ADRENAL MEDULLA

The adrenal medulla is developmentally, functionally, and structurally distinct from the adrenal cortex. It is composed of specialized neural crest (neuroendocrine) cells, termed *chromaffin* cells, and their supporting (sustentacular) cells. The adrenal medulla is a major source of catecholamines (epinephrine, norepinephrine) in the body, though norepinephrine also serves as a neurotransmitter, particularly for

sympathetic nerves. Neuroendocrine cells similar to chromaffin cells are widely dispersed in an extra-adrenal system of clusters and nodules that, together with the adrenal medulla, make up the *paraganglion system*.

The most important diseases of the adrenal medulla are neoplasms, which include neoplasms of chromaffin cells (*pheochromocytomas*) and neuronal neoplasms (*neuroblastic tumors*). Neuroblastomas and other neuroblastic tumors are discussed in Chapter 10. Tumors of sympathetic ganglia are called *paragangliomas*, and these can also secrete catecholamines.

Pheochromocytoma

Pheochromocytomas are neoplasms composed of chromaffin cells, which synthesize and release catecholamines and, in some instances, peptide hormones. It is important to recognize these tumors because they are a rare cause of surgically correctable hypertension.

- *Extra-adrenal pheochromocytomas (paragangliomas)* occurring in sites such as the organs of Zuckerkandl and the carotid body are discussed in Chapter 16.
- *Bilateral adrenal pheochromocytomas* may be present in up to 50% of patients with a familial tumor syndrome (see later).
- *All pheochromocytomas and paragangliomas should be considered potentially malignant*, and the adjectives benign and malignant are no longer recommended. The term *metastatic pheochromocytoma/paraganglioma* is used for cases with metastatic disease. Metastatic behavior is more common (20% to 40%) in extra-adrenal paragangliomas, and in tumors arising in the setting of certain germline mutations (see later).
- *Ten percent of adrenal pheochromocytomas are not associated with hypertension*. Of the 90% that present with hypertension, approximately two-thirds have “paroxysmal” episodes associated with a sudden rise in blood pressure and palpitations, which can, on occasion, be fatal.

It is now recognized that as many as 30%–40% of individuals with pheochromocytomas and paragangliomas harbor an oncogenic germline mutation. These mutations involve at least a dozen genes, the most common of which are listed in Table 24.10. Patients with germline mutations are typically younger at presentation than those with sporadic tumors and more often have bilateral tumors. The affected genes fall into two broad classes, those that enhance growth factor receptor pathway signaling (e.g., *RET*, *NF1*) and those that increase the activity and stability of two hypoxia-induced transcription factors, HIF-1 α and HIF-2 α (thus, creating a so-called “pseudohypoxia” phenotype). *VHL* encodes a tumor-suppressor protein that is needed for the oxygen-dependent degradation of HIF-1 α and that is disrupted by loss-of-function mutations in patients with *von Hippel-Lindau (VHL) syndrome*. VHL syndrome is associated with a number of tumors, including pheochromocytomas and paragangliomas. Similarly, *EPAS1* encodes HIF-2 α , and gain-of-function mutations are found in individuals with the “polycythemia paraganglioma syndrome” that stabilize HIF-2 α by mutating amino acid residues that are required for its degradation. Other familial cases of pheochromocytoma are associated with germline mutations in genes encoding several components of the succinate dehydrogenase complex (most commonly, *SDHB*, *SDHC*, and *SDHD*). This complex is involved in mitochondrial electron transport and oxygen sensing, and it is believed that these mutations also lead to up-regulation of HIF protein, recapitulating “pseudohypoxia.” Recent integrated molecular analyses of pheochromocytomas and paragangliomas have shown that many of the genes implicated in a germline predisposition to these neoplasms (e.g., *NF1*, *VHL*, *EPAS1*, and *RET*) are also involved by somatic mutations in sporadic tumors.

TABLE 24.10 Familial Syndromes Associated With Pheochromocytoma and Extra-adrenal Parangliomas

Syndrome	Gene	Associated Lesion	Other Features
Multiple endocrine neoplasia, type 2A (MEN-2A)	<i>RET</i>	Pheochromocytoma/paranglioma	Medullary thyroid carcinoma Parathyroid hyperplasia
Multiple endocrine neoplasia, type 2B (MEN-2B)	<i>RET</i>	Pheochromocytoma/paranglioma	Medullary thyroid carcinoma Marfanoid habitus Mucocutaneous GNs
Neurofibromatosis, type 1 (NF1)	<i>NF1</i>	Pheochromocytoma	Neurofibromatosis Café-au-lait spots Optic nerve glioma
von Hippel-Lindau (VHL)	<i>VHL</i>	Pheochromocytoma/paranglioma	Renal cell carcinoma Hemangioblastoma Pancreatic neuroendocrine neoplasm
Hereditary paraganglioma 1	<i>SDHD</i>	Pheochromocytoma, paraganglioma	GIST
Hereditary paraganglioma 3	<i>SDHC</i>	Paranglioma	GIST
Hereditary paraganglioma 4	<i>SDHB</i>	Pheochromocytoma, paraganglioma	GIST
Polycythemia paraganglioma syndrome	<i>EPAS1</i>	Pheochromocytoma, paraganglioma	Polycythemia

Additional rare syndromes associated with pheochromocytoma or paragangliomas are not included in this table.

GIST, Gastrointestinal stromal tumor; *GN*, ganglioneuroma; *NF1*, neurofibromin; *SDHB*, succinate dehydrogenase complex, subunit B; *SDHC*, succinate dehydrogenase complex, subunit C; *SDHD*, succinate dehydrogenase complex, subunit D; *EPAS1*, endothelial PAS-domain protein 1 (encodes for HIF-2 α).

MORPHOLOGY

Pheochromocytomas range from small, circumscribed lesions confined to the adrenal (Fig. 24.54) to large hemorrhagic masses weighing kilograms. The average weight of a pheochromocytoma is 100 g, but weights from just over 1 g to almost 4000 g have been reported. The larger tumors are well demarcated by either connective tissue or compressed cortical or medullary tissue. Richly vascularized fibrous trabeculae within the tumor produce a lobular pattern. In many tumors, remnants of the adrenal gland can be seen, stretched over the surface or attached at one pole. On section, the cut surfaces of smaller pheochromocytomas are yellow-tan. Larger lesions tend to be hemorrhagic, necrotic, and cystic and typically efface the adrenal gland. Incubation of fresh tissue with a potassium dichromate solution turns the tumor a dark brown color due to oxidation of stored catecholamines, thus the term *chromaffin*.

The histologic pattern in pheochromocytoma is quite variable. The tumors are composed of clusters of polygonal to spindle-shaped chromaffin cells or chief cells that are surrounded by supporting sustentacular cells, creating small nests or alveoli (**zellballen**) that are supplied by a rich vascular network (Fig. 24.55). Uncommonly, the dominant cell type is a spindle or small cell; various patterns can be found in any one tumor. The cytoplasm has a finely granular appearance, best demonstrated with silver stains, due to the presence of granules containing catecholamines. The nuclei are usually round to ovoid, with a stippled

“salt-and-pepper” chromatin that is characteristic of neuroendocrine tumors. Electron microscopy reveals variable numbers of membrane-bound, electron-dense secretory granules (Fig. 24.56). Immunoreactivity for neuroendocrine markers (chromogranin and synaptophysin) is seen in the chief cells, while the peripheral sustentacular cells stain with antibodies against S-100, a calcium-binding protein expressed by a variety of mesenchymal cell types.

Determining malignancy in pheochromocytomas has been attempted but proven futile. Several histologic features, such as numbers of mitoses, confluent tumor necrosis, and spindle cell morphology, have been associated with an aggressive behavior and increased risk of metastasis, but are not entirely reliable. Tumors with “benign” histologic features may metastasize, while bizarrely pleomorphic tumors may remain confined to the adrenal gland. In fact, cellular and nuclear pleomorphism, including the presence of giant cells, and mitotic figures are often seen in benign pheochromocytomas, while cellular monotony is paradoxically associated with aggressive behavior. Even capsular and vascular invasion may be encountered in these lesions. Therefore, the definitive diagnosis of malignancy in pheochromocytomas is based exclusively on the presence of metastases. These may involve regional lymph nodes as well as more distant sites, including liver, lung, and bone.

Clinical Features

The dominant clinical manifestation of pheochromocytoma is *hypertension*, observed in 90% of patients. Approximately two-thirds of pheochromocytoma patients with hypertension demonstrate *paroxysmal episodes*, which are described as an abrupt, precipitous elevation in blood pressure associated with tachycardia, palpitations, headache, sweating, tremor, and a sense of apprehension. These episodes may also be associated with pain in the abdomen or chest, nausea, and vomiting. Isolated paroxysmal episodes of hypertension occur in less than one-half of patients; more commonly, patients demonstrate chronic, sustained elevation in blood pressure punctuated by the aforementioned paroxysms. The paroxysms may be precipitated by

emotional stress, exercise, changes in posture, and palpation in the region of the tumor; urinary bladder paragangliomas occasionally precipitate a paroxysm during micturition. The elevations of blood pressure are induced by the sudden release of catecholamines that may acutely precipitate congestive heart failure, pulmonary edema, myocardial infarction, ventricular fibrillation, and cerebrovascular accident.

The cardiac complications have been attributed to what has been called *catecholamine cardiomyopathy*, or catecholamine-induced myocardial instability and ventricular arrhythmias. Nonspecific myocardial changes, such as focal necrosis, mononuclear infiltrates, and interstitial fibrosis, have been attributed either to ischemic damage secondary to

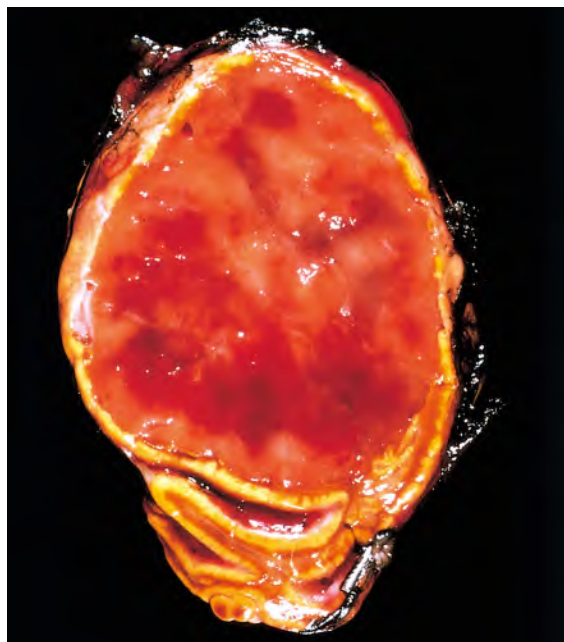


Fig. 24.54 Pheochromocytoma. The tumor is enclosed within an attenuated cortex and demonstrates areas of hemorrhage. The comma-shaped residual adrenal is seen below. (Courtesy Dr. Jerrold R. Turner, Department of Pathology, Brigham and Women's Hospital, Boston.)

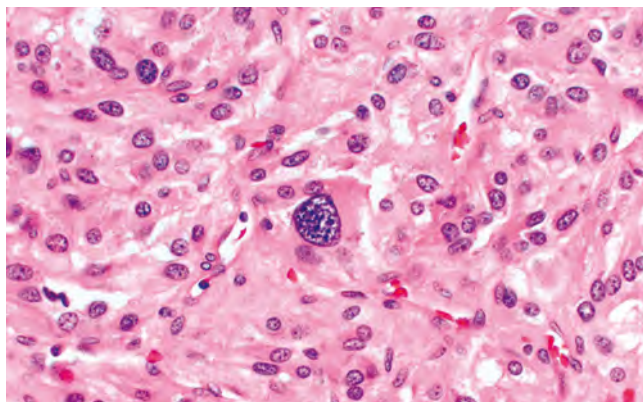


Fig. 24.55 Pheochromocytoma demonstrating characteristic nests of cells ("zellballen") with abundant cytoplasm. Granules containing catecholamine are not visible in this preparation. It is not uncommon to find bizarre cells, even in pheochromocytomas that are biologically benign. (Courtesy Dr. Jerrold R. Turner, Department of Pathology, Brigham and Women's Hospital, Boston.)

catecholamine-induced vasospasm or to direct catecholamine toxicity. In some cases, pheochromocytomas secrete other hormones, such as corticotropin and somatostatin, and may therefore be associated with clinical features related to the secretion of these or other peptide hormones. Laboratory diagnosis of pheochromocytoma is based on the demonstration of increased urinary excretion of free catecholamines and their metanephrine metabolites, or elevated plasma levels of metanephrines.

Isolated benign tumors are treated with surgical excision after preoperative and intraoperative medication with adrenergic-blocking agents to prevent a hypertensive crisis. Multifocal lesions require long-term medical treatment for hypertension.

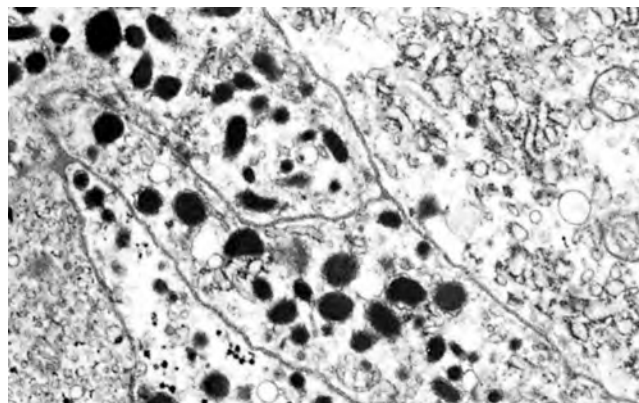


Fig. 24.56 Electron micrograph of pheochromocytoma. This tumor contains membrane-bound secretory granules in which catecholamines are stored (30,000 $\times$).

MULTIPLE ENDOCRINE NEOPLASIA SYNDROMES

The MEN syndromes are a group of inherited diseases resulting in proliferative lesions (hyperplasia, adenomas, and carcinomas) of multiple endocrine organs. Like other inherited cancer disorders (Chapter 7), endocrine tumors arising in the context of MEN syndromes have certain distinct features that contrast with their sporadic counterparts.

- Tumors occur at a *younger age* than sporadic tumors.
- They arise in *multiple endocrine organs*, either *synchronously* (at the same time) or *metachronously* (at different times).
- Even in one organ, the tumors are often *multifocal*.
- The tumors are usually preceded by an *asymptomatic stage of hyperplasia* involving the cell of origin. For example, individuals with MEN-2 almost universally demonstrate C-cell hyperplasia in the thyroid parenchyma adjacent to medullary thyroid carcinomas.
- These tumors are usually *more aggressive* and *recur* in a higher proportion of cases than do similar sporadic endocrine tumors.

MULTIPLE ENDOCRINE NEOPLASIA, TYPE 1

MEN-1, or *Wermer syndrome*, is a rare heritable disorder with a prevalence of about 2 per 100,000. MEN-1 is characterized by abnormalities involving the parathyroid, pancreas, and pituitary gland; thus the mnemonic device, the three Ps:

- **Parathyroid:** Primary hyperparathyroidism is the most common manifestation of MEN-1 (80% to 95% of patients) and is the initial manifestation of the disorder in most patients, appearing in almost all patients by 40 to 50 years of age. Parathyroid abnormalities include both hyperplasia and adenomas.
- **Pancreas:** Endocrine tumors of the pancreas are a leading cause of morbidity and mortality in persons with MEN-1. These tumors are usually aggressive and often present with metastatic disease. It is not uncommon to find multiple "microadenomas" scattered throughout the pancreas in conjunction with one or two dominant lesions. MEN-1-associated pancreatic endocrine tumors are often functional; however, because pancreatic polypeptide is the most commonly secreted product, many tumors fail to produce an endocrine hypersecretion syndrome. Among those that do, Zollinger-Ellison syndrome (associated with gastrinomas) and hypoglycemia and neurologic manifestations (associated with insulinomas) are most common.
- **Pituitary:** The most frequent anterior pituitary tumor encountered in MEN-1 is a prolactinoma; some patients develop acromegaly from somatotropin-secreting tumors.

It is now recognized that the spectrum of this disease extends beyond the three *Ps*. The duodenum is the most common site of gastrinomas in individuals with MEN-1 (far in excess of the frequency of pancreatic gastrinomas), and synchronous duodenal and pancreatic tumors may be present in the same individual. In addition, carcinoid tumors, thyroid and adrenocortical adenomas, and lipomas are more frequent than in the general population.

MEN-1 syndrome is caused by germline mutations in the *MEN1* tumor suppressor gene, which encodes a protein called *menin*. *Menin* is a component of several different transcription factor complexes, which, depending on the specific binding partner and the cell context, may either promote or inhibit tumorigenesis.

The dominant manifestations of MEN-1 usually result from the peptide hormones that are overproduced and include such abnormalities as recurrent hypoglycemia due to insulinomas, intractable peptic ulcers in persons with Zollinger-Ellison syndrome, nephrolithiasis caused by PTH-induced hypercalcemia, or symptoms of prolactin excess from a pituitary tumor. As expected, malignant behavior by one or more of the endocrine tumors arising in these patients is often the proximate cause of death.

MULTIPLE ENDOCRINE NEOPLASIA, TYPE 2

MEN-2 is subclassified into three distinct syndromes: MEN-2A, MEN-2B, and MEN-4.

- **MEN-2A, or *Sipple syndrome*, is characterized by pheochromocytoma, medullary carcinoma of the thyroid, and parathyroid hyperplasia (see Table 24.10).** Medullary carcinomas of the thyroid occur in almost 100% of patients. They are usually multifocal and are virtually always associated with foci of C-cell hyperplasia in the adjacent thyroid. Medullary carcinomas may elaborate calcitonin and other active products and are usually clinically aggressive. Among individuals with MEN-2A, 40% to 50% have pheochromocytomas, which are often bilateral and may arise in extra-adrenal sites. Parathyroid hyperplasia and evidence of hypercalcemia or renal stones occur in 10% to 20% of patients. **MEN-2A is clinically and genetically distinct from MEN-1 and is caused by germline gain-of-function mutations in the *RET* proto-oncogene.** As noted earlier, the *RET* proto-oncogene encodes a receptor tyrosine kinase that binds glial-derived neurotrophic factor and other related ligands and transmits growth and differentiation signals (Chapter 7). Loss-of-function mutations in *RET* result in intestinal aganglionosis and Hirschsprung disease (Chapter 17). In contrast, in MEN-2A (as well as in MEN-2B), germline mutations constitutively activate the RET receptor. Families with isolated medullary thyroid carcinomas and germline *RET* mutations without additional features of MEN-2 are often classified as a separate entity (“familial medullary thyroid carcinoma”), but it is important to look carefully for other aspects of a MEN2 phenotype.
- **MEN-2B has significant clinical overlap with MEN-2A.** Patients develop medullary thyroid carcinomas, which are usually multifocal and more aggressive than in MEN-2A, and pheochromocytomas. However, unlike in MEN-2A, primary hyperparathyroidism is not present. In addition, MEN-2B is accompanied by neuromas or ganglioneuromas involving the skin, oral mucosa, eyes, respiratory tract, and gastrointestinal tract, and a marfanoid habitus, with long axial skeletal features and hyperextensible joints (see Table 24.10). A germline missense mutation leading to a single amino acid change in RET, distinct from the mutations that are seen in MEN-2A, seems to be responsible for virtually all cases of MEN-2B. This point substitution affects a critical region of the tyrosine kinase domain of the protein and leads to constitutive activation of RET in the absence of ligand.

In contrast to MEN-1, in which the long-term benefit of early diagnosis by genetic screening is more controversial, diagnosis via screening of at-risk family members in MEN-2 kindreds is clearly important because medullary thyroid carcinoma is a life-threatening disease that can be prevented by early thyroidectomy. According to recommendations issued in 2015, all patients with medullary thyroid carcinoma are advised to undergo germline *RET* mutation testing, since even “apparently sporadic” cases might harbor a germline mutation, which has implications for genetic counseling and testing for unaffected family members. Routine genetic testing identifies *RET* mutation carriers earlier and more reliably in MEN-2 kindreds; all individuals carrying germline *RET* mutations are advised to undergo prophylactic thyroidectomy to prevent the inevitable development of medullary carcinomas.

- **MEN-4** has emerged as a new entity in the past decade. These patients have clinical features that phenocopy MEN-1 patients, but in contrast to that syndrome, harbor germline *CDKN1B* mutations, leading to reduced levels of the cell-cycle checkpoint protein p27.

PINEAL GLAND

The French philosopher-scientist Rene Descartes called the pineal the “seat of rational thought (the soul),” but this gland gets much less attention in clinical medicine. Pineal gland disorders (virtually only tumors) are rare. The pineal gland is a minute, pinecone-shaped organ (hence its name), weighing 100 to 180 mg and lying between the superior colliculi at the base of the brain. It is composed of a loose, neuroglial stroma that encloses nests of epithelial-appearing *pinocytes*, cells with photosensory and neuroendocrine functions (hence the designation of the pineal gland as the “third eye”). Silver impregnation stains reveal that these cells have long, slender processes reminiscent of primitive neuronal precursors intermixed with the processes of astrocytic cells. The principal secretory product of the pineal gland is melatonin, which is involved in the control of circadian rhythms, including the sleep-wake cycle; hence the popular use of melatonin for the treatment of jet lag.

All tumors involving the pineal gland are rare; most arise from sequestered embryonic germ cells (Chapter 28). Pinealomas are divided into two main categories, *pineoblastomas* and *pineocytomas*, based on their level of differentiation, which, in turn, correlates with their aggressiveness. These tumors are rare and are described in specialized texts.

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SELECTED READINGS

- Acquaviva G, Visani M, Repaci A et al: Molecular pathology of thyroid tumors of follicular cells: a review of genetic alterations and their clinic-pathologic relevance, *Histopathology* 72:6–31, 2018. [A review on genetics of thyroid neoplasms on how the landscape of thyroid neoplasms has changed over time, with excellent summary figures on the multistep pathogenesis of invasive thyroid cancers from precursor lesions.]
- Asa SL, Mete O, Perry A, Osamura RY: Overview of the 2022 WHO Classification of Pituitary Tumors, *Endocrine Pathology* 33, 6–26, 2022. [A review on the updated WHO classification of pituitary tumors and other nonneoplastic entities.]
- Bilezikian JP, Khan AA, Silverberg, et al: Evaluation and management of primary hyperparathyroidism: summary statement and guidelines from the

- fifth international workshop, J Bone Miner Res*, 37 (11): 2293-2314, 2022. [The most recent set of international guidelines on the evaluation and management of primary hyperparathyroidism.]
- Broome DT, Pantalone KM, Kashyap SR, and Philipson LH: *Approach to the patient with MODY-monogenic diabetes, J Clin Endocrinol Metab*, 106 (1): 237-250, 2021. [This review focuses on clinically important aspects of the evaluation and management of patients with certain genetic forms of diabetes.]
- Charmandari E, Nicolaides NC, Chrousos GP: *Adrenal insufficiency, Lancet* 383:2152-2167, 2014. [A comprehensive review on the epidemiology, pathogenesis, clinical features, diagnosis, and management of adrenal insufficiency.]
- Chung WK, Erion K, Florez JC, et al: *Precision medicine in diabetes: a consensus report from the American Diabetes Association (ADA) and the European Association for the Study of Diabetes (EASD), Diabetes Care* 43(7): 1617-1635, 2020. [A report on precision medicine as it relates to diabetes, including prospects for the future.]
- Czech MP: *Insulin action and resistance in obesity and type 2 diabetes, Nat Med* 23:804-814, 2017. [A comprehensive review on the pathophysiology of obesity and insulin resistance in type 2 diabetes with some wonderful summary figures on the interplay between the pancreas, liver, adipose tissue, and muscle.]
- Davies TF, Andersen S, Latif R, et al: *Graves' disease, Nat Rev Dis Primers* 2020;6(1): 52. [A review on the pathogenesis and treatment of Graves' disease.]
- Drucker DJ, Habener JF, Holst JJ: *Discovery, characterization, and clinical development of glucagon-like peptides, J Clin Invest* 127:4217-4227, 2017. [GLP-1 receptor agonists have emerged as one of the most important therapeutic options in diabetes, and this review provides a historical timeline and mechanisms of action of GLP agonists in human diseases. This is worth reading to understand how complex and drawn-out the process of successful drug development can be!]
- Fagin JA and Nikiforov YE: *Progress in thyroid cancer genomics: a 40-year journey, Thyroid*, 2023;33(11):1271-1286. [A very nice discussion of the history and discoveries that led to our current understanding of the genetics of thyroid cancer.]
- Ho KKY, Kaiser UB, Chanson P, et al: *Pituitary adenoma or neuroendocrine tumour: the need for an integrated prognostic classification, Nat Rev Endocrinol* 19: 671-678, 2023. [Report of a Pituitary Society workshop to understand how the WHO 2022 classification guidelines will affect patient care.]
- Husebye ES, Anderson MS, Kampe O: *Autoimmune polyendocrine syndromes, N Engl J Med* 378:1132-1141, 2018. [A review on the genetics, pathophysiology, and clinical manifestations of autoimmune polyendocrine syndromes.]
- Loriaux DL: *Diagnosis and differential diagnosis of Cushing syndrome, N Engl J Med* 376:1451-1459, 2017. [A fairly recent clinically oriented review on the etiopathogenesis, laboratory diagnosis, and treatment of Cushing syndrome. This review is a great place to get started.]
- Mete O, Erickson LA, Juhlin CC, et al: *Overview of the 2022 WHO classification of adrenal cortical tumors. Endocr Pathol* 33:155-196, 2022. [This commentary summarizes the major changes to the classification and genetics of adrenal neoplasms.]
- Minisola S, Arnold A, Belaya Z, et al: *Epidemiology, pathophysiology, and genetics of primary hyperparathyroidism, J Bone Min Res*, 37(11): 2315-2329, 2022. [A comprehensive review on the underlying pathophysiology and genetic abnormalities contributing to hyperparathyroidism.]
- Pacheco MC: *Multiple endocrine neoplasia: a genetically diverse group of familial tumor syndromes, J Pediatr Genet* 5:89-97, 2016. [A review on the subtypes, genetics, and pathology of MEN syndromes.]
- Rindi G, Mete O, Uccella, et al: *Overview of the 2022 WHO classification of neuroendocrine neoplasms, Endocrine Pathology*, 33: 115-154, 2022. [The WHO has recently updated the classification of neuroendocrine neoplasms, including well-differentiated neuroendocrine tumors (NETs) and poorly differentiated neuroendocrine carcinomas (NECs).]
- Rushworth RL, Torpy DJ, Falhammar H: *Adrenal crisis, N Engl J Med* 381:852-861, 2019. [A review on the causes, pathophysiology, and treatment of acute adrenal crises.]
- Schlumberger M and Leboulleux: *Current practice in patients with differentiated thyroid cancer, Nat Rev Endocrinol*, 17(3): 176-188, 2021. [A clinical review focusing on patients with thyroid cancer.]
- Szczerbinski L and Florez JC: *Precision medicine of obesity as an integral part of type 2 diabetes management – past, present, and future, Lancet Diabetes Endocrinol*, 11(11):861-878, 2023. [This review discusses the genetic underpinnings of obesity and the possible implications for precision medicine.]